Groups and Rings

Labix

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Contents

1 Introduction to Groups

Abstract algebra is the study of symmetry, and the most basic form of symmetry is refined into the definition of a group. However the notion of a group was not explicitly defined until set theory and the formalization of a group. Despite this, the concept of groups and group actions had been made use by a number of Mathematicians, such as Galois in his study of the symmetries of the roots of a polynomial.

1.1 Basic Concepts of Groups

Formally, the definition of a group involves a set and an operation on the set that gives rise to symmetry.

Definition 1.1.1 (Binary Operation)
A binary operation $*$ on a set G is a function

$$* : G \times G \to G$$

Definition 1.1.2 (Groups)
A group is an ordered pair $(G, *)$ where G is a set and $* : G \times G \to G$ is a binary operation on G satisfying the following axioms.

- Associativity: $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$
- Identity element: There exists an element $e \in G$, called an identity of G, such that for all $a \in G$ we have $a * e = e * a = a$
- Inverse element: For each $a \in G$ there is an element a^{-1} of G, called an inverse of a, such that $a * a^{-1} = a^{-1} * a = e$.

We often call the binary operation the group operation of $(G, *)$ to distinguish other algebraic structures on $(G, *)$ if it exists. It is common to omit the binary operation when defining a group.

Example 1.1.3
The following are groups.

- Let $n \in \mathbb{N}$. Let $G = \{e^{2\pi ik/n} \in \mathbb{C} \mid 0 \leq k \leq n-1\}$. Then G is a group with complex multiplication.
- Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{N}\}$. Then G is a group with complex multiplication.
- Let $G = \{a + bi \in \mathbb{C} \mid \sqrt{a^2 + b^2} = 1\}$. Then G is a group with complex multiplication.

Definition 1.1.4 (Abelian Groups)
Let $(G, *)$ be a group. We say that G is abelian if for all $a, b \in G$, we have

$$a * b = b * a$$

Example 1.1.5
The following are true.

- $(\mathbb{N}, +)$ is not a group.
- $(\mathbb{Z}, +)$ is an abelian group.
- $(\mathbb{R} \setminus \{0\}, \times)$ is an abelian group.

Proof
No element in $\mathbb{N}$ is an inverse for $1 \in \mathbb{N}$.

Addition in $\mathbb{Z}$ is associative, commutative and has an identity as proved in Number Systems.

Multiplication in $\mathbb{R} \setminus \{0\}$ is associative, commutative and has an identity as proved in Number Systems. ■

Lemma 1.1.6
Let G be a group. Then the following are true regarding the identity and inverses within G.

- The identity of G is unique.
- For each $a \in G$, a^{-1} is unique.
- For all $a \in G$, we have $(a^{-1})^{-1} = a$.
- For all $a, b \in G$, we have $(ab)^{-1} = (b^{-1})(a^{-1})$.

Proof
Let e, f be identities of G. Let $a \in G$. Since e is the identity, $ef = e$. Since f is the identity, $ef = f$. Thus $e = ef = f$.

Suppose that $b, c \in G$ are inverses of a. Since groups are associative, $(ba)c = b(ac)$. From the left, $(ba)c = ec = c$. From the right, $b(ac) = be = b$. Thus $b = (ba)c = b(ac) = c$.

Suppose that $a^{-1} = b$. Since the inverse of a is b, we have $ab = e$. But since $ab = e$, the inverse of b is a.

Suppose that the inverse of a is c and the inverse of b is d. Then $(ab)(dc) = a(bd)c = ac = e$. Thus the inverse of ab is $dc = b^{-1}a^{-1}$. ∎

Definition 1.1.7 (Order of an Element)
For a group G and an element $x \in G$ define the order of x to be the smallest integer n such that $x^n = 1$, and denote this integer by

$$|x| = \min\{n \in \mathbb{N} \mid x^n = 1\}$$

If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order.

Definition 1.1.8 (Order of a Group)
Let G be a group. The order of G is the number of elements that G has, denoted by $|G|$. We say that G is a finite group if $|G|$ is a finite number. Otherwise G is an infinite group.

A group does not necessarily have to contain an element of the same order of the group.

Example 1.1.9 (The Klein Four Group)
Let $K = \{1, a, b, c\}$. Define a binary operation $\cdot : K \times K \to K$ by the following rules.

- For all $x \in K$, we have $1 \cdot x = x = x \cdot 1$.
- For all $x \in K$, we have $x \cdot x = 1$.
- $a \cdot b = b \cdot a = c$.
- $a \cdot c = c \cdot a = b$.
- $b \cdot c = c \cdot b = a$.

Then $(K, \cdot)$ is a group, and $|a| = |b| = |c| = 2$.

Lemma 1.1.10
Let G be a group. Let $k \in \mathbb{N}$. Then $a^k = 1$ if and only if $|a|$ divides k.

Proof
If $k = n|a|$ for some $n \in \mathbb{N}$ then we have

$$a^k = a^{|a|n} = (a^{|a|})^n = 1$$

Conversely if $a^k = 1$, then since $|a|$ is by definition the smallest number such that the $a^{|a|} = 1$, we can apply division algorithm to obtain $k = |a|q + r$ for some $q \in \mathbb{N}$ and $0 \leq r < n$. Now $a^k = a^{|a|q+r}$ implies that $1 = a^r$, which contradicts the minimality of $|a|$. Thus we conclude. ∎

Lemma 1.1.11
Let G be a group and $g \in G$ an element of finite order n. Then

$$|g^k| = \frac{n}{\gcd(k,n)} = \frac{\operatorname{lcm}(k,n)}{k}$$

for any $k \in \mathbb{N}$

Proof
The equality between the two expressions is trivial since $\gcd(m,n) \times \operatorname{lcm}(m,n) = m \times n$. Now let $b = \frac{n}{\gcd(k,n)}$ and $c = \frac{k}{\gcd(k,n)}$. Then we must have $\gcd(b,c) = 1$. Now

$$\begin{aligned}(g^k)^b &= g^{kb} \\ &= g^{\gcd(k,n)cb} \\ &= g^{nc} \\ &= (g^n)^c \\ &= 1\end{aligned}$$

Thus we must have $|g^k|$ divides $\frac{n}{\gcd(k,n)}$. Now we know that $g^{k|g^k|} = (g^k)^{|g^k|} = 1$. Thus we must have n divides $k|g^k|$, meaning $\gcd(k,n)b$ divides $\gcd(k,n)c|g^k|$ and b divides $c|g^k|$. We know that $\gcd(b,c) = 1$ thus b divides $|g^k|$. From the fact that $b = \frac{n}{\gcd(k,n)}$ we have $|g^k|$ divides b as well and thus $b = |g^k|$ and

$$|g^k| = \frac{n}{\gcd(k,n)}$$

■

Lemma 1.1.12
Let G be a group. Let $a, b \in G$ be elements that commute with each other. Then $|ab|$ divides $\operatorname{lcm}(|a|, |b|)$.

Proof
Let $l = \operatorname{lcm}(|a|, |b|)$. Since a and b commutes, we have that

$$(ab)^l = a^l b^l = 1$$

and so $|ab|$ divides l. ■

1.2 Subgroups

The idea of a subgroup is that some (not all) elements in a group can also form a group in and of itself. Most of the structure of a group carries on to its subgroup, while the converse may not necessarily be true.

Definition 1.2.1 (Subgroups)
Let $(G, *)$ be a group. A subgroup of G is a subset $H \subseteq G$ such that $(H, *)$ is a group. We denote that H is a subgroup of G by $H \leq G$.

Example 1.2.2
Let $n \in \mathbb{N}^+$. Then $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

Subgroups are still part of the larger group. This means that if H is a subgroup of G, then certain properties of H are inherited from G. For example, the uniqueness of the identity element ensures that H and G share the same identity element.

Lemma 1.2.3
Let G be a group and $H \leq G$. Then $1_H = 1_G$.

Proof
For all $h \in H$, we have $h \cdot 1_G = h$ since $h \in G$. Hence 1_G is the unique identity element of H. ∎

To check that a subset of a group is a subgroup, it is not necessary to go through the three axioms of a group as it is rather tedious. Instead, we may turn to an easier criterion which involves less work.

Proposition 1.2.4 (The Subgroup Criterion)
Let $(G, *)$ be a group. A subset H of G is a subgroup if and only if the following are true.

- If $a, b \in H$ then $a * b \in H$
- If $a \in H$ then $a^{-1} \in H$

Proof
If H is a subgroup of G, then by definition of a group the above are all true.
Suppose that H is satisfies the above two conditions. We show that it implies the four definitions of a group. Since $a, b \in H$ implies $ab \in H$, closure is satisfied. Since $H \subseteq G$, and G is associative, H is also associative. If $a \in H$ then $a^{-1} \in H$ thus inverse exists. Finally since H is closed, $aa^{-1} = 1 \in H$ thus identity is in H. Thus H is a group. ∎

Proposition 1.2.5
Let G be a group. Let $H, K \leq G$ be subgroups. Then $H \cap K \leq G$ is a subgroup.

Proof
Suppose that H, K are subgroups of G. We show that $H \cap K$ with the subgroup criterion. We have that

- $1 \in H$ and $1 \in K$ thus $1 \in H \cap K$
- Suppose that $a, b \in H \cap K$. Then since $ab \in H$ and $ab \in K$ we have $ab \in H \cap K$
- Suppose that $a \in H \cap K$ since H and K are subgroups respectively, $a^{-1} \in H$ and $a^{-1} \in K$ thus $a^{-1} \in H \cap K$.

The subgroup criterion is satisfied thus $H \cap K$ is a subgroup of G. ∎

1.3 Homomorphisms and Isomorphisms

Homomorphisms and isomorphisms are important concepts not only in terms of the ability to classify groups with similar structure, but is also useful in the creation of new groups, such as subgroups and quotient groups. One common idea throughout all of mathematics is to not only study the object itself, but to also study functions in and out of the object. In the case of groups, a meaningful function would be one that preserve the study of a group. This leads to the definition of a group homomorphism.

Definition 1.3.1 (Group Homomorphisms)
Let $(G, *)$ and $(H, \circ)$ be groups. A map $\phi : G \to H$ such that

$$\phi(x * y) = \phi(x) \circ \phi(y)$$

for all $x, y \in G$ is called a homomorphism.

As a result of preserving group structures, group homomorphisms are more rigid compared to a regular function.

Lemma 1.3.2
Let G, H be groups. Let $\phi : G \to H$ be a group homomorphism. Then the following are true regarding ϕ.

- $\phi(1) = 1$

- $\phi(g^{-1}) = \phi(g)^{-1}$ for all $g \in G$
- $\phi(g^n) = \phi(g)^n$ for all $n \in \mathbb{N}$ and all $g \in G$

Proof
We have that

$$\begin{aligned} \phi(1) &= \phi(1 \cdot 1) \\ \phi(1) &= \phi(1) \cdot \phi(1) \\ \phi(1)\phi(1)^{-1} &= \phi(1)\phi(1)\phi(1)^{-1} \\ 1 &= \phi(1) \end{aligned}$$

We have that

$$\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1) = 1$$

and

$$\phi(g)\phi(g)^{-1} = 1$$

Also since the inverse is unique, we have that $\phi(g^{-1}) = \phi(g)^{-1}$.

We have that $\phi(g)\cdots\phi(g) = \phi(g\cdots g)$ by applying the definition of homomorphism $n-1$ times.
■

Lemma 1.3.3
The following are true regrading group homomorphisms.

- Let G, H, K be groups. Let $\phi : G \to H$ and $\psi : H \to K$ be group homomorphisms. Then $\psi \circ \phi$ is a group homomorphism.
- Let G be a group. Then $\text{id}_G : G \to G$ is a group homomorphism.

Proof
Let $g_1, g_2 \in G$. Then we have

$$\psi(\phi(g_1g_2)) = \psi(\phi(g_1)\phi(g_2)) = \psi(\phi(g_1))\psi(\phi(g_2))$$

so that $\psi \circ \phi$ is a group homomorphism.

Let $g_1, g_2 \in G$. Then we have

$$\text{id}_G(g_1g_2) = g_1g_2 = \text{id}_G(g_1)\text{id}_G(g_2)$$

so that id_G is a group homomorphism. ■

Definition 1.3.4 (Kernel)
Let G, H be groups. Let $\phi : G \to H$ be a homomorphism. Define the kernel of ϕ to be

$$\ker(\phi) = \{g \in G | \ker(g) = 1_H\}$$

Lemma 1.3.5
Let G, H be groups. Let $\phi : G \to H$ be a homomorphism. Then the following are true regarding the kernel.

- Then $\ker(\phi)$ is a subgroup of G.
- ϕ is injective if and only if $\ker(\phi) = \{1_G\}$.

Proof
We prove closure, identity and inverse as stated by the subgroup criterion.

- $\phi(1_G) = 1_H$, thus we have $1_G \in \ker(\phi)$
- If $a, b \in \ker(\phi)$, then $\phi(a) = 1$ and $\phi(b) = 1$ and
$$\phi(ab) = \phi(a)\phi(b) = 1$$
thus $ab \in \ker(\phi)$.
- If $a \in \ker(\phi)$ then
$$1 = \phi(1) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1}) = \phi(a^{-1})$$
By the subgroup criterion $\ker(\phi) \leq G$.

Suppose that ϕ is injective. Let $g \in \ker(\phi)$. Then we have that $\phi(g) = 1_H = \phi(1_G)$. By injectivity of ϕ we know that $g = 1_G$. Hence $\ker(\phi) = 1_G$. Now suppose that $\ker(\phi) = \{1_G\}$. Suppose that $\phi(g_1) = \phi(g_2)$. Then we have

$$\begin{aligned}\phi(g_1) &= \phi(g_2)\\ \phi(g_1)\phi(g_2)^{-1} &= 1_H\\ \phi(g_1g_2^{-1}) &= 1_H\end{aligned}$$

Then $g_1g_2^{-1} \in \ker(\phi) = \{1_G\}$, we have that $g_1g_2^{-1} = 1_G$ so that $g_1 = g_2$. Hence ϕ is injective. ■

When a map of sets have an inverse, we say that the two sets are bijective. When a group homomorphisms has an inverse, we have the corresponding notion called isomorphic.

Definition 1.3.6 (Group Isomorphisms)
Let G, H be groups. Let $\phi : G \to H$ be a group homomorphism.
- We say that ϕ is a group isomorphism if ϕ is bijective.
- We say that G and H are isomorphic if there exists a group isomorphism $\phi : G \to H$. In this case, we write $G \cong H$.

Proposition 1.3.7 (Properties of Isomorphism)
Let G, H be groups. Let $\phi : G \to H$ be a group isomorphism. Then the following are true.
- ϕ^{-1} is an isomorphism
- $|G| = |H|$
- G is abelian if and only if H is abelian
- $|x| = |\phi(x)|$

Proof
Since ϕ is bijective, ϕ^{-1} is also bijective thus it is well defined. But we also need to prove that ϕ^{-1} is a homomorphism. Let $h_1, h_2 \in H$. Then there exists unique $g_1, g_2 \in G$ such that $\phi(g_1) = h_1$ and $\phi(g_2) = h_2$. Then

$$\phi^{-1}(h_1h_2) = \phi^{-1}(\phi(g_1)\phi(g_2)) = \phi^{-1}(\phi(g_1g_2)) = g_1g_2 = \phi^{-1}(h_1)\phi^{-1}(h_2)$$

thus we are done.

Since ϕ is bijective, $|G| = |H|$.

Suppose that G is abelian. $\phi(a)\phi(b) = \phi(ab) = \phi(ba) = \phi(b)\phi(a)$. Thus H is abelian. Since ϕ is bijective it has a bijective inverse. Thus the backwards implication is proved.

Suppose that $|x|$ and $|\phi(x)|$ are finite. Then we have

$$1_H = \phi(1_G) = \phi(x^{|x|}) = \phi(x)^{|x|}$$

so that $|\phi(x)| \mid |x|$. By a similar argument using ϕ^{-1}, we conclude that $|x| \mid |\phi(x)|$. Hence $|\phi(x)| = \phi(x)$. Now suppose that $|\phi(x)|$ is infinite. Then for all $n \in \mathbb{N}$, we have $\phi(x^n) = \phi(x)^n \neq 1_H$ and so $x^n \neq \phi^{-1}(1_H) = 1_G$ by applying bijection ϕ^{-1} on both sides. By a similar argument, if $|x|$ is infinite, then $|\phi(x)|$ is infinite. Hence $|x| = |\phi(x)|$. ■

When two groups are isomorphic, they share any and all properties of each other. We think of the isomorphic groups as two representations of the same abstract group. This is similar to how sets are uniquely determined by their cardinality, and elements in two bijective sets are only a renaming of one another. However, for groups, not only do we care about the number of elements, we also care about whether the group operation between the two sets are structurally the same.

Example 1.3.8
The groups $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are not isomorphic to each other.

Proof
Suppose that $\phi : \mathbb{Z} \to \mathbb{Q}$ is a group isomorphism. We know that $\phi(n) = n\phi(1)$ for all $n \in \mathbb{Z}$. Suppose that $\phi(1) = \frac{r}{s} \in \mathbb{Q}$ where $\gcd(r, s) = 1$. Let p be a prime number that does not divide s. I claim that $\frac{1}{p} \notin \text{im}(\phi)$. Indeed if it does, then there exists $n \in \mathbb{Z}$ such that $\frac{nr}{s} = n\phi(1) = \phi(n) = \frac{1}{p}$, and $\frac{nr}{s} = \frac{1}{p}$ if and only if $nrp = s$. But this is impossible since p does not divide s. Hence ϕ is not surjective and this is a contradiction.
We also have $\mathbb{Q}$ and $\mathbb{R}$ are not isomorphic as groups since there is no bijection between $\mathbb{Q}$ and $\mathbb{R}$ by Cantor's diagonalization argument. Similarly $\mathbb{Z}$ and $\mathbb{R}$ have different cardinalities since $\mathbb{Z}$ and $\mathbb{Q}$ have the same cardinality. ■

Definition 1.3.9 (The Automorphism Group)
Let G be a group. An automorphism of G is a group isomorphism $\phi : G \to G$. Define the automorphism group G to be

$$\text{Aut}(G) = \{\phi : G \to G \mid \phi \text{ is an isomorphism } \}$$

Lemma 1.3.10
Let G be a group. Then the automorphism group $(\text{Aut}(G), \circ)$ is a group.

Proof
Composing bijective maps also gives bijective maps and so composition is a binary operation. This operation is associative since composition of functions is associative. The identity map id_G of G is the identity of the automorphism group since for any automorphism $\phi : G \to G$, we have that

$$\phi \circ \text{id}_G = \phi = \text{id}_G \circ \phi$$

Finally, each automorphism is a bijection and so is invertible. The inverse of an automorphism is again an automorphism and so we conclude. ■

2 Generating a Group

2.1 Generating Sets

Definition 2.1.1 (Subgroups Generated by a Set)
Let G be a group and A a subset of G. Define the subgroup generated by A to be

$$\langle A\rangle = \{a_1^{e_1}\cdots a_n^{e_n} \mid n\in\mathbb{N}^+, a_1,\dots,a_n\in A, e_1,\dots,e_n\in\{\pm 1\}\}$$

the subgroup of all elements of G that can be expressed as the finite product of elements in A and their inverses.

If $A=\{a_1,\dots,a_r\}$ is finite, we often write the generated subgroup as $\langle A\rangle = \langle a_1,\dots,a_r\rangle$, omitting the brackets.

Proposition 2.1.2
Let G be a group and A a subset of G. Then we have that

$$\langle A\rangle = \bigcap_{A\subseteq H\leq G} H$$

Proof
Since $\langle A\rangle$ is a subgroup of G containing A, we know that $\bigcap_{A\subseteq H\leq G} H\subseteq\langle A\rangle$.

Let $a_1^{e_1}\cdots a_n^{e_n}\in\langle A\rangle$. Let $H\leq G$ be a subgroup such that $A\subseteq H$. Since $a_1,\dots,a_n\in A\subseteq H$ and H is group, we have that $a_1^{e_1}\cdots a_n^{e_n}\in H$. Hence $a_1^{e_1}\cdots a_n^{e_n}\in\bigcap_{A\subseteq H\leq G} H$ and so $\langle A\rangle\subseteq\bigcap_{A\subseteq H\leq G} H$. ■

This means that the generated subgroup is the smallest subgroup of G containing A.

Definition 2.1.3 (Generating Set of a Group)
Let G be a group and S a subset of G. We say that S is a generating set of G if $G=\langle S\rangle$.

Example 2.1.4
The subset $\{1\}$ generates the group $\mathbb{Z}$.

Proof
Every number $n\in\mathbb{N}$ is a finite sum of n ones. Similarly every number $-n$ for $n\in\mathbb{N}$ is a finite sum of n negative ones. ■

2.2 Cyclic Groups

Definition 2.2.1 (Cyclic Groups)
Let G be a group. We say that G is cyclic if there exists $g\in G$ such that

$$G=\langle g\rangle$$

Lemma 2.2.2
Let G be a group. If G is cyclic, then G is abelian.

Proof
Let $\langle g\rangle$ be a cyclic group. Then for any $g^m, g^n\in\langle g\rangle$, we have that

$$g^m\cdot g^n = g^{m+n} = g^{n+m} = g^n\cdot g^m$$

and so we conclude. ■

Proposition 2.2.3
Let G be a cyclic group. Then any subgroup H of G is also cyclic.

Proof
Let $H \leq G = \langle g \rangle$. Every element in H is of the form g^n for some $n \in \mathbb{Z}$. Take the smallest positive k such that $g^k \in H$. Consider another element $g^n \in H$. By the division algorithm, $n = qk + r$ for some $r \in \{0, \dots, k-1\}$. Thus we have

$$g^n = g^{kq+r} \iff g^r = (g^k)^{-q}(g^n)$$

Now since $g^k \in H$, $(g^k)^{-1} \in H$ and $(g^k)^q \in H$ thus $(g^k)^q \in H$ and $(g^k)^q(g^n) \in H$ by inverse and closure of a group. Thus $g^r \in H$. However this is a contradiction since k is the least power of g such that $g^k \in H$ but g^r with $r < k$ is now in H. Thus we must have $r = 0$ and $g^n = (g^k)^q$. This applies for every $g^n \in H$ thus $H = \langle g^k \rangle$. ■

One can also show that in the case of a cyclic group, the order of the group is equal to the order of the generator. However the converse is not always true. That is, the order of the group and the order of one particular element being the same does not imply that the group is cyclic.

Proposition 2.2.4
Let G be a cyclic group. Let g be a generator of G. Then $|G| = |g|$.

Proof
Suppose that $|g| = n$ Note that $g^{n+k} = g^k$ for any $k \in \{0, \dots, n-1\}$. Thus $\{g^n | n \in \mathbb{Z}\}$ has n distinct elements. Thus $|G| = n$.
If $|g| = \infty$ then $g^n \neq 1$ for all n not equal to 0. Thus $\{g^n | n \in \mathbb{Z}\}$ are all distinct and $|G| = \infty$. ■

Using cyclic subgroups, we can impose a stronger condition so that the order of the product of two elements can be given an explicit formula.

Proposition 2.2.5
Let G be a group. Let $a, b \in G$ such that the following are true.

- $ab = ba$.
- $\langle a \rangle \cap \langle b \rangle = \{1\}$.

Then we have

$$|ab| = \operatorname{lcm}(|a|, |b|)$$

Proof
Let $n = |ab|$. Since a and b commute, we have that

$$1 = (ab)^n = a^n b^n$$

so that $a^n = b^{-n}$. Thus $a^n \in \langle b \rangle$. But $a^n \in \langle a \rangle \cap \langle b \rangle = \{1\}$ implies that $a^n = 1$. Similarly, we have that $b^n = 1$. This means that we have $|a|, |b|$ divides n. and thus $\operatorname{lcm}(|a|, |b|)$. By lemma 1.1.12 we know that $|ab|$ divides $\operatorname{lcm}(|a|, |b|)$ and so we conclude. ■

2.3 Classifying Cyclic Groups

Definition 2.3.1 (Finite Cyclic Groups)
Let $n \in \mathbb{N}^+$. Define the cyclic group of order n to be the set

$$C_n = \{e^{2\pi i k/n} \in \mathbb{C} \mid 0 \leq k \leq n-1\}$$

together with complex multiplication.

Proposition 2.3.2
Let $n, m \in \mathbb{N}^+$. Then $C_n \cong C_m$ if and only if $n = m$.

Lemma 2.3.3
Let G be a cyclic group. Then G is isomorphic to one of the following.

- C_n for some $n \in \mathbb{N}$.
- $\mathbb{Z}$

Proposition 2.3.4
Let $n \in \mathbb{N}$ and $0 \leq k \leq n-1$ be an integer. Then g^k is a generator of C_n if and only if $\gcd(k, n) = 1$.

Proof Suppose that g^k is a generator of C_n. Then $g = g^{ak}$ for some $a \in \mathbb{Z}$. Since $g^n = 1$, there exists $b \in \mathbb{Z}$ such that $g^{ak-bn} = g$ and $0 \leq ak - bn \leq n-1$. Then $ak - bn = 1$ has a solution if and only if $\gcd(k, n) = 1$ thus we are done.

Now suppose that $\gcd(k, n) = 1$. Then $\left|g^k\right| = \frac{n}{\gcd(k,n)=n}$ thus we are done. ■

3 Normal Subgroups and Quotient Groups

3.1 Lagrange's Theorem

Definition 3.1.1 (Left and Right Cosets)
Let G be a group. Let H be a subgroup of G.
- A left coset of H is a set of the form $gH = \{gh \mid h \in H\}$ for some $g \in G$.
- A right coset of H is a set of the form $Hg = \{hg \mid h \in H\}$ for some $g \in G$.

Example 3.1.2
Let $n \geq 2$. Then $1 + n\mathbb{Z} = \{1 + nk \mid k \in \mathbb{Z}\}$ is a coset of $n\mathbb{Z}$ in $\mathbb{Z}$.

Proposition 3.1.3
Let G be a group. Let $H \leq G$ be a subgroup. Let $g_1, g_2 \in G$. Then the following are equivalent.
- $g_1H = g_2H$
- $g_2 \in g_1H$
- $g_1^{-1}g_2 \in H$

Proof
(1) $\implies$ (2): Suppose that $g_1H = g_2H$. Let $g_1h \in g_1H$. Then there exists k such that $g_1h = g_2k$. Then $g_2 = g_1hk^{-1}$ and since H is a subgroup, $hk^{-1} \in H$ and thus $g_2 \in g_1H$.

(2) $\implies$ (3): Suppose that $g_2 \in g_1H$. Then there exists $h \in H$ such that $g_2 = g_1h$. Then $g_1^{-1}g_2 = h$. Hence $g_1^{-1}g_2 \in H$.

(3) $\implies$ (1): Let $g_2h \in g_2H$. By (3) there exists $k \in H$ such that $g_1^{-1}g_2 = k$. We then have

$$\begin{aligned} g_1^{-1}g_2 &= k \\ g_1^{-1}g_2h &= kh \\ g_2h &= g_1kh \\ g_2h &\in g_1H \end{aligned}$$

Since $kh \in H$. Thus $g_2H \subseteq g_1H$. Mirror the argument for $g_2^{-1}g_1 \in H$ and we have that $g_1H \subseteq g_2H$ and we are done. ∎

Proposition 3.1.4
Let G be a group. Let $H \leq G$ be a subgroup. Let $g_1, g_2 \in G$. Then $g_1H = g_2H$ if and only if $Hg_1 = Hg_2$.

Proof
Let $g_1H = g_2H$. Let $hg_1 \in Hg_1$. Since we know that $g_1g_2^{-1} \in H$, then $hg_1g_2^{-1} \in H$. Let $hg_1g_2^{-1} = k \in H$. Then $hg_1 = kg_2 \in Hg_2$ thus we have that $Hg_1 \subseteq Hg_2$. The argument is similar for $Hg_2 \subseteq Hg_1$ thus we have that $Hg_1 = Hg_2$. For the reverse, the argument is also similar. ∎

Proposition 3.1.5
Let G be a group. Let $H \leq G$ be a subgroup. Then the left (right) cosets of G partition G.

Proof
We first show that if $g_1H \neq g_2H$, then $g_1H \cap g_2H = \emptyset$. Suppose that $k \in g_1H \cap g_2H$, then by the above we have $g_1H = kH = g_2H$ thus a contradiction. Thus we must have $g_1H \cap g_2H = \emptyset$. Now we must have that

$$\bigcup_{g \in G} gH = G$$

since every gH must at least have $g \in gH$. Thus we are done. ∎

Proposition 3.1.6
Let G be a group. Let $H \leq G$ be a subgroup. Then the number of distinct left cosets is equal to the number of distinct right cosets.

Proof
Since all the distinct left cosets partition G and $g_1H = g_2H$ if and only if $Hg_1 = Hg_2$, we must have the same number of distinct right cosets as well in order to partition G. ■

Proposition 3.1.7
Let G be finite a group. Let $H \leq G$ be a subgroup. Then for any $g \in G$, $|gH| = |H|$.

Proof
Let $g \in G$. We prove that the map $\phi : H \to gH$ defined by $\phi(h) = gh$ is a bijection. For injectivity, $gh_1 = gh_2$ implies $h_1 = h_2$ by applying g^{-1} on both sides. Let $k \in gH$, then there exists $h \in H$ such that $k = gh \in gH$. This proves surjectivity and we are done. ■

Definition 3.1.8 (Number of Distinct Cosets)
Let G be a group. Let $H \leq G$ be a subgroup. Define $[G : H]$ to be the number of distinct left cosets of H in G.

The set of all left (right) cosets is also important for the construction of quotient groups as well as one of the major players in group actions.

Definition 3.1.9 (Set of Cosets)
Let G be a group. Let $H \leq G$ be a subgroup. Define

$$\frac{G}{H} = \{gH \mid g \in G\}$$

the set of cosets of H in G.

Lagrange's theorem is a powerful theorem that has many applications. Some indirect results include the class equation and orbit stabilizer theorem as we will later see.

Theorem 3.1.10 (Lagrange's Theorem)
Let G be a finite group. Let $H \leq G$ be a subgroup. Then

$$|G| = [G : H]|H|$$

Proof
We know that distinct left cosets partition G and every left coset has $|H|$ elements thus $|G| = [G : H]|H|$. ■

An immediate result is that orders of subgroups must divide the order of the group. Notice that this does not imply the converse: given any number n that divides the order of G, it does not necessarily mean that there exists a subgroup of G with order n. As always, it is good to think of counterexamples. A prototypical one would be A_4 (see chapter 5 for definition) that does not have subgroups of order 6. This is the smallest example.

Another immediate result of Lagrange's theorem is the following.

Corollary 3.1.11
Let G be a finite group. Let $g \in G$. Then the order of g divides the order of G.

Proof
For every $g \in G$, $\langle g \rangle \leq G$ thus by Lagrange's Theorem $|g| = |\langle g \rangle| \mid |G|$ ■

Corollary 3.1.12
Let G be a group of prime order p. Then G is cyclic.

Proof
For any $1 \neq g \in G$, $\langle g \rangle$ is a subgroup of G. By Lagrange's theorem, it must divide the order of G, which is p. Since p is a prime, we must have $\langle g \rangle = G$. ∎

3.2 Normal Subgroups

Given a group G and a subgroup H of G, it is natural to ask whether we can define a group structure on the set of cosets G/H. One natural way to defining multiplication in cosets is via the formula

$$g_1H \cdot g_2H = g_1g_2H$$

However, this formula is not well defined because different representatives of a coset does not give the same coset on multiplication, and this is because left and right cosets in general do not coincide.

Definition 3.2.1 (Normal Subgroups)
Let G be a group. Let $N \leq G$ be a subgroup. We say that N is normal if

$$gN = Ng$$

for all $g \in G$. We write $N \trianglelefteq G$ in this case.

Definition 3.2.2 (Conjugates)
Let G be a group. Let $H \leq G$ be a subgroup. Let $g \in G$. Define the conjugate of H in g to be the set

$$gHg^{-1} = \{ghg^{-1} \mid h \in H\}$$

Proposition 3.2.3
Let G be a group. Let $N \leq G$ be a subgroup. The following are equivalent.

- N is normal in G
- $gNg^{-1} \subseteq N$ for all $g \in G$
- $gNg^{-1} = N$ for all $g \in G$

Proof
(1) $\implies$ (2) Let N be normal. Let $gng^{-1} \in gNg^{-1}$. Since N is normal there exists $k \in N$ such that $gn = kg$. Then $gng^{-1} = kgg^{-1} = k \in N$.

(2) $\implies$ (1) Let $gNg^{-1} \subseteq N$. Then for all $n \in N$ there exists $k \in N$ such that $gng^{-1} = k$. Then $gn = kg$ for all n thus $gN = Ng$ and we are done.

(2) $\implies$ (3) We only need to show that $N \subseteq gNg^{-1}$. Let $n \in N$. Then $ng \in Ng$. Since N is normal, there exists $k \in N$ such that $gk = ng$ and $k = g^{-1}ng$. But then $n = gkg^{-1}$ thus $n \in gNg$ and we are done.

(3) $\implies$ (2) is trivial. ∎

Example 3.2.4
Let G be an abelian group. Then any subgroup of G is normal.

Proof
Let H be a subgroup of G. Since G is abelian, we have $gHg^{-1} = gg^{-1}H = H$ for any $g \in G$. Hence H is normal. ∎

Proposition 3.2.5
Let G be a group. Let $H \leq G$ be a subgroup. Write $g_1H \cdot g_2H = \{ab \mid a \in g_1H, b \in g_2H\}$. Then

$$g_1H \cdot g_2H = (g_1g_2)H$$

for all $g_1, g_2 \in G$ if and only if $H \trianglelefteq G$.

Proof
Suppose first that $g_1H \cdot g_2H = (g_1g_2)H$. Let $g \in G$. We want to show that $gHg^{-1} \subseteq H$. Notice that $gH \cdot g^{-1}H = H$. For any $h \in H$, we have $gh \in gH$ and $g^{-1} \in g^{-1}H$ so that $ghg^{-1} \in G$.

Now suppose that N is normal. Let $g_1h_1 \in g_1H$ and $g_2h_2 \in g_2H$. Then we have

$$g_1h_1g_2h_2 = g_1g_2h_3h_2 \in (g_1g_2)H$$

since N is normal and $h_1g_2 = g_2h_3$. Thus $g_1H \cdot g_2H \subseteq (g_1g_2)H$. Now suppose that $g_1g_2h \in (g_1g_2)H$. Then in particular this element lies in $g_1H \cdot g_2H$ because $g_1g_2h = g_1 \cdot 1 \cdot g_2 \cdot h$. And so we conclude. ■

Definition 3.2.6 (Quotient Groups)
Let G be a group. Let N be a normal subgroup of G. Define the quotient group of G by N to be the set

$$\frac{G}{N} = \{[g] = gN \mid g \in G\}$$

together with the binary operation $\cdot : G/N \times G/N \to G/N$ defined by

$$(gN) \cdot (hN) = (gh)N$$

Proposition 3.2.7
Let G be a group. Let N be a normal subgroup of G. Then G/N is a group.

Proof
We check the group axioms:

- Associativity: Let gN, hN and kN be in G/N. Then by two applications of the above theorem, we have that $((gN) \cdot (hN)) \cdot (kN) = (ghN) \cdot (kN) = (gh)kN$ By associativity of G, we conclude.
- Identity: Let $gN \in G/N$. Then it is clear that $(gN) \cdot (N) = (g \cdot 1)N = gN$ so that N is the identity in G/N.
- Inverse: Let $gN \in G/N$. Then we can see that $(gN) \cdot (g^{-1}N) = (gg^{-1})N = N$ so that each element in G/N has an inverse.

All the group axioms are satisfied and so we conclude. ■

Example 3.2.8
Let $n \in \mathbb{N}^+$. Then $\mathbb{Z}/n\mathbb{Z}$ is a group isomorphic to C_n.

Normal subgroups are also closely related to kernels. This is because for any group G and normal subgroup N, there is a canonical projection map $p : G \to G/N$.

Proposition 3.2.9
Let G be a group. Let N be a subgroup of G. Then N is a normal subgroup if and only if N is the kernel of some group homomorphism.

Proof
Suppose that N is a normal subgroup of G. Then consider the homomorphism $\phi : G \to G/N$

defined by $\phi(g) = gN$. Then

$$\begin{aligned}\ker(\phi) &= \{g \in G | \phi(g) = N\} \\ &= \{g \in G | gN = N\} \\ &= \{g \in G | g \in N\} \\ &= N\end{aligned}$$

Thus we are done.

Let N be the kernel of the homomorphism $\phi : G \to H$. Then $\ker(\phi) = \{g \in G | \phi(g) = 1\} = N$. By the normality test, showing that $gng^{-1} \in N$ is sufficient for $n \in N$ and $g \in G$. We have that

$$\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g^{-1}) = \phi(g)\phi(g)^{-1} = 1$$

thus $gng^{-1} \in \ker(\phi) = N$ thus we are done. ■

3.3 Normalizers

Sometimes, not every element in a group allows a subgroup H to normalize. Therefore can obtain a subset of G that contains all the elements that allows H to be normal.

Definition 3.3.1 (Normalizers)
Let G be a group. Let $S \subseteq G$. Then the normalizer is defined to be

$$N_G(S) = \{g \in G \mid gS = Sg\} = \{g \in G \mid gSg^{-1} = S\}$$

Proposition 3.3.2
Let G be a group. Let $S \subseteq G$. Then $S \subseteq N_G(S) \leq G$.

Proof
Let $g, h \in N_G(S)$. Then $h^{-1} \in N_G(S)$ since $h^{-1}S = Sh^{-1}$ if and only if $hS = Sh$ thus $h^{-1} \in N_G(S)$. Now we want $gh^{-1} \in N_G(S)$. We have

$$gh^{-1}S = g(Sh^{-1}) = (gS)h^{-1} = Sgh^{-1}$$

and so we have that $N_G(S) \leq G$. For the remaining part, if $s \in S$, then $sS = S = Ss$ and so $S \subseteq N_G(S)$. ■

Proposition 3.3.3
Let G be a group and H a subgroup of G. Then we have $H \trianglelefteq N_G(H)$.

Proof
We know that $H \subseteq N_G(H)$. Since H and $N_G(H)$ are both subgroups of G, we must have $H \leq N_G(H)$. By definition of $N_G(H)$, if $g \in N_G(H)$ then $gHg^{-1} = H$ and so we conclude. ■

Proposition 3.3.4
Let G be a group and $N \subseteq G$. Then $N \trianglelefteq G$ if and only if $N_G(N) = G$.

Proof
Let $N \trianglelefteq G$. Trivially we already have $N_G(N) \leq G$. Let $g \in G$. Then since N is normal, we have $gN = Ng$ thus $g \in N_G(N)$ and we are done.
Let $N_G(N) = G$. Then for all $g \in G$, $gN = Ng$ thus N is normal. ■

As motivated from above, $N_G(S)$ is just all the elements in G so that S is allowed to be normal. Then obviously if $N_G(S)$ is the entirety of G then S would be allowed to be normal.

3.4 The Isomorphism Theorems for Groups

Theorem 3.4.1 (The First Isomorphism Theorem)
Let G, H be groups. Let $\phi : G \to H$ be a homomorphism. Then the following are true regarding the kernel and the image of the homomorphism.

- $\ker(\phi) \trianglelefteq G$
- $\operatorname{im}(\phi) \leq H$
- There is a group isomorphism
$$\frac{G}{\ker(\phi)} \cong \operatorname{im}(\phi)$$
given by the function $g\ker(\phi) \mapsto \phi(g)$.

Proof
Proposition 2.3.7 shows that $\ker(\phi)$ is a normal subgroup of G.

Let $\phi(g), \phi(h) \in \operatorname{im}(\phi)$. Then $\phi(g) \cdot \phi(h) = \phi(gh) \in \operatorname{im}(\phi)$ so that closure is satisfied. Now clearly $\phi(g)^{-1} \in \operatorname{im}(\phi)$ since $\phi(g^{-1}) = \phi(g)$ and so we conclude.

We construct an isomorphism f between cosets of G of $\ker(\phi)$ and elements of $\phi(G)$. Write N for $\ker(\phi)$ Let $gN \in G/\ker(\phi)$. Define f by $f(gN) = \phi(g)$. I claim that this is the isomorphism we need.
We first show that this map is well defined. The point here is to show that any representative is fine. In particular, if $gH = hN$ are two representations of $the same coset$ and g, h is distinct, then we want $f(gN) = f(hN)$. Well $gN = hN$ if and only if $gh^{-1} \in N$. This means that $gh^{-1} \in \ker(\phi)$ and $\phi(gh)^{-1} = 1$. This means that $\phi(g)\phi(h)^{-1} = 1$ and $\phi(g) = \phi(h)$. This means that the map is well defined.
We now show that it is a homomorphism. Let gN and hN be cosets. Then

$$f((gN)(hN)) = f((gh)N) = \phi(gh) = \phi(g)\phi(h) = f(gN)f(hN)$$

thus we are done. Finally we show that it is an isomorphism. Firstly let $f(gN) = f(hN)$. Then

$$\begin{aligned} f(gN) &= f(hN) \\ \phi(g) &= \phi(h) \\ \phi(g)\phi(h)^{-1} &= 1 \\ \phi(gh^{-1}) &= 1 \end{aligned}$$

This means that $gh^{-1} \in \ker(\phi)$ thus $gN = hN$. Now let $\phi(g) \in \phi(G)$. Then quite obviously $f(gN) = \phi(g)$ thus it is surjective and bijective and we are done. ■

The first isomorphism theorem is one of the first substantial and fundamental result in group theory. The theorem itself may not seem surprising, but it has a wide application across different areas of Mathematics that make use of group theory. In particular, the following two isomorphism theorems are in part due to the first isomorphism theorem.

Theorem 3.4.2 (The Second Isomorphism Theorem)
Let G be a group, let $A \leq G$ and $B \trianglelefteq G$. Then the following are true.

- AB is a subgroup of G
- $B \trianglelefteq AB$ and $A \cap B \trianglelefteq A$
- There is an isomorphism of groups
$$\frac{A}{A \cap B} \cong \frac{AB}{B}$$
given by the function $a + A \cap B \mapsto aB$.

Proof
We first prove that AB is a subgroup of G. Let $a_1b_1, a_2b_2 \in AB$. Since $b_1a_2 \in Ba_2$ and B is a normal subgroup, there exists $b_3 \in B$ such that $b_1a_2 = a_2b_3$. Then we have $a_1b_1a_2b_2 = a_1a_2b_3b_2 \in AB$. Hence products of elements in AB lie in AB. Now let $ab \in AB$. Its inverse in G is given by $b^{-1}a^{-1}$. Then $b^{-1}a^{-1} \in Ba^{-1}$ implies that there exists $b' \in B$ such that $b^{-1}a^{-1} = a^{-1}b'$. Hence the inverse of an element in AB also lies in AB. By the subgroup criterion, we conclude that AB is a subgroup of G.

To show that $B \trianglelefteq AB$, we show that $abb'(ab)^{-1} \in B$ for any $b' \in B$ and $ab \in AB$. But this is trivial since $ab \in G$ and we know that since $B \trianglelefteq G$, $abb'(ab)^{-1} \in B$ thus we are done.
For $A \cap B \trianglelefteq A$, we want $kak^{-1} \in A$ for $k \in A \cap B$ and $a \in A$. But since $k \in A \cap B$ implies $k \in A$ and A is a subgroup, $kak^{-1} \in A$ thus we are done.

Define $\phi : A \to \frac{AB}{B}$ by $\phi(a) = aB$. Then ϕ is a homomorphism. Indeed $\phi(a_1a_2) = a_1a_2B$ and $\phi(a_1)\phi(a_2) = a_1B \cdot a_2B = a_1a_2B$. Also, we have that $a \in \ker(\phi)$ if and only if $a \in B$. This shows that $\ker(\phi) = A \cap B$. Finally, given $aB \in \frac{AB}{B}$, it is clear that $\phi(a) = aB$ such that ϕ is surjective. By the first isomorphism theorem, we have that

$$\frac{A}{A \cap B} = \frac{A}{\ker(\phi)} \cong \phi(A) = \frac{AB}{B}$$

and so we conclude. ■

Theorem 3.4.3 (The Third Isomorphism Theorem)
Let G be a group. Let H, K be normal subgroups of G such that $H \leq K$. Then the following are true.

- $K/H \trianglelefteq G/H$.
- There is an isomorphism of groups

  $$\frac{G/H}{K/H} \cong \frac{G}{K}$$

  given by the function $gH + K/H \mapsto gK$.

Proof
Let $kH \in K/H$. Let $gH \in G/H$ I want to show that $(gH)(kH)(gH)^{-1} \in \frac{K}{H}$. But

$$(gH)(kH)(gH)^{-1} = gkg^{-1}H = k'H \in \frac{K}{H}$$

for some $k' = gkg^{-1}$. This is true since K/H is normal.

Define $\phi : \frac{G}{H} \to \frac{G}{K}$ by $\phi(gH) = gK$. This is a group homomorphism since

$$\phi(g_1g_2H) = g_1g_2K = g_1K \cdot g_2K = \phi(g_1H)\phi(g_2H)$$

Moreover, notice that $gH \in \ker(\phi)$ if and only if $g \in K$. This means that $gH \in \ker(\phi)$ if and only if $gH \in K/H$. Thus $\ker(\phi) = K/H$. Finally, given $gK \in G/K$, it is clear that $\phi(gH) = gK$ so that ϕ is surjective. By the first isomorphism theorem, we have that

$$\frac{G/H}{K/H} = \frac{G/H}{\ker(\phi)} \cong \phi(G/H) = \frac{G}{K}$$

and so we conclude. ■

Theorem 3.4.4 (The Correspondence Theorem for Groups)
Let G be a group. Let $N \trianglelefteq G$ be normal. Let $\phi : G \to \frac{G}{N}$ be the quotient map. Then the following are true.

- There is an inclusion preserving bijection
$$\left\{ H \subseteq G \;\middle|\; \begin{matrix} H \text{ is a subgroup of } G \\ \text{containing } N \end{matrix} \right\} \overset{1:1}{\longleftrightarrow} \left\{ H \subseteq G/N \;\middle|\; \begin{matrix} H \text{ is a subgroup} \\ \text{of } G/N \end{matrix} \right\}$$
given by $H \mapsto \phi(H)$.
- The above bijection restricts to a bijection
$$\left\{ H \subseteq G \;\middle|\; \begin{matrix} H \text{ is a normal subgroup} \\ \text{of } G \text{ containing } N \end{matrix} \right\} \overset{1:1}{\longleftrightarrow} \left\{ H \subseteq G/N \;\middle|\; \begin{matrix} H \text{ is a normal} \\ \text{subgroup of } G/N \end{matrix} \right\}$$

4 Permutation and Symmetries

Évariste Galois first defined a group to something similar to symmetric groups. Indeed they encompass a high amount of symmetrical information, which is the at the heart of group theory. They arise in most situations and surprisin GLy, we will see that every group embeds into a symmetric group of some order, finite or infinite (we will not see infinite symmetric groups here, but the result is still true) as an immediate application of the notion of group actions.

4.1 The Symmetric Groups

We begin with the definition of a symmetric group. Notice that we will only define the finite case here, but there is also a version in which the symmetric group is infinite.

Definition 4.1.1 (The Symmetric Group)
Let X be a finite set. Denote

$$\text{Sym}(X) = \{f : X \to X \mid f \text{ is bijective}\}$$

the set of all bijections from X to itself. The symmetric group of X is defined to be X together with the usual composition of functions. Elements of $\text{Sym}(X)$ are called permutations.

Note that the symmetric group is indeed a group by the following reasons.

- The composition of functions are associative.
- The identity element: id_X which is the identity function on X has the property that $f \circ \text{id}_X = \text{id}_X = \text{id}_X \circ f$ for any $f \in \text{Sym}(X)$.
- For each $f \in \text{Sym}(X)$, f is bijective and so it has an inverse $f^{-1} : X \to X$ which is also bijective and thus lies in $\text{Sym}(X)$.

We can represent each element in the symmetric group in the following way.

Definition 4.1.2 (Matrix Notation)
Let X be a finite set. Each $f : X \to X$ in $\text{Sym}(X)$ is represented by a $2 \times |X|$ matrix as follows: Label the elements of X as $1, \dots, n$ for $n = |X|$. Then we write $f : X \to X$ as

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ \phi(1) & \phi(2) & \cdots & \phi(n) \end{pmatrix}$$

It is easy to see that the order of the symmetric group is just how many rearrangements of the elements of X.

Lemma 4.1.3
Let X be a finite set with $|X| = n$. Then the order of $\text{Sym}(X)$ is $n!$.

Proof
Using the matrix form, we see that $\phi(1), \dots \phi(n)$ is a permutation of $1, \dots, n$. In particular, if $\phi(1)$ has n possibilities, then $\phi(2)$ has $n-1$ options etc. Then there will be $n!$ different permutations. ■

Proposition 4.1.4
Let X be a finite set of cardinality n. Then $\text{Sym}(X)$ is non-abelian if and only if $n \geq 3$.

Proof
Suppose that $n \geq 3$. By fixing $n-3$ elements to map to themselves in $\text{Sym}(X)$, we see that $S_3 \leq S_n$. It remains to show that S_3 is non-abelian. Write $X = \{1, 2, 3\}$. Let $f : X \to X$ be defined by $f(1) = 2$, $f(2) = 1$ and $f(3) = 3$. Let $g : X \to X$ be defined by $g(1) = 3$, $g(2) = 2$ and $g(3) = 1$. Then it is clear that $f(g(3)) = f(1) = 2$ and $g(f(3)) = g(3) = 1$ so that f and g does not commute.

For $n = 1$ and 2, it is clear by inspection that S_1 and S_2 must be abelian. ■

Proposition 4.1.5
Let X and Y be finite sets. If $|X| = |Y|$ then $\text{Sym}(X) \cong \text{Sym}(Y)$.

Proof
Since $|X| = |Y|$, there exists a bijection $\phi : X \to Y$. Define a function $\rho : \text{Sym}(X) \to \text{Sym}(Y)$ by $f \mapsto \phi \circ f \circ \phi^{-1}$. It is a group homomorphism since

$$\rho(f \circ g) = \phi \circ f \circ g \circ \phi^{-1} = \phi \circ f \circ \phi^{-1} \circ \phi \circ g \circ \phi^{-1} = \rho(f) \circ \rho(g)$$

for any $f, g \in \text{Sym}(X)$. It is a bijection since it has an inverse given by $f \mapsto \phi^{-1} \circ f \circ \phi$. ■

Since $\text{Sym}(X)$ is invariant under the number of elements of X, we denote S_n once and for all for $\text{Sym}(\{1, \dots, n\})$.

4.2 Cycle Notation

Notice that matrix presentation is unique since each element $f \in \text{Sym}(X)$ is uniquely defined by where it sends each element to. While this presentation makes it clear the images of each element in X, it is often rather cumbersome to write. Therefore we have an alternate and more compact notation.

Definition 4.2.1 (Cycle Notation)
Let X be a finite set. Let $a_1, \dots, a_r \in X$ be distinct. Then denote $\begin{pmatrix} a_1 & \dots & a_r \end{pmatrix}$ the permutation in $\text{Sym}(X)$ where
- $f(a_i) = a_{i+1}$ for $1 \leq i \leq r - 1$
- $f(a_r) = a_1$
- $f(b) = b$ for $b \notin \{a_1, \dots, a_r\}$

If $f \in \text{Sym}(X)$ is equal to $\begin{pmatrix} a_1 & \dots & a_r \end{pmatrix}$ for some distinct elements in $a_1, \dots, a_r$ then we say that f is a cycle in $\text{Sym}(X)$.

We also have the notion of disjoint cycles.

Definition 4.2.2 (Disjoint Cycles)
Let X be a finite set. Two cycles $\begin{pmatrix} a_1 & \dots & a_r \end{pmatrix}$ and $\begin{pmatrix} b_1 & \dots & b_s \end{pmatrix}$ in $\text{Sym}(X)$ are said to be disjoint if $\{a_1, \dots, a_r\} \cap \{b_1, \dots, b_s\} = \emptyset$.

Since a cycle fixes any element not within the cycle, disjoint cycles should commutes.

Proposition 4.2.3
Let X be a finite set. Then disjoint cycles in $\text{Sym}(X)$ commute.

Proof Suppose that $f = (a_1, \dots, a_r)$ and $g = (b_1, \dots, b_s)$ are disjoint cycles in $\text{Sym}(X)$. Let $x \in X$. If $x \in \{b_1, \dots, b_s\}$, then $g(x) \in \{b_1, \dots, b_s\}$ so that $g(x) \notin \{a_1, \dots, a_r\}$ and $f(g(x)) = g(x)$. On the other hand, we have $f(x) = x$ which implies $g(f(x)) = g(x)$.

If $x \notin \{b_1, \dots, b_s\}$, then the same argument gives $f(g(x)) = f(x) = g(f(x))$ and so we conclude. ■

Compared to matrix presentations, the cycle notation is far from being unique. In particular, we can represent the same element in $\text{Sym}(X)$ by either swapping the order of the individual cycles or by changing the starting element of each cycle by another element in the same cycle. We prove that the cycle notation is unique up the following sense.

Theorem 4.2.4
Let X be a finite set. Then the following are true.

- Every element in $\text{Sym}(X)$ can be written as a product of disjoint cycles.
- The product is unique in the following sense: If $f \in \text{Sym}(X)$ has the form
$$f = f_1 \cdots f_p = g_1 \cdots g_q$$
where each f_i are disjoint cycles of length > 1 from each other for $1 \leq i \leq p$ and similarly for g_i, then $p = q$ and
$$\{f_1, \dots, f_p\} = \{g_1, \dots, g_q\}$$

Proof
If id_X is the identity element then we can write it as () the empty cycle. Now assume that $\text{id}_X \neq f \in \text{Sym}(X)$. Let $Y = \{x \in X \mid f(x) \neq x\}$. Let $a_1 \in Y$. Since $\text{Sym}(X)$ has finite order, f has finite order, say n. Let
$$m_1 = \min\{m \in \mathbb{N} \mid (f^m)(a_1) = a_1\}$$
Then for $2 \leq i \leq m_1$, define $a_i = f(a_{i-1})$. If $Y = \{a_1, \dots, a_{m_1}\}$ then f is now a cycle and we are done. So suppose that $Y \setminus \{a_1, \dots, a_{m_1}\}$ is non-empty with the element $a_{m_1+1} \in Y \setminus \{a_1, \dots, a_{m_1}\}$. Set
$$m_2 = \{m \in \mathbb{N} \mid (f^m)(a_{m_1+1}) = a_{m_1+1}\}$$
Then for $m_1+2 \leq i \leq m_2$, define $a_i \in X$ by $a_i = f(a_{i-1})$. Since f is injective, the sets $\{a_1, \dots, a_{m_1}\}$ and $\{a_{m_1+1}, \dots, a_{m_2}\}$ have empty intersection. If $Y = \{a_1, \dots, a_{m_2}\}$ then f is a product of two disjoint cycles and we are done.

If not, this process can be repeated. It also terminates in finite steps since $\text{Sym}(X)$ is finite. When it terminates, we will have obtained a product of disjoint cycles representing f. Uniqueness follows because if $f = f_1 \cdots f_m$ we have seen by construction that each f_i completely determines f. ■

Cycle notations make multplication of elements in S_n easier. Suppose that we want to find the following product
$$\tau \cdot \psi = \begin{pmatrix} 2 & 4 & 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 3 \end{pmatrix} \begin{pmatrix} 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 & 7 \end{pmatrix} \begin{pmatrix} 2 & 5 & 6 & 9 \end{pmatrix} \begin{pmatrix} 4 & 10 \end{pmatrix}$$
We start with any element, say 1. Then observe that ψ maps 1 to 3. We then see that 3 maps to 1 in τ so 1 maps to itself and we can omit the trivial cycles. Now for 2, we see that 2 maps to 5 in ψ and 5 maps to 6 in τ. So we can write down $\begin{pmatrix} 2 & 6 & \cdots \end{pmatrix}$. Now instead of choosing another element to begin. We follow up on 6 to see that 6 maps to 9. This means we can continue to chain and write $\begin{pmatrix} 2 & 6 & 9 & \cdots \end{pmatrix}$. Once you reach the end, you begin a new cycle with another element not in previous cycles.

4.3 The Permutation Groups

Definition 4.3.1 (Permutation Groups)
Let X be a set. A permutation group is a subgroup of $\text{Sym}(X)$.

In particular, $\text{Sym}(X)$ is also a permutation group.

Definition 4.3.2 (Support)
Let X be a set. Let $g \in \text{Sym}(X)$. Define the support of g to be
$$\text{supp}(g) = \{x \in X | g(x) \neq x\}$$
Let G be a permutation group on X. Define the support of G to be
$$\text{supp}(G) = \{x \in X | g(x) \neq x \text{ for some } g \in G\}$$

Lemma 4.3.3

Let G be a permutation group on a finite set X. Let $g \in G$. Then

$$\text{supp}(\langle g \rangle) = \text{supp}(g)$$

Proof
Let $x \in \text{supp}(g)$. Then $g(x) \neq x$ so that $x \in \text{supp}(\langle g \rangle)$. Now suppose that $x \in \text{supp}(\langle g \rangle)$. Then there exists some $g^n \in \langle g \rangle$ such that $g^n(x) \neq x$. But then $g(x) \neq x$. Otherwise if $g(x) = x$ then $g^n(x) = x$, a contradiction. Thus $x \in \text{supp}(g)$ and so we conclude. ■

Lemma 4.3.4
Let G be a permutation group on a finite set X and $H \leq G$. Then $\text{supp}(H) \subseteq \text{supp}(G)$.

Proof
Let $x \in \text{supp}(H)$. Then there exists some $h \in H$ such that $h(x) \neq x$. Then in particular $h \in G$ so that $x \in \text{supp}(G)$. ■

Proposition 4.3.5
Let G, H be permutation groups on a finite set X. If $\text{supp}(G) \cap \text{supp}(H) = \emptyset$ then $G \cap H = \{1\}$.

Proof
We show the contrapositive. Suppose that $G \cap H$ has a non-trivial element, say g. Then since $\langle g \rangle \leq G, H$, we have that $\text{supp}(\langle g \rangle) \subseteq \text{supp}(G), \text{supp}(H)$ so that

$$\text{supp}(g) = \text{supp}(\langle g \rangle) \subseteq \text{supp}(G) \cap \text{supp}(H)$$

But $\text{supp}(g)$ cannot be empty because if it is, then g fixes all $x \in X$ so that g is the identity, a contradiction. Thus $\text{supp}(G) \cap \text{supp}(H)$ is non-empty. ■

4.4 Cycle Types

Definition 4.4.1 (Cycle Types)
Let $n \in \mathbb{N}^+$. Let $\phi \in S_n$. We say that ϕ has cycle type $2^{r_2}3^{r_3}\cdots$ if it has exactly r_k k cycles for $k \geq 2$.
A transposition is a cycle of type 2.

Notice that the following proposition means that we will have a third notation for elements of S_n.

Proposition 4.4.2
Let $n \in \mathbb{N}^+$. Let τ be a cycle of type k in S_n. Then the following are true.

- τ has order k
- τ can be decomposed into a product of $k - 1$ transpositions

Proof

- Let τ be a cycle of order k. Notice that multiplying n times of the same element means that we are taking the ith element in τ to the $i + n$ modulo k element in the cycle. If $n = k$ then every element goes to itself, which means that τ^k is just the permutation that maps every element to iself, which is the identity of the permutation group.
- For a cycle $(x_1, \dots, x_k)$. Notice that the product

  $$\begin{pmatrix} x_{k-1} & x_k \end{pmatrix} \cdots \begin{pmatrix} x_2 & x_3 \end{pmatrix} \begin{pmatrix} x_1 & x_2 \end{pmatrix}$$

  returns the original cycle.

Thus we are done. ■

Proposition 4.4.3
Let X be a finite set. Let $f \in \text{Sym}(X)$ be such that $f = f_1 \cdots f_k$ is a product of k-cycles. Then

$$|f| = \text{lcm}(|f_1|, \dots, |f_k|)$$

Proof
We proceed by induction. We have done the case $n = 1$ above. Now suppose that for all $1 \geq m < k$, we have $|f_1 \cdots f_m| = \text{lcm}(n_1, \dots, n_m)$. Let $g = f_1 \cdots f_{k-1}$. Write $f_i = \begin{pmatrix} a_{i,1} & \cdots & a_{i,t_i} \end{pmatrix}$. Then

$$\text{supp}(g) = \{a_{i,j} \mid 1\ leqi < k, 1 \leq j \leq t_i\}$$

and $\text{supp}(f_k) = \{a_{k,1}, \dots, a_{k,t}\}$ so that by lemma 3.4.4, we have $\langle g \rangle \cap \langle f_k \rangle = \{1\}$. By proposition 1.3.6 and since disjoint cycles commute, we have that

$$|f| = \text{lcm}(|g|, |f_k|) = \text{lcm}(|f_1|, \dots, |f_k|)$$

and so we conclude. ■

Proposition 4.4.4
Let $n \in \mathbb{N}^+$. Let $f, g \in S_n$ be permutations such that $f = (a_1, \dots, a_r)$, then

$$gfg^{-1} = \begin{pmatrix} g(a_1) & \cdots & g(a_r) \end{pmatrix}$$

Proof
For $1 \leq i \leq r$, let $b_i = g(a_i)$. For $1 \leq i \leq r - 1$, we have that

$$\begin{aligned} (gfg^{-1})(b_i) &= g(f(a_i)) \\ &= g(a_{i+1}) \\ &= b_{i+1} \end{aligned}$$

Similarly, we have that $(gfg^{-1})(b_r) = a_1$. For $b \in X \setminus \{b_1, \dots, b_r\}$, suppose that $c \in X$ is such that $g(c) = b$. Then we have that $(gfg^{-1})(b) = g(f(c)) = g(c) = b$ and so we conclude. ■

Proposition 4.4.5
Let $n \in \mathbb{N}^+$. Two permutations in S_n are conjugate if and only if they have the same cycle types.

Proof
If f and g are conjugate in S_n, then by the above theorem they have the same cycle type. ■

Definition 4.4.6 (The Sign Function)
Let $n \in \mathbb{N}^+$. Define the sign function $\text{sgn} : S_n \to \{-1, 1\} \leq \mathbb{Q}^\times$ by

$$\text{sgn}(\phi) = \begin{cases} 1 & \text{if } \phi \text{ is even} \\ -1 & \text{if } \phi \text{ is odd} \end{cases}$$

Lemma 4.4.7
Let $n \in \mathbb{N}^+$. Then the sign function is a group homomorphism.

4.5 The Alternating Groups

Based on the number of transpositions an element can be decomposed to, we classify them into odd and even permutations. This motivates the definition of the alternating group.

Definition 4.5.1 (Odd/Even Permutations)
Let $n \in \mathbb{N}^+$. A permutation $\phi \in S_n$ is said to be even if it has an even number of transpositions, and odd if it has an odd number of transpositions.

Definition 4.5.2 (The Alternating Group A_n)
Let $n \in \mathbb{N}^+$. The alternating group A_n is defined to be

$$A_n = \{\phi \in S_n | \phi \text{ is even}\}$$

Lemma 4.5.3
Let $n \in \mathbb{N}^+$. Then the following are true.
- $\ker(\text{sgn}) = A_n$.
- $|A_n| = \frac{n!}{2}$.

Proof
$\ker(\text{sgn}) = A_n$ is true by definition of the alternating group. Then by the first isomorphism theorem we have

$$\frac{S_n}{A_n} \cong \{-1, 1\}$$

By Lagrange's theorem, we have $|S_n|/|A_n| = 2$. Hence $|A_n| = \frac{S_n}{2} = \frac{n!}{2}$. ∎

4.6 The Dihedral Groups

Definition 4.6.1 (The Dihedral Group)
Let $n \geq 3$. Define

$$\sigma = \begin{pmatrix} 1 & \cdots & n \end{pmatrix} \in S_n \quad \text{and} \quad \tau = \prod_{i=1}^{\lfloor n/2 \rfloor} \begin{pmatrix} i & n-i+1 \end{pmatrix} \in S_n$$

Define the dihedral group to be

$$D_{2n} = \langle \sigma, \tau \rangle$$

Lemma 4.6.2
Let $n \geq 3$. Then the following are true regarding the dihedral group $D_{2n} = \langle \sigma, \tau \rangle$.
- $|\sigma| = n$.
- $|\tau| = 2$.
- $\sigma\tau = \tau\sigma^{-1}$.

Proof
It is clear that $|\sigma| = n$ since it is an n-cycle. Similarly, τ is a product of 2-cycles and so has order 2. If n is even, then we have

$$\sigma\tau = \begin{pmatrix} 1 & \cdots & n \end{pmatrix} \prod_{i=1}^{\lfloor n/2 \rfloor} \begin{pmatrix} i & n-i+1 \end{pmatrix} = \prod_{i=2}^{n/2} (i, n-i+2)$$

and

$$\tau\sigma^{-1} = \prod_{i=1}^{\lfloor n/2 \rfloor} \begin{pmatrix} i & n-i+1 \end{pmatrix} \begin{pmatrix} 1 & n & n-1 & \cdots & 2 \end{pmatrix} = \prod_{i=2}^{n/2} (i, n-i+2)$$

If n is odd, we have

$$\sigma\tau = \begin{pmatrix} 1 & \cdots & n \end{pmatrix} \prod_{i=1}^{\lfloor n/2 \rfloor} \begin{pmatrix} i & n-i+1 \end{pmatrix} = \prod_{i=2}^{(n-1)/2} (i, n-i+2)$$

and

$$\tau\sigma^{-1} = \prod_{i=1}^{\lfloor n/2 \rfloor} \begin{pmatrix} i & n-i+1 \end{pmatrix} \begin{pmatrix} 1 & n & n-1 & \cdots & 2 \end{pmatrix} = \prod_{i=2}^{(n-1)/2} (i, n-i+2)$$

∎

Proposition 4.6.3
Let $n \geq 3$. Then $|D_{2n}| = 2n$.

Proof
Since $\sigma\tau = \tau\sigma^{-1}$, it suffices to count all elements in D_{2n} of the form $\sigma^k\tau^m$ for $k, m \in \mathbb{N}$. Since $|\sigma| = n$ and $|\tau| = 2$, we conclude that there are $2n$ elements in D_{2n}. ■

Proposition 4.6.4
Let $n \geq 3$. Write $D_{2n} = \langle \sigma, \tau \rangle$. Then $\langle \sigma \rangle$ is a normal subgroup of D_{2n}.

Proof
Let $\sigma^k\tau^m \in D_{2n}$. Then we have

$$\sigma^k\tau^m\sigma^p\tau^{-m}\sigma^{-k} = \tau^m\sigma^{-k}\sigma^p\sigma^k\tau^{-m} = \tau^m\sigma^p\tau^{-m} = \sigma^{-p}\tau^{m-m} = \sigma^p \in \langle \sigma \rangle$$

Hence $\langle \sigma \rangle$ is normal in D_{2n}. ■

5 Group Actions

Group actions are in a sense more fundamental then the notion of groups. One of the pioneers of group theory, Galois first used techniques in group actions to manipulate roots of a polynomial by permutations.

5.1 Group Actions

Within group theory, there are also interesting applications of group actions by considering groups acting on particular sets such as the set of conjugacy classes and the set of cosets.

Definition 5.1.1 (Groups Actions)
Let G be a group. Let X be a set. An action of G on X is a map $\cdot : G \times X \to X$ such that the following properties are satisfied.

- Associativity: $(g_1g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$
- Identity: $1 \cdot x = x$

Example 5.1.2
Let G be a group. Then G acts on itself by left multiplication.

Example 5.1.3
Let G be a group. Let $H \leq G$ be a subgroup. Then G acts on the set of cosets $G/H = \{gH \mid g \in G\}$ by the action defined by

$$g_1 \cdot g_2H = g_1g_2H$$

Depending on how one looks at action of the group on the set (element wise, or group wise), we obtain two viewpoints of the same idea.

Proposition 5.1.4
Let G be a group. Let X be a set. Suppose that G acts on X. Let $\phi : G \to \text{Sym}(X)$ be the function defined by $\phi(g)(x) = g \cdot x$. Then the following are true.

- ϕ is a well defined map.
- The mapping $\phi : G \to \text{Sym}(X)$ is a homomorphism

Proof

- To show that $\phi(g)$ is a permutation, we simply need to show that it is a bijection on itself. $\phi(g)$ is a bijection if and only if it has an inverse. For all $x \in X$, we have that

$$\begin{aligned}(\phi(g^{-1}) \circ \phi(g))(x) &= \phi(g^{-1})(\phi(g)(x)) \\ &= \phi(g^{-1}g)(x) \\ &= \phi(1)(x) \\ &= x\end{aligned}$$

  Thus we have that $\phi(g^{-1}) = \phi(g)^{-1}$.

- We have that

$$\phi(g_1g_2)(x) = (g_1g_2) \cdot x = g_1 \cdot (g_2 \cdot x) = \phi(g_1)(g_2 \cdot x) = (\phi(g_1) \circ \phi(g_2))(x)$$

  so that $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$

And so we conclude. ∎

In particular, the theorem also shows that G is isomorphic to a subgroup of $\text{Sym}(X)$ provided that ϕ is injective. This is the content of Cayley's theorem.

Theorem 5.1.5 (Cayley's Theorem)
Every group G is isomorphic to a subgroup of $\text{Sym}(G)$.

Proof
Consider the map $\phi_g : G \to G$ defined by $\phi_g(x) = gx$ for any $g \in G$. $\phi_{g^{-1}}$ is an inverse of ϕ_g and thus ϕ_g is bijective and thus $\phi_g \in \text{Sym}(G)$ by definition.
Consider the function $T : G \to \text{Sym}(G)$ defined by $T(g) = \phi_g$. We have that

$$\begin{aligned}(\phi_g \circ \phi_h)(x) &= \phi_g(\phi_h(x)) \\ &= \phi_g(hx) \\ &= ghx \\ &= \phi_{gh}(x)\end{aligned}$$

and thus T is a homomorphism.
Now we show that T is injective. Suppose that $T(g) = \text{id}_G$. This means that $gx = x$ for all $x \in G$. But by definition of a group this means that $g = 1$ and thus $\ker(T) = 1$. Thus G is isomorphic to the image of T, which is a subgroup of $\text{Sym}(G)$. ■

Using the fact that ϕ gives a homomorphism from G to the symmetric group $\text{Sym}(X)$ of the acting set X, we can define the kernel of a group action, which are precisely elements of g that fix the set X.

Definition 5.1.6 (Kernel)
Let G be a group. Let X be a set. Suppose that G acts on X. Define the kernel of the action to be the kernel of $\phi : G \to \text{Sym}(X)$, which is

$$\ker(G, X, \cdot) = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\}$$

We say that the group action is faithful if $\ker(G, X, \cdot) = \{1\}$.

This idea of a kernel is also employed in the proof of Cayley's theorem above. We can also improve on the result of Cayley's theorem using the kernel of an action. Moreover, notice that the kernel of a group action is a normal subgroup of G just because the definition of it is just a special case of the kernel of a general group homomorphism.

Proposition 5.1.7
Let G be a group. Let X be a set of cardinality $n \in \mathbb{N}$. Suppose that G acts on X. If G acts on X faithfully then G is isomorphic to a subgroup of S_n.

Proof
Write $\phi : G \to \text{Sym}(X)$ to be the isomorphism given in theorem 4.1.2. By the first isomorphism theorem, we have that

$$G \cong \frac{G}{\ker(G, X, \cdot)} \cong \text{im}(\phi) \leq \text{Sym}(X)$$

where the last inequality follows because $\text{im}(\phi)$ is a permutation group on X. ■

This is a stronger result then Cayley's theorem because for example, the group G could act on a set X with strictly lower cardinality than G so that the above proposition gives a stronger result.

5.2 Orbits and Stabilizers

Orbits and stabilizers form the core of group actions. Lagrange's theorem applies to orbits and stabilizers to mimic a very similar result to Lagrange's theorem itself. Similar to how Lagrange's theorem is fundamental in group theory, the orbit-stabilizer theorem is fundamental in the theory of group actions.

Definition 5.2.1 (Orbits)
Let G be a group. Let X be a set. Suppose that G acts on X. Define the orbit of G containing x to be the set
$$\text{Orb}_G(x) = \{g \cdot x \mid g \in G\}$$
for any $x \in X$.

Different $x \in X$ could give the same orbit. Indeed if $g \cdot x = y$ then $g^{-1} \cdot y = x$. In fact, this gives an equivalence relation on X defining the orbits of G on X. For $x, y \in X$, suppose that $x \sim y$ if there exists $g \in G$ such that $g \cdot x = y$. It is easy to check that $\sim$ defines an equivalence relation on X and its equivalence classes are in fact the orbits.

Definition 5.2.2 (Stabilizer)
Let G be a group. Let X be a set. Suppose that G acts on X. Let $x \in X$. Define the stabilizer of x to be
$$\text{Stab}_G(x) = \{g \in G \mid g \cdot x = x\}$$

Example 5.2.3
Let G be a group. Let $H \leq G$ be a subgroup. Then the stabilizer of H the action of G on G/H is given by
$$\text{Stab}_G(H) = H$$

The stabilizer cam be seem as a generalization of normalizers. Indeed one can consider the group action of G on the set $\{xHx^{-1} \mid x \in G\}$ for H a subgroup of G. This action is given by $g \cdot xHx^{-1} = gxHx^{-1}g^{-1}$. Then notice that the set of all $g \in G$ that stabilizes H is precisely
$$N_G(H) = \text{Stab}_G(H)$$

We will also see important types of stabilizers when considering particular applications of group actions to some sets.

Proposition 5.2.4
Let G be a group. Let X be a set. Suppose that G acts on X. Then the following are true.
- $\text{Stab}_G(x) \leq G$
- $\bigcap_{x \in X} \text{Stab}_G(x) = \ker(G, X, \cdot)$

Proof
Suppose that $\cdot : G \times X \to X$ is a group action.

- Let $g, h \in \text{Stab}_x$. Then $h^{-1} \in \text{Stab}_G(x)$ since $h^{-1} \cdot x = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = x$. Then
$$\begin{aligned}(gh^{-1}) \cdot x &= g \cdot (h^{-1} \cdot x)\\ &= g \cdot x\\ &= x\end{aligned}$$
Thus $gh^{-1} \in \text{Stab}_x$ and the subgroup criterion is satisfied.
- We show inclusion of the sets. Let $g \in \bigcap_{x \in X} \text{Stab}_G(x)$. Then for any $x \in X$, $g \cdot x = x$ and thus $g \in \ker(G, X)$. Let $g \in \ker(G, X)$, then for any $x \in X$, $g \cdot x = x$ which means that $g \in \text{Stab}_G(x)$ for all $x \in X$. Thus $g \in \bigcap_{x \in X} \text{Stab}_G(x)$.

This gives the desired result. ■

Theorem 5.2.5 (Orbit Stabilizer Theorem)
Let G be a finite group. Let X be a set. Suppose that G acts on X. If $x \in X$, then
$$|\text{Orb}_G(x)| = [G : \text{Stab}_G(x)] \quad \text{and} \quad |G| = |\text{Orb}_G(x)||\text{Stab}_G(x)|$$

Proof
The goal is to define a bijection between cosets of $\text{Stab}_G(x)$ and elements in $\text{Orb}_G(x)$. Define $f : \text{Orb}_G(x) \to G/G_x$ by $f(g) = g\text{Stab}_G(x)$. This map is clearly well defined.
Injectivity:
Let $g\text{Stab}_G(x) = h\text{Stab}_G(x)$. Then $g^{-1}h \in \text{Stab}_G(x)$ which means that $(g^{-1}h) \cdot x = x$. Then applying the action of g on both sides, we have

$$h \cdot x = g \cdot x$$

which proves that $g = h$.
Surjectivity:
For any $g\text{Stab}_G(x)$, just choose $g \in G$. Then clearly $f(g) = g\text{Stab}_G(x)$.
Thus $|\text{Orb}_G(x)| = [G : \text{Stab}_G(x)]$. ■

The following corollary is an important consequence of the Orbit Stabilizer theorem.

Corollary 5.2.6
Let G be a finite group. Let X be a set. Suppose that G acts on X. Then the following are true regarding the orbits of the action.
- Either $\text{Orb}_G(x) = \text{Orb}_G(y)$ or $\text{Orb}_G(x) \cap \text{Orb}_G(y) = \emptyset$
- The set of all orbits form a partition of X

Proof
Define a relation where for $x, y \in X$, we say that $x \sim y$ if and only if $g \cdot x = y$ for some $g \in G$. It is clear that this is an equivalence relation and thus the corollary follows. ■

5.3 The Action of Conjugation

Conjugation marks one of the most important group actions that one should consider, with the other two being rather trivial: the action of left multiplication and the action of G on the set of cosets of H in G. In particular, Cauchy's theorem is a major application of such a group action.

Definition 5.3.1 (Conjugation)
Let G be a group. Define conjugation to be an action of G on G where $\cdot : G \times G \to G$ is defined by $g \cdot x = gxg^{-1}$. Define the conjugacy classes of a group to be the orbits of conjugation:

$$\text{Cl}(x) = \{gxg^{-1} \mid g \in G\} = \text{Orb}_G(x)$$

Two elements are said to be conjugate to each other if they lie in the same conjugacy classes.

It is easy to see that the action of conjugation is indeed a group action. Aside from the conjugacy classes of a group, we also have the notion of centralizers.

Definition 5.3.2 (Centralizer and Center) Let G be a group. Define the centralizer of a group to be the stabilizer of conjugation:

$$C_G(x) = \{g \in G \mid gx = xg\} = \text{Stab}_G(x)$$

Define the center of a group to be the kernel of conjugation:

$$Z(G) = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in X\} = \ker(G, G, \cdot)$$

Since $Z(G)$ consists of all elements of G that commute with other elements, $Z(G) = G$ naturally implies that G is abelian.

Note that conjugacy classes and centralizers and centers can be defined independent of the notion of group action. Indeed they can be defined by the set of conditions. However, by defining these notions through the use of group actions show that group actions are indeed a rather general theory and applies to all kinds of subgroups one can imagine.

Lemma 5.3.3
Let G be a group. Then $Z(G) \trianglelefteq G$ and $C_G(x) \leq G$.

Proof
$Z(G)$ is the kernel by definition and so it is a normal subgroup of G. $C_G(x)$ is a special case of the stabilizer and so it is a subgroup of G. ■

The following proposition is a non-trivial result of what one can do with the center of a group.

Proposition 5.3.4
Let G be a group. If $\frac{G}{Z(G)}$ is cyclic, then G is abelian.

Proof
Suppose that $\langle xZ(G)\rangle = G/Z(G)$. Let $g \in G$. Then $gZ(G) = x^a Z(G)$ for some $a \in \mathbb{N}$. This means that $x^{-a}g \in Z(G)$. This means that there is some $z \in Z(G)$ such that $x^{-a}g = z$ which implies that $g = x^a z$. Now suppose that we go through the same process with $h \in G$ to obtain $h = x^b z'$. Then we have

$$\begin{aligned} gh &= x^a z x^b z' \\ &= x^{a+b} z z' && (z \in Z(G)) \\ &= x^{b+a} z' z \\ &= x^b z' x^a z \\ &= hg \end{aligned}$$

Thus we are done. ■

Proposition 5.3.5
Let p be a prime and let $n \in \mathbb{N}$. Let G be a finite group with $|G| = p^n$. Then $|Z(G)| > 1$.

Proof
Consider the action of G on itself by conjugation. Then corollary 4.2.5 shows that the order of $\text{Cl}(x)$ is a power of p. A rewrite of definition shows that $Z(G) = \{x \in G \mid |\text{Cl}(x)| = 1\}$. Suppose that $|Z(G)| = 1$. Then corollary 4.2.5 again implies that $p^n = |G| = 1 + p^{a_1} + \cdots + p^{a_m}$ since all the conjugacy classes partition G. However this is a contradiction since the left hand side is divisible by p while the right hand side is not. ■

Corollary 5.3.6
Let p be a prime and $n \in \mathbb{N}$. Let G be a finite group with $|G| = p^n$. Then the following are true.

- If $n = 2$, then $|G|$ is abelian.
- If $n = 3$, then either G is abelian or $|Z(G)| = p$

Proof
Suppose that $|G| = p^2$. By Lagrange's theorem, $|Z(G)|$ is either p or p^2. If $|Z(G)| = p$, then $\frac{G}{Z(G)}$ has order p a prime. This means that $\frac{G}{Z(G)}$ is cyclic. By proposition 4.3.4, G is abelian. If $|Z(G)| = p^2$, then $|Z(G)| = G$ thus G is again abelian.

Now suppose that $|G| = p^3$. By Lagrange's theorem, $|Z(G)|$ is either p, p^2 or p^3. If it is p^2, then $\frac{G}{Z(G)}$ has order p and thus is cyclic. By proposition 4.3.4, G is abelian. If $|Z(G)| = p^3$, then $|Z(G)| = G$ and thus G is again abelian. Finally, the case $|Z(G)| = p$ is left. ■

We end the section with two important consequence of the Orbit Stabilizer theorem applied to the action of conjugacy.

Theorem 5.3.7 (Cauchy's Theorem)
Let G be a finite group. Let p be a prime divisor of $|G|$. Then G contains an element of order p. Moreover, the number of elements of order p is congruent to -1 modulo p.

Proof
Define $X = \{(x_1, \dots, x_p) | x_i \in G \text{ and } x_1 \cdots x_p = 1\}$. Notice that

$$\begin{aligned}(x_1, \dots, x_p) \in X &\implies x_1 \cdots x_p = 1 \\ &\implies x_2 \cdots x_p x_1 = 1 \\ &\implies (x_2, \dots, x_p, x_1) \in X\end{aligned}$$

Repeatedly applying the above proves that

$$(x_i, \dots, x_p, x_1, \dots, x_{i-1}) \in X$$

for all $i \leq p$. Now let $C = \langle (1, \dots, p) \rangle \leq S_p$ be the subgroup of the symmetric group S_p generated by the p-cycle $(1, \dots, p)$. Define an action $\cdot : C \times X \to X$ by

$$(\sigma, (x_1, \dots, x_p)) = (x_{\sigma(1)}, \dots, x_{\sigma(x_p)})$$

for $\sigma \in C$. This is indeed a group action: Associativity and Identity is easy to check in conjunction with the above fact that $(x_i, \dots, x_p, x_1, \dots, x_{i-1}) \in X$ for $(x_1, \dots, x_p) \in X$.

By corollary 4.2.5, the orbits of this action partition X. There are now three types of orbits.

- Type 1: Orbits of size 1.
  Notice that $f = (1, \dots, p) \in \text{Stab}_C((1, \dots, 1))$. Since $C = \langle f \rangle$ has prime order and $\text{Stab}_C(1, \dots, 1) \leq C$, we conclude that $\text{Stab}_C((1, \dots, 1)) = C$ which implies that $|\text{Orb}_C((1, \dots, 1))| = 1$ by the orbit stabilizer theorem. A similar argument shows that the p-tuple $(x, \dots, x)$ has also orbit size 1.

  In fact, every orbit of size 1 must be one of the above two cases. Suppose that S is an orbit of size 1. Let $(x_1, \dots, x_p) \in X$. Then the orbit stabilizer theorem implies that $\text{Stab}_C((x_1, \dots, x_p)) = C$. Since $f = (1, \dots, p) \in C$, we have

  $$\begin{aligned}(x_1, \dots, x_p) &= f \cdot (x_1, \dots, x_p) && (f \in \text{Stab}_C((x_1, \dots, x_p))) \\ &= (x_2, \dots, x_p, x_1)\end{aligned}$$

  which means that $x_1 = \cdots = x_p$. This means that $x_1^p = 1$. Since p is a prime this means that $|x_1| = 1$ or p. In which either way it falls into one of the two above cases.
- Type 2: Orbits of size p.
  Any orbit of size greater than 1 must have orbit size p. Indeed if $T_1, \dots, T_n$ are all the distinct orbits of size greater than 1, the fact that $|C| = p$ together with the orbit stabilizer theorem implies that each T_i must have order p.

Now we show that $|X| = |G|^{p-1}$. Define a map $\phi : X \to G^{p-1}$ by $\phi((x_1, \dots, x_p)) = (x_1, \dots, x_{p-1})$. Injectivity is clear since x_p is uniquely determined: $x_p = (x_1 \cdots x_{p-1})^{-1}$. Surjectivity is also clear since for $(x_1, \dots, x_{p-1}) \in G^{p-1}$, we can just choose $x_p = (x_1 \cdots x_{p-1})^{-1}$. Thus $|X| = \left|G^{p-1}\right| = |G|^{p-1}$.

Now let $\mathcal{P} = \{x \in G | |x| = p\}$. Then since the orbits partition X, we have

$$|G|^{p-1} = |X| = |\text{Orb}_G((x_1, \dots, x_p))| + \sum_{x \in \mathcal{P}} |\text{Orb}_G((x, \dots, x))| + \sum_{i=1}^{n} |T_i| = 1 + |\mathcal{P}| + np$$

Since $p | |G|$, we deduce that $|\mathcal{P}| \equiv -1 \pmod p$. In particular, G must contain an element of order p. ∎

Cauchy's theorem is a useful tool on analysing the the elements of a group.

Theorem 5.3.8 (The Class Equation)
Let G be a finite group and let $g_1, g_2, \dots, g_r$ be representatives of the distinct conjugacy classes of G not contained in $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r}[G : C_G(g_i)] = |Z(G)| + \sum_{i=1}^{r}|\text{Orb}_G(x_i)|$$

Proof
Suppose that G acts on itself by conjugation. We know that the orbits partition the group G since it is a quotient of an equivalence relation. However, for any $g \in Z(G)$, $gxg^{-1} = x$ for any $x \in X$. Thus every element in $Z(G)$ is itself an orbit. Combining these facts, we have that

$$|G| = |Z(G)| + \sum_{i=1}^{r}|G : C_G(g_i)| = |Z(G)| + \sum_{i=1}^{r}|\text{Cl}(g_i)|$$

■

5.4 Free Actions and Fixed Points

Definition 5.4.1 (Fixed Points)
Let G be a group. Let X be a set. Suppose that G acts on X. Let $g \in G$. We say that $x \in X$ is a fixed point of g if $g \cdot x = x$. The set of all fixed points of $g \in G$ is

$$\text{Fix}_X(g) = \{x \in X \mid g \cdot x = x\}$$

We say that $g \in G$ is fixed point free if $\text{Fix}_X(g) = \emptyset$.

Lemma 5.4.2 (The Not Burnside Lemma)
Let G be a finite group. Let X be a finite set. Suppose that G acts on X. Then

$$|\{\text{Orb}_G(x) \mid x \in X\}| = \frac{1}{|G|}\sum_{g \in G}|\text{Fix}_X(g)|$$

Proof
Define the set

$$A = \{(g, x) \in G \times X \mid g \cdot x = x\}$$

It is clear that $|A| = \sum_{g \in G}|\text{Fix}_X(g)| = \sum_{x \in X}|\text{Stab}_G(x)|$. Let $\text{Orb}_G(x_1), \dots, \text{Orb}_G(x_r)$ denote all the distinct orbits in X. We have seen that

$$X = \coprod_{i=1}^{r}\text{Orb}_G(x_i)$$

Also if $x \in \text{Orb}_G(x_i)$, then $\text{Stab}_G(x) = \text{Stab}_G(x_i)$ by the orbit stabilizer theorem. Thus we have

$$\begin{aligned}
\sum_{g \in G}|\text{Fix}_X(g)| &= \sum_{x \in X}|\text{Stab}_G(x)| \\
&= \sum_{x \in X}|\text{Stab}_G(x)| \\
&= \sum_{i=1}^{r}|\text{Stab}_G(x)| \cdot |\text{Orb}_G(x_i)| \\
&= \sum_{i=1}^{r}|G| \\
&= r|G|
\end{aligned}$$

and so we conclude. ■

Corollary 5.4.3
Let G be a finite group. Let X be a finite set such that $|X| > 1$. Suppose that G acts on X. Suppose that G has precisely one orbit. Then G contains fixed point free element.

Proof
Suppose for a contradiction that G contains no fixed point free elements. Then $|\text{Fix}_X(g)| \geq 1$ for all $g \in G$. Then since $\text{Fix}_X(1_G) = |X|$, the above lemma implies that

$$|G| = \sum_{g \in G} |\text{Fix}_X(g)| = |X| + \sum_{g \in G \setminus \{1\}} |\text{Fix}_X(g)| \geq |X| + |G| - 1$$

which is a contradiction since $|X| \geq 1$. Thus we conclude. ■

Notice that it is important that G is a finite group in this corollary, otherwise it would not hold.

Definition 5.4.4 (Free Actions)
Let G be a group. Let X be a set. Suppose that G acts on X. We say that G is a free action on X if $g \in G$ is such that there exists some $x \in X$ such that $g \cdot x = x$, then $g = 1$.

Lemma 5.4.5
Let G be a group. Let X be a set. Suppose that G acts on X. Then G is a free action on X if and only if $\text{Fix}_X(g) = \emptyset$ for all $g \neq 1$.

Proof
Suppose that G is a free action. Then for all $g \neq 1$, $g \cdot x \neq x$ for all x. Thus $\text{Fix}_X(g) = \emptyset$. Now suppose that the converse is true. Then $g \cdot x \neq x$ for all $x \in X$ and all $g \neq 1$. Thus G is a free action on G. ■

5.5 Transitive Actions

Definition 5.5.1 (Transitive Actions)
Let G be a group. Let X be a set. Suppose that G acts on X. We say that G is a transitive action on X if for all $x, y \in X$, there exists $g \in G$ such that $g \cdot x = y$.

Lemma 5.5.2
Let G be a group. Let X be a set. Suppose that G acts transitively on X. Then $\text{Orb}_G(x) = X$.

Proof
Immediate from definitions. ■

6 Products of Groups

Groups are defined as a set with additional constraints. Since sets have the Cartesian product, it is natural to also consider group structures on the Cartesian product. There are, however, three different notions of products of groups.

6.1 External Direct Products

We first have the external product. It is the most direct and natural way to construct multiplication between two groups.

Definition 6.1.1 (External Product)
Let $(G, *)$ and $(H, \circ)$ be groups. Define a group operation of G and H called the external (direct) product $G \times H$ with the operation

$$(g_1, h_1) \times (g_2, h_2) = (g_1 * g_2, h_1 \circ h_2)$$

for $g_1, g_2 \in G$ and $h_1, h_2 \in H$.

It is easy to see that the group operation indeed defines a group structure on $G \times H$. Indeed we can check the group operations element wise. Moreover, from set theory we know that the order of $G \times H$ is equal to $|G| \times |H|$. There is nothing mysterious about external direct products.

Proposition 6.1.2
Let $(g, h) \in G \times H$. If g and h has finite order r and s respectivelty, then

$$|(g, h)| = \text{lcm}(r, s)$$

Proof
Trivially we have that $|(g,h)|| \text{lcm}(r,s)$. Suppose that $|(g,h)| < \text{lcm}(r,s)$. Then by definition of the lcm, either $r||(g,h)|$ or $s||(g,h)|$ but not both. But then maximally we only have one of the elements in (g,h) be the identity. Thus we must have $|(g,h)| = \text{lcm}(r,s)$. ■

Proposition 6.1.3
Let G_1, G_2 be groups. Then the following are true.

- $G_1 \times \{1\}$ and $\{1\} \times G_2$ are normal subgroups of $G_1 \times G_2$.
- $\frac{G_1 \times G_2}{G_1 \times \{1\}} \cong G_2$, and $\frac{G_1 \times G_2}{\{1\} \times G_2} \cong G_1$

Proof
Without loss of generality we only consider the subgroup G_1 in $G_1 \times G_2$. Checking that $G_1 \times \{1\}$ is trivial. For any $(g_1, g_2) \in G_1 \times G_2$ and $(h, 1) \in G_1 \times \{1\} \leq G_1 \times G_2$, we have

$$(g_1, g_2) \cdot (h, 1) \cdot (g_1^{-1}, g_2^{-1}) = (g_1 h g_1^{-1}, 1) \in G_1 \times \{1\}$$

Hence $G_1 \times \{1\}$ is normal in $G_1 \times G_2$.

Define a function $\phi : \frac{G_1 \times G_2}{G_1 \times \{1\}} \to G_2$ by

$$\phi((g_1, g_2)G_1 \times \{1\}) = g_2$$

To show that this function is well defined, we need to show that different representatives of the same element in the quotient maps to the same element. So suppose that $(g_1, g_2)G_1 \times \{1\} = (h_1, h_2)G_1 \times \{1\}$. Then we have that $(h_1^{-1}, h_2^{-1}) \cdot (g_1, g_2) \in G_1 \times \{1\}$. Hence $g_2 = h_2$, and so

$$\phi((g_1, g_2)G_1 \times \{1\}) = g_2 = h_2 = \phi((h_1, h_2)G_1 \times \{1\})$$

We now show that ϕ is bijective. Suppose that $\phi((g_1, g_2)G_1 \times \{1\}) = \phi((h_1, h_2)G_1 \times \{1\})$. Then we have $g_2 = h_2$. But $g_2 = h_2$ implies that $(g_1, g_2)G_1 \times \{1\} = (h_1, h_2)G_1 \times \{1\}$. Hence ϕ is injective. Now suppose that $g_2 \in G_2$. Then $\phi((1, g_2)G_1 \times \{1\}) = g_2$. Hence ϕ is surjective. Thus the two groups are isomorphic. ■

Taking $G_2 = \{1\}$ the trivial group, we see that $G_1 \times \{1\}$ is isomorphic to G_1. Hence we usually write the subgroup $G_1 \times \{1\}$ in $G_1 \times G_2$ simply as G_1.

Proposition 6.1.4
Let $m, n \in \mathbb{N}$. Then
$$C_m \times C_n \cong C_{mn}$$
if and only if $\gcd(m, n) = 1$.

Proof
Let $C_m \times C_n \cong C_{mn}$. Let g, h be the generator of C_m, C_n respectively. Then (g, h) is the generator of $C_{m,n}$ since isomorphisms map generators to generators. We have proved that $|\gcd(g, h)| = \text{lcm}(m, n) = \frac{mn}{\gcd(m,n)}$. But $|C_n \times C_m| = mn$ thus $mn = \frac{mn}{\gcd(m,n)}$ and $\gcd(m, n) = 1$.
Now let $\gcd(m, n) = 1$. Define a function $\phi : C_m \times C_n \to C_{m\times n}$ where $\phi(g^a, h^b) = g^a h^b$. I will show that it is an isomorphism. Firstly ϕ is a homomorphism since
$$\phi((g^{a_1}, h^{b_1})()) =$$
■

6.2 Internal Direct Products

While the external product is concerned of the product between two groups, the internal direct product is interested in how two subgroups of a group could be "multiplied" to form the group itself. Essentially, we are trying to decompose a group into its internal direct product.

Definition 6.2.1 (Product of Subgroups)
Let G be a group. Let $H, K \leq G$ be subgroups. Define the product of H and K to be
$$HK = \{hk \mid h \in H, k \in K\}$$

Notice that the set HK is not necessarily a group by its own right. Even if it is, it could happen that HK is a subgroup of G. In particular, we have a necessary and sufficient criterion for HK to be a subgroup of G.

Proposition 6.2.2
Let G be a group. Let $H, K \leq G$. Then HK is a subgroup if and only if $HK = KH$.

Proof First suppose that HK is a subgroup. Since $K \subseteq HK$ and $H \subseteq HK$, we have that $KH \subseteq HK$. by closure. To show the reverse. Let $hk \in HK$. Then there exists some $a \in HK$ such that $a^{-1} = hk$ since HK is a group. Then $a \in HK$ implies that $a = h'k'$ for some $h'k' \in HK$. Then
$$hk = a^{-1} = k'^{-1}h'^{-1} \in KH$$
thus we are done.
Now suppose that $HK = KH$. Let $h_1k_1 \in HK$ and $h_2k_2 \in HK$. We show that it satisfies the subgroup criterion.

- $1 \in HK$ is trivial
- For closure, Note that $k_1h_2 = h'k'$ since $KH = HK$. Then
$$h_1k_1h_2k_2 = h_1h'k'k_2 = HK$$
The last part is true since H, K are subgroups.
- Let $hk \in HK$. Then $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$

Thus we are done. ■

Applying the theory of group actions gives us to important propositions to proceed in finding the criteria for $HK \cong G$.

Proposition 6.2.3
Let G be a finite group. Let H, K be subgroups of G. Then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

Proof
Consider K acting on G/H be left multiplication defined by $k \cdot gH = (kg)H$. We have that

$$KH = \coprod_{i=1}^{s} k_i H$$

for each k_iH and k_jH disjoint if $i \neq j$. Thus $|KH| = \sum_{i=1}^{s} |k_iH| = \sum_{i=1}^{s} |H| = s|H|$. On the other hand, notice that $gH \in \text{Orb}_K(H)$ if and only if $gH = kH$ for some $k \in K$. Thus $\text{Orb}_K(H) = \{kH \mid k \in K\} = \{k_1H, \dots, k_sH\}$ and $|\text{Orb}_K(H)| = s = \frac{|KH|}{|H|}$.

Now if $k \in K$, then $k \in \text{Stab}_K(H)$ if and only if $kH = H$ which is true if and only if $k \in H$. Thus $\text{Stab}_K(H) = H \cap K$. By the orbit stabilizer theorem, we have that

$$|K| = |\text{Orb}_K(H)| \cdot |\text{Stab}_K(H)|$$

implies that $|K| = \frac{|KH|}{|H|}|H \cap K|$ and so we conclude. ■

Proposition 6.2.4
Let G be a finite group. Let $H, K \leq G$ be subgroups of G. Then

$$[G : H \cap K] \leq [G : H][G : K]$$

Proof
Let $X_1 = G/H$ and $X_2 = G/K$ be the set of left cosets of H and K respectively. It is clear that G acts on X_1 and X_2 so that G acts on $X_1 \times X_2$. Let $x = (H, K) \in X_1 \times X_2$. For $g \in G$, we have that

$$g \in \text{Stab}_G(x) \iff gH = H \text{ and } gK = K$$

But this is true if and only if $g \in H$ and $g \in K$. Thus $\text{Stab}_G(x) = H \cap K$. By the orbit stabilizer theorem, we have that

$$[G : H \cap K] = |\text{Orb}_G(x)| \leq |X| = [G : H][G : K]$$

and so we conclude. ■

Proposition 6.2.5
Let G be a finite group. Let H and K be subgroups of G such that $\gcd([G : H], [G : K]) = 1$. Then $G = HK$.

Proof
Notice that since the gcd is 1, we have that

$$\text{lcm}([G : H], [G : K]) = \frac{|G|^2}{|H||K|}$$

Also we have that $[G : H \cap K] = [G : H][H : H \cap K]$ and $[G : H \cap K] = [G : K][K : H \cap K]$ which means that $[G : H \cap K]$ is a common multiple of $[G : H]$ and $[G : K]$ so that

$$\text{lcm}([G : H], [G : K]) \leq [G : H \cap K]$$

Then we have that

$$\frac{|G|^2}{|H||K|} \leq [G : H \cap K] = \frac{|G|}{|H \cap K|}$$

which shows that

$$|G| \leq \frac{|H||K|}{|H \cap K|} = |HK|$$

We also have trivially that $|HK| \leq |G|$. Thus we conclude. ■

Definition 6.2.6 (Internal Direct Products)
Let G be a group. Let $H, K \leq G$ be subgroups. We say that G is the internal direct product of G if $G \cong H \times K$.

To recognize a group as a product of subgroups, we first learn what the essence of external direct products are. Let G_1, G_2 be groups. Then $G_1 \times G_2$ has the following important properties:

- $G_1 \cap G_2 = \{1\}$.
- Every element of $G_1 \times G_2$ can be expressed uniquely as the product of an element in G_1 and an element in G_2.
- G_1, G_2 are normal subgroups of $G_1 \times G_2$ (under the assumptions of the above two, these condition is equivalent to saying elements of G_1 commute with elements of G_2).

In the start of this subsection we introduced the product of two subgroups. This is so that we may recognize condition two in the above.

Proposition 6.2.7
Let G be a group. Let $H, K \leq G$ be subgroups. Then G is the internal direct product of H and K if the following are true.

- $H \cap K = \{1\}$.
- $G = HK$.
- H, K are normal subgroups of G.

Proof
Since H and K are normal subgroups of G, we have that $HK = KH$. Indeed, let $hk \in HK$. Then $hk \in hK = Kh$ hence there exists $k' \in K$ such that $hk = k'h \in KH$. Similarly, let $kh \in KH$. Then $kh \in Kh = hK$ hence there exists $k' \in K$ such that $kh = hk' \in HK$. Hence by 6.2.2, we know that HK is a subgroup of G.

Define a function $f : H \times K \to G$ by $f(h, k) = hk$. I claim that f is a group isomorphism. Firstly, I claim that f is a group homomorphism. Let $h_1k_1 \in HK$ and $h_2k_2 \in HK$. Then we have

$$f(h_1, k_1)f(h_2, k_2) = f((h_1, k_1)(h_2, k_2)) \iff h_1k_1h_2k_2 = h_1h_2k_1k_2 \iff h_1k_1h_2k_1^{-1}h_2^{-1}h_1^{-1} = 1$$

Since H is normal in G, we have that $k_1h_2k_1^{-1} \in H$. Hence $h_1(k_1h_2k_1^{-1})h_2^{-1}h_1^{-1} \in H$. Similarly, since K is normal in G, we have that $k' = h_2k_1^{-1}h_2^{-1} \in K$ and hence $h_1k_2k'h_1^{-1} \in K$. Hence $h_1k_1(h_2k_1^{-1}h_2^{-1})h_1^{-1} \in K$. Then we have

$$h_1k_1h_2k_1^{-1}h_2^{-1}h_1^{-1} \in H \cap K = \{1\}$$

so that f is a group homomrphism.

For injectivity, suppose that $(h, k) \in \ker(f)$. Then $hk = 1$ implies $h = k^{-1}$. Then $h \in H$ and $h = k^{-1} \in K$ implies that $h \in H \cap K = \{1\}$. Hence $h = 1$. Similarly, we have that $k = 1$. Hence $(h, k) = 1$ and f is injective. For surjectivity, suppose that $g \in G$. Then $G = HK$ implies that there exists $hk \in HK$ such that $hk = g$. Then $f(h, k) = hk = g$ so that f is surjective. ■

6.3 Semidirect Products

We would like to generalize the direct product of two finite groups.

Proposition 6.3.1
Let H, K be groups. Let ϕ be a homomorphism from K into $\text{Aut}(H)$. Let G be the set of all ordered pairs (h, k) where $h \in H$ and $k \in K$. Then the operation

$$(h_1, k_1)(h_2, k_2) = (h_1 \cdot \phi(k_1)(h_2), k_1k_2)$$

allows G to be a group.

Proof
Clearly closure is satisfied. We check associativity since it is asymetrical. We have that

$$\begin{aligned}((h_1, k_1)(h_2, k_2))(h_3, k_3) &= (h_1 \cdot \phi_{k_1}(h_2), k_1k_2)\,(h_3, k_3) \\ &= (h_1 \cdot \phi_{k_1}(h_2) \cdot (\phi_{k_1k_2}(h_3)), k_1k_2k_3) \\ &= (h_1 \cdot \phi_{k_1}(h_2) \cdot (\phi_{k_1} \circ \phi_{k_2})(h_3), k_1k_2k_3) \\ &= (h_1 \cdot \phi_{k_1}(h_2 \cdot \phi_{k_2}(h_3)), k_1k_2k_3) \\ &= (h_1, k_1)(h_2 \cdot \phi_{k_2}(h_3), k_2k_3) \\ &= (h_1, k_1)((h_2, k_2)(h_3, k_3))\end{aligned}$$

The identity in this group will be $(1_H, 1_K)$. Indeed we have that $(h, k)(1_H, 1_K) = (h \cdot \phi_k(1_H), k) = (h, k)$ and $(1_H, 1_K)(h, k) = (\phi_{1_K}(h), k) = (h, k)$. The inverse of (h, k) is $(\phi_{k^{-1}}(h^{-1}), k^{-1})$ ■

Definition 6.3.2 (Semidirect Product)
Let H, K be groups. Let $\phi : K \to \text{Aut}(H)$ be a group homomorphism. Define the semidirect product of H and K to be the set

$$H \rtimes_\phi K = H \times K$$

equipped with the operation

$$(h_1, k_1)(h_2, k_2) = (h_1 \cdot \phi(k_1)(h_2), k_1k_2)$$

Some authors define semidirect products alternatively as $H \ltimes_\psi K$ for $\psi : H \to \text{Aut}(K)$. It is easy to see that the two definitions are equivalent by establishing an isomorphism

$$H \rtimes_\phi K \cong K \ltimes_\phi H$$

by swapping the order of the cartesian pair.

Proposition 6.3.3
Let H, K be groups. Let $\phi : K \to \text{Aut}(H)$ be a group homomorphism. Then the following are true regarding the semidirect product.

- $H \cong \{(h, 1) \mid h \in H\} \trianglelefteq H \rtimes_\phi K$ is a normal subgroup of G.
- $K \cong \{(1, k) \mid k \in K\} \leq H \rtimes_\phi K$.
- $H \cap K = 1$.

Proof
Clearly $H \cong \{(h, 1)|h \in H\}$. Notice that $(h_1, 1)(h_2, 1) = (h_1 \cdot (\phi(1))(h_2), 1) = (h_1h_2, 1)$ since $\phi(1)$ is the identity automorphism. This means that the multiplication structure of H is preserved. Since H itself is a group, this makes the set a subgroup of G. Let $(h, k) \in G$. Let $(h_0, 1) \in H$. Then

$$(h, k)(h_0, 1)(\phi_{k^{-1}}(h^{-1}), k^{-1}) = (h, k)(h_0 \cdot (\phi_{k^{-1}}(h^{-1})), k^{-1})$$

Notice that the second term multiplies to 1 which means that the entire product lies in H. Thus we have proven $(h, k)H(h, k)^{-1} \subseteq H$.

Similarly, notice that $(1, k_1)(1, k_2) = (\phi(k_1)(1), k_1k_2) = (1, k_1k_2)$. The results then follow.

From the above identification we clearly see that they are disjoint except from the identity. ■

These are the fundamental results that we want the semidirect product to have when we construct it.

Proposition 6.3.4
Let H, K be groups. Let $\phi : K \to \text{Aut}(H)$ be a group homomorphism. Then the following are equivalent.

- The identity map between $H \rtimes_\phi K \cong H \times K$ is a group isomorphism.
- $K \trianglelefteq H \rtimes_\phi K$.
- ϕ is the trivial homomorphism from K to $\text{Aut}(H)$.

Proof
(1) $\implies$ (2): We have seen that $K \trianglelefteq H \times K$ so we are done.

(2) $\implies$ (3): Assume that K is normal in $H \rtimes_\phi K$. Then we know that $(h, 1)(1, k)(h^{-1}, 1) \in K$. But also

$$(h^{-1}, 1)(1, k)(h, 1) = (h^{-1}\phi(1)(1), k)(h, 1) = (h^{-1}, k)(h, 1) = (h^{-1}\phi(k)(h), k) \in K$$

Hence $h^{-1}\phi(k)(h) = 1$ implies that $\phi(k)(h) = h$ for all $k \in K$ and $h \in H$. Hence ϕ is the trivial homomorphism.

(3) $\implies$ (1): If K is the trivial homomorphism, then multiplication in $H \rtimes_\phi (K)$ is given by $(h_1, k_1)(h_2, k_2) = (h_1h_2, k_1k_2)$. Hence $H \rtimes_\phi K \cong H \times K$ via the identity homomorphism. ■

The above proposition gives a necessary and sufficient condition for the semi direct product to be isomorphic to the external direct product. The following theorem gives a sufficient criterion for the semi direct product to be isomorphic to the internal direct product.

Proposition 6.3.5
Let G be a group with subgroups H, K such that $H \trianglelefteq G$ and $H \cap K = 1$. Let $\phi : K \to \text{Aut}(H)$ be the homomorphism defined by $\phi(k)(h) = khk^{-1}$. Then $HK \cong H \rtimes_\phi K$.

Proof
Firstly note that if $h_1k_1 = h_2k_2$ in the product HK, then $h_2^{-1}h_1 = k_2k_1^{-1}$ and together with $H \cap K = 1$, this implies that $h_2^{-1}h_1 = 1$ and $k_2k_1^{-1}$ and $h_1 = h_2$ and $k_1 = k_2$.

Define a map $\psi : HK \to H \rtimes_\phi K$ by $\psi(hk) = (h, k)$. It is well defined because every element in HK can be written uniquely as an element h times an element in K by the above. It is a group homomorphism because we have

$$\begin{aligned}
\psi(h_1k_1h_2k_2) &= \psi(h_1k_1h_2k_1^{-1}k_1k_2) \\
&= \psi(h_1h_3k_1k_2) && (H \text{ is normal}, h_3 = k_1h_2k_1^{-1}) \\
&= (h_1h_3, k_1k_2) \\
&= (h_1\phi(h_2)(k_1), k_1k_2) \\
&= (h_1, h_2)(k_1, k_2) \\
&= \psi(h_1, k_1)\psi(h_2, k_2)
\end{aligned}$$

It remains to show that this is a bijection. But $|HK| = \frac{|H||K|}{|H\cap K|} = |H||K|$ by assumption and $|H \rtimes_\phi K| = |H||K|$. Also it is clear that ϕ is surjective. Thus ψ is a bijection and we are done. ■

As a direct result, if in addition to the above conditions, we have $G = HK$ then $G \cong H \rtimes_\phi K$.

7 Introduction to Rings

7.1 Basic Concept of Rings

Definition 7.1.1 (Rings)
A ring consists of an abelian group $(R, +)$ together with a binary operation $\cdot : R \times R \to R$ such that the following are true.

- Associativity: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$.
- Unitality: There exists an element $1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in R$.
- Left distributivity: $(a + b) \cdot c = (a \cdot c) + (b \cdot c)$ for all $a, b, c \in R$.
- Right distributivity: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ for all $a, b, c \in R$.

Example 7.1.2
The following are rings.

- $(\mathbb{Z}, +, \cdot)$.
- $(\mathbb{Q}, +, \cdot)$.
- $(\mathbb{R}, +, \cdot)$.
- $(\mathbb{C}, +, \cdot)$.
- Let R be a ring. Let X be a set. Then the set

$$\{f : X \to R \mid f \text{ is a function }\}$$

together with pointwise addition and multiplication is a ring.

Lemma 7.1.3
Let $(R, +, \cdot)$ be a ring.

- $0 \cdot a = a \cdot 0 = 0$ for all $a \in R$.
- $-a = (-1) \cdot a = a \cdot (-1)$.
- $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$ for all $a, b \in R$.
- $(-1) \cdot (-1) = 1$.
- $(-a) \cdot (-b) = a \cdot b$ for all $a, b \in R$.

Proof
We have $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$. Hence $0 \cdot a = 0$.

We have $(-1) \cdot a + a = ((-1) + 1) \cdot a = 0 \cdot a = 0$. Hence $(-1) \cdot a = -a$. Similarly, we have $a \cdot (-1) + a = a((-1) + 1) = a \cdot 0 = 0$. Hence $a \cdot (-1) = -a$.

We have $(-a) \cdot b = (-1) \cdot a \cdot b = a \cdot (-1) \cdot b = a \cdot (-b)$. Also, we have $(-a) \cdot b = (-1) \cdot a \cdot b = (-1)(a \cdot b)$.

We have $(-1) \cdot (-1) + (-1) = (-1) \cdot (-1) + 1 \cdot (-1) = ((-1) + 1) \cdot (-1) = 0 \cdot (-1) = 0$. Hence $(-1) \cdot (-1) = 1$.

We have $(-a) \cdot (-b) = a \cdot (-1) \cdot (-1) \cdot b = a \cdot 1 \cdot b = a \cdot b$. ■

Lemma 7.1.4
Let R be a ring. Then $R \neq \{0\}$ if and only if $1 \neq 0$.

Proof
If $1 \neq 0$ then clearly R cannot only have one element. Let R be a ring such that $1 = 0$. Let $a \in R$. Then $a = 1 \cdot a = 0 \cdot a = 0$. ■

As with subgroups and groups, we have subrings of a ring.

Definition 7.1.5 (Subring)
Let R be a ring. Let $S \subseteq R$ be a subset. We say that S is a subring of R if S is a ring in its own right.

Proposition 7.1.6 (Subring Criterion)
Let R be a ring and S a subset of R. Then S is a subring of R if and only if

- $a, b \in S \implies ab \in S$
- $a, b \in S \implies a - b \in S$
- $1 \in S$

Proof
Suppose that R is a ring and S is a subring. Then the subgroup criterion is already satisfied. If $a, b \in S$ then $ab \in S$ since S is a ring. Identity also exists since it is a ring.
Now suppose that S is a subset that satisfies the three criterion. By the subgroup criterion S is an abelian subgroup. Associativity and distributivity is automatically satisfied. The closure property is also satisfied by the first item. The identity is also satisfied by the third item thus we are done. ∎

Lemma 7.1.7
Let R be a ring. Let $A, B \subseteq R$ be subrings. Then $A \cap B$ is a subring of R.

Proof
We show it using the subring criterion. The subgroup criterion is met since A, B are subgroups. If $a, b \in A \cap B$ then $ab \in A$ and $ab \in B$ by closure of rings thus $ab \in A \cap B$. Since A, B are subrings, $1 \in A$ and $1 \in B$ thus we are done. ∎

Similar to the order of a group, we have the characteristic of a ring.

Definition 7.1.8 (Characteristic)
Let R be a ring. If there exists a positive integer n such that $nx = 0$ for $x \in R$, then the smallest of such integer is called the characteristic of R. If there is no such integer then it has characteristic 0.

Definition 7.1.9 (Commutative Rings)
Let R be a ring. We say that R is commutative if $ab = ba$ for all $a, b \in R$.

7.2 The Cancellation Law

Proposition 7.2.1 (Cancellation Law)
Let R be a ring. Then the following are equivalent.

- If $ab = 0$ in R then $a = 0$ or $b = 0$
- If $ab = ac$ and $a \neq 0$ then $b = c$
- If $ba = ca$ and $a \neq 0$ then $b = c$

Proof
$(1) \implies (2)$: We have that $ab = ac$ implies $ab - ac = 0$ by additive inverse and $a(b - c) = 0$ by distributivity and we are done.

$(2) \implies (1)$: Choose $c = 0$.

$(1) \iff (3)$: Similar to the above argument. ∎

Definition 7.2.2 (Domain)
Let $R \neq 0$ be a ring. We say that R is a domain if the above cancellation law holds.

Definition 7.2.3 (Integral Domain)
Let $R \neq 0$ be a ring. We say that R is an integral domain if R is commutative and a domain.

7.3 Division in a Ring

Definition 7.3.1 (Units of a Ring)
Let R be a ring. An element $a \in R$ is a unit in R if there is some $b \in R$ such that $ab = ba = 1$. The set of units in R is denoted $R^\times$.

Example 7.3.2
We have $\mathbb{Z}^\times = \{\pm 1\}$.

Lemma 7.3.3
Let R be a ring. Let $a \in R$ be a unit. Then there exists a unique $b \in R$ such that $ab = ba = 1$.

Proof
Suppose that $b, c \in R$ are two elements satisfying $ab = ba = 1$ and $ac = ca = 1$. Then we have that

$$c1 \cdot c = (ba)c = b(ac) = b \cdot 1 = b$$

so we conclude. ■

Proposition 7.3.4
Let $(R, +, \cdot)$ be a ring. Then $(R^\times, \cdot)$ is a group.

Proof
By the unitality axiom of rings, 1_R is the identity element of the group. By definition of units, every element of $R^\times$ has an inverse. Associativity follows from the associativity axiom of rings.
■

Definition 7.3.5 (Division Rings)
Let $R \neq 0$ be a ring. We say that R is a division ring if $R^\times = R \setminus \{0\}$.

In particular, division ring as the name suggests, allows division in the ring. This is because since every element in R is a unit, for any $a, b \in R$, we have that $a = a(b^{-1}b) = (ab^{-1})(b)$. It looks u GLy but if we let $k = ab^{-1}$, then we are in fact expressing $a = kb$ for any $a, b \in R$. Unfortunately we do not have the notion of size in a general division ring so it is not particularly useful that we can do this.

Proposition 7.3.6
Let R be a ring. If R is a division ring, then R is a domain.

Proof
Let R be a division ring and $a, b \in R$ such that $ab = 0$. Then $a^{-1}ab = a^{-1}0$ thus $b = 0$ and we are done. ■

Definition 7.3.7 (Fields)
Let R be a ring. We say that R is a field if R is commutative and a division ring.

Fields are studied extensively in field theory, where its unique properties as a field is discussed. In groups and rings, we perform treatment on fields by considering its properties as a ring, rather than a field.

Example 7.3.8
The following are true.

- $\mathbb{Z}$ is not a field.

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields.

Proposition 7.3.9
Let $(R, +, \cdot)$ be a ring. Then R is a field if and only if $(R, +)$ and $(R \setminus \{0\}, \cdot)$ are abelian groups and the distributive law $a(b + c) = ab + ac$ holds.

Proof
Let R be a field. Then $(R, +)$ is already an abelian group. Since every field is a division ring, every element except 0 is a unit and thus $(R, \cdot)$ is a group. Since a field is also commutative, $(R, \cdot)$ is abelian. Since every field is a ring, the distributive law already holds and we are done.
Suppose that the three criteria is satisfied. Then R is a division ring since every nonzero element has an inverse in the group $(R, \cdot)$. It is also commutative since $(R, \cdot)$ is abelian thus we are done. ∎

I give the distributive law here although it seems redundant. Sometimes we want to check that a set with two binary operations is a field. In this case, we will need the distributive law to link the two binary operations in to action.

Proposition 7.3.10
Let R be a ring. If R is a field, then R is an integral domain.

Proof
Let R be a field. Since R is a division ring, it is a domain. Since R is also commutative, we thus have that R is an integral domain. ∎

Proposition 7.3.11
Let R be a ring. If R is an integral domain and $|R|$ is finite, then R is a field.

Proof
Let $\{0, a_1, \dots, a_n\}$ be elements of a finite integral domain. We simply show that every element has a mupltiplicative inverse. Fix $1 \leq i \leq n$. Notice that the n products $a_i a_j$ for $1 \leq j \leq n$ are all distinct and nonzero. This means that $a_i a_1, \dots, a_i a_n$ is a permutation of $a_1, \dots, a_n$. Since one of $a_1, \dots, a_n$ is the multiplicative identity, we must have that for some j, $a_i a_j = 1$ thus we are done. ∎

7.4 Products of Rings

Definition 7.4.1 (Direct Product of Rings)
Let $\{R_i \mid i \in I\}$ be a collection of rings. Define the direct product of the collection of rings to be the set
$$\prod_{i \in I} R_i = \{(r_i)_{i \in I} \mid r_i \in R_i\}$$
together with pointwise addition and multiplication.

7.5 Elements of a Ring

Definition 7.5.1 (Zero Divisors)
Let R be a ring. Let $0 \neq a \in R$. We say that a is a zero divisor if there exists $0 \neq b \in R$ such that $ab = 0$.

Lemma 7.5.2
Let R be a ring. Then R is a domain if and only if R has no zero divisors.

Proof
Suppose that R is a domain. Then if $a, b \neq 0$, we must have $ab \neq 0$ by the cancellation law. Hence

R has no zero divisors. Conversely, if R has no zero divisors, then $ab = 0$ and $a \neq 0$ implies that $b = 0$. Hence the cancellation law is satisfied. ■

Definition 7.5.3 (Nilpotents)
Let R be a ring. An element $x \in R$ is said to be nilpotent if $x^n = 0$ for some $n \in \mathbb{N}$.

Lemma 7.5.4
Let R be a ring. Let $x \in R$. If x is nilpotent, then x is a zero divisor.

Proof
Suppose that $n \in \mathbb{N}$ is smallest for which $x^n = 0$. Then $x \cdot x^{n-1} = 0$ but x and x^{n-1} is non-zero. Hence x is a zero divisor. ■

Example 7.5.5
The rings $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ has no nilpotent elements and zero divisors.

Definition 7.5.6 (Idempotents)
Let R be a ring. An element $x \in R$ is said to be idempotent if $x^2 = x$.

8 Ideals and their Properties

8.1 Basic Definition of Ideals

Definition 8.1.1 (Left, Right and Two-Sided Ideals)
Let $(R, +, \cdot)$ be a ring. Let $I \subseteq R$.
- We say that I is a left ideal of R if $(I, +)$ is a subgroup of $(R, +)$ and $r \cdot i \in I$ for all $r \in R$ and $i \in I$.
- We say that I is a right ideal of R if $(I, +)$ is a subgroup of $(R, +)$ and $i \cdot r \in I$ for all $r \in R$ and $i \in I$.
- We say that I is a two-sided ideal of R is I is both a left ideal and a right ideal.

Lemma 8.1.2
Let R be a commutative ring. Let $I \subseteq R$ be a subset. Then I is a left ideal of R if and only if I is a right ideal of R.

Proof
Trivial. ■

Example 8.1.3
Let $I \subseteq \mathbb{Z}$ be a subset. Then I is an ideal of $\mathbb{Z}$ if and only if there exists $n \in \mathbb{N}^+$ such that $I = n\mathbb{Z}$.

Proof
Clearly if $n \in \mathbb{N}^+$, then $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Let $kn \in n\mathbb{Z}$. Since $n \in I$ and $k \in \mathbb{Z}$ and I is an ideal, we know that $kn \in I$. Hence $n\mathbb{Z} \subseteq I$.

Conversely, let $I \subseteq \mathbb{Z}$ be an ideal of $\mathbb{Z}$. Then by the well ordering principle the set of non-negative elements of I has a minimal element, say $n \in \mathbb{N}^+$. Then I claim that $I = n\mathbb{Z}$. Let $x \in I$. Since I is an ideal, we have $x - kn \in I$ for all $k \in \mathbb{Z}$. Then the set of positive elements of the set $\{x - kn \mid k \geq 0\}$ has a minimal element by the well ordering principle, say $x - k_0 n$. If $0 < x - k_0 n < n$, then it contradicts the minimality of n. If $x - k_0 n > n$, then $x - (k_0 + 1)n > 0$, contradicting the minimality of $x - k_0 n$. Hence $x - k_0 n = n$. Then $x - (k_0 + 1)n = 0$ implies that $x = (k_0 + 1)n$. Hence $x \in n\mathbb{Z}$ and $I \subseteq n\mathbb{Z}$. ■

Lemma 8.1.4
Let R be a ring. Let I be an ideal of R. Then $I = R$ if and only if I contains a unit of R.

Proof
Let $I = R$. Then trivially $1 \in I$. Let $x \in R$. Let $u \in I$ be a unit. Now $x = x(u^{-1}u) = (xu^{-1})u$. Since $xu^{-1} \in R$ and $u \in I$, their multiplication will also be in I, thus $x \in I$. ■

This lemma is also very useful if you consider the fact that the multiplicative identity is also a unit.

Definition 8.1.5 (Maximal Ideals)
Let R be a ring. Let I be a left ideal of R. We say that I is a maximal ideal of R if the following are true.
- $I \neq R$.
- No proper ideal of R contains I.

Becareful that maximal ideals are not necessarily unique. A typical example would be the fact that (2) and (3) are principle ideals that are both maximal in $\mathbb{Z}$.

Proposition 8.1.6
Let $R \neq 0$ be a ring. Then R has a maximal left ideal.

Proof

Let $\mathcal{P}$ be the set of all proper left ideals of R ordered by inclusion. Since R is non-zero, the ideal (0) is proper and so belongs to $\mathcal{P}$. Thus $\mathcal{P} \neq \emptyset$. Let $\mathcal{C}$ be a chain in $\mathcal{P}$. Define

$$Z = \bigcup_{X \in \mathcal{C}} X$$

If $\mathcal{C}$ is empty then $Z = \{0\}$. We show that Z is a left ideal. Clearly $0 \in Z$. If $a \in Z$ and $r \in R$, then $a \in X$ for some $X \in \mathcal{C}$ so that $ra \in X \subseteq Z$. Now suppose that $a, b \in Z$. Then $a \in X$ and $b \in Y$ for some $X, Y \in \mathcal{C}$. Since $\mathcal{C}$ is a chain, without loss of generality assume that $X \subseteq Y$. Then $a \in Y$ so that $a + b \in Y \subseteq Z$. Thus Z is a left ideal.

Since all $X \in \mathcal{C}$ are proper ideals with $1 \notin X$, then $1 \notin Z$. Then Z is proper and $Z \in \mathcal{P}$. Z is then an upper bound of $\mathcal{C}$. By Zorn's lemma, $\mathcal{P}$ has a maximal element. Then the maximal element is a maximal left ideal of R. ■

Definition 8.1.7 (Prime Ideals)
Let R be a ring. Let $P \subseteq R$ be a proper ideal of R. We say that P is a prime ideal of R if $ab \in P$ implies that $a \in P$ or $b \in P$.

Example 8.1.8
Let $n \in \mathbb{N}^+$. Then $n\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$ if and only if n is prime.

Proof

Suppose that n is prime. Let $ab \in n\mathbb{Z}$. Then $n \mid ab$. Since n is prime, we must have $n \mid a$ or $n \mid b$. Hence $a \in n\mathbb{Z}$ or $b \in n\mathbb{Z}$. Hence $n\mathbb{Z}$ is prime.

Suppose that $n = xy$ is not prime. Then $xy \in n\mathbb{Z}$, but $x \notin n\mathbb{Z}$ and $y \notin n\mathbb{Z}$. Hence $n\mathbb{Z}$ is not prime. ■

8.2 Ideals and Generating Sets

Definition 8.2.1 (Elements Generating an Ideal)
Let R be a ring. Let I be an indexing set. Let $S = \{a_i \mid i \in I\} \subseteq R$ be a set of elements of R. Define

$$(S) = (a_i \mid i \in I) = \left\{ \sum_{k=1}^{n} r_k a_k \mid r_k \in R \right\}$$

Lemma 8.2.2
Let R be a ring. Let I be an indexing set. Let $\{a_i \mid i \in I\} \subseteq R$ be a set of elements of R. Then $(a_i \mid i \in I)$ is a left ideal of R.

Proof

Let $s_1, s_2 \in (r_i \mid i \in I)$. We can write $s_1 = \sum_{k=1}^{n_1} r_k a_k$ and $s_2 = \sum_{j=1}^{n_2} r_j a_j$. Then every term in $s_1 + s_2$ is the product of $r \in R$ and $a \in \{a_i \mid i \in I\}$. Hence the additivity rule is satisfied. For $a = \sum_{k=1}^{n} r_k a_k \in (a_i \mid i \in I)$, we can use distributivity to see that

$$r \cdot a = r \cdot \sum_{k=1}^{n} r_k a_k \in (a_i \mid i \in I) = \sum_{k=1}^{n} (r \cdot r_k) \cdot a_k \in (a_i \mid i \in I)$$

for any $r \in R$. Hence we conclude. ■

Definition 8.2.3 (Generators of an Ideal)
Let R be a ring. Let I be a left ideal of R. Let $S \subseteq R$ be a subset. We say that S are generators of I if
$$I = (S)$$

Definition 8.2.4 (Principal Ideals)
Let R be a ring. Let I be a left ideal of R. We say that I is a principal ideal if
$$I = (a)$$
is generated by a single element $a \in R$.

8.3 Operations on Ideals

Definition 8.3.1 (Sums and Intersections of Ideals)
Let R be a ring. Let I, J be left ideals of R.

- Define the sum of I and J by
$$I + J = \{i + j \mid i \in I, j \in J\}$$
- Define the intersection of I and J by
$$I \cap J = \{r \in R \mid r \in I, r \in J\}$$

Proposition 8.3.2
Let R be a ring. Let I, J be left ideals of R. Then $I + J$ and $I \cap J$ are both left ideals of R.

Proof
Let R be a ring and I, J its ideals.

- We first show that $I + J$ is an ideal. Let $i_1 + j_1, i_2 + j_2 \in I + J$. Then since R is an abelian group an I, J are subgroups of R,
$$i_1 + j_1 + i_2 + j_2 = (i_1 + i_2) + (j_1 + j_2) \in I + J$$
thus closure is satisfied. Associativity is satisfied since it inherits from R. We also have $0 \in I, J$ thus $0 \in I + J$. The inverse of $i + j \in I + J$ is $-i - j \in I + J$ since $-i \in I$ and $-j \in J$. Thus $I + J$ is a subgroup of R.
Now let $r \in R$ and $i + j \in I + J$. Then $r(i + j) = ri + rj$ by distributivity. Since I, J are ideals, there exists $s \in I$ such that $s = ri$ and $t \in J$ such that $t = rj$. Then $r(i+j) = ri+rj = s + t \in I + J$ thus we are done.
- $I \cap J$ are already subgroups of R. Let $k \in I \cap J$. Let $r \in R$. Then $rk \in r(I \cap J)$. Then there exists $i \in I$ and $j \in J$ such that $i = rk = j$. Then $i = j \in I \cap J$ thus we are done.

■

The following definition is only useful when the ring is commutative.

Example 8.3.3
Let $n, m \in \mathbb{N}^+$. Then the following are true.

- $n\mathbb{Z} + m\mathbb{Z} = \gcd(n, m)\mathbb{Z}$.
- $n\mathbb{Z} \cap m\mathbb{Z} = \text{lcm}(n, m)\mathbb{Z}$.

Definition 8.3.4 (Product of Ideals)

Let R be a ring. Let I, J be two-sided ideals of R. Define the product of I and J to be

$$IJ = (ij \mid i \in I, j \in J) = \left\{ \sum_{k=1}^{n} a_k b_k \mid a_k \in I, b_k \in J \right\}$$

Lemma 8.3.5
Let R be a ring. Let I_1, I_2 be two-sided ideals of R. Then the following are true.
- $I_1 I_2$ is a two-sided ideal of R.
- $I_1 I_2 \subseteq I_1, I_2$.

Proof
Clearly $I_1 I_2$ is a subgroup of R. Let $\sum_{k=1}^{n} a_k b_k \in I_1 I_2$. Let $r \in R$. Then $ra_i \in I_1$ for all i, hence $r \sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n} (ra_k) b_k \in I_1 I_2$ and $I_1 I_2$ is a left ideal. Similarly, because I_2 is a right ideal, we have $I_1 I_2$ is a right ideal.

Let $\sum_{k=1}^{n} a_k b_k \in I_1 I_2$. Since I_1 is a right ideal, we have $a_k b_k \in I_1$. Hence the sum also lies in I_1. Similarly, because I_2 is a left ideal, we have $a_k b_k \in I_2$ and the sum lies in I_2. ■

Example 8.3.6
Let $n, m \in \mathbb{N}^+$. Then $(n\mathbb{Z})(m\mathbb{Z}) = nm\mathbb{Z}$.

Lemma 8.3.7
Let R be a commutative ring. Let I_1, I_2, I_3 be ideals of R. Then the following are true.
- Product is Commutative: $I_1 I_2 = I_2 I_1$.
- Product is Associative: $(I_1 I_2) I_3 = I_1 (I_2 I_3)$.
- Distributivity: $I_1 (I_2 + I_3) = I_1 I_2 + I_1 I_3$.

9 The Isomorphism Theorem for Rings

9.1 Ring Homomorphisms

Definition 9.1.1 (Ring Homomorphism)
Let R, S be rings. A ring homomorphism is a map $\phi : R \to S$ such that the following are true:
- $\phi(a+b) = \phi(a) + \phi(b)$ for all $a, b \in R$
- $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in R$
- $\phi(1_R) = 1_S$

A bijective ring homomorphism is called a ring isomomorphism.

Example 9.1.2
Let R be a ring. Then the function $\phi : \mathbb{Z} \to R$ defined by $\phi(n) = n \cdot 1_R$ is a ring homomorphism.

Lemma 9.1.3
Let R, S be rings. Let $\phi : R \to S$ be a ring homomorphism. Then the following are true.
- $\phi(0_R) = 0_S$.
- $\phi(-a) = -\phi(a)$.
- $\phi|_{R^\times} : R^\times \to S^\times$ is a group homomorphism. Moreover, $\phi(a^{-1}) = \phi(a)^{-1}$.

Proof
We have that
$$\phi(0_R) = \phi(0_R + 0_R) = \phi(0_R) + \phi(0_R)$$
Since all elements of S has an additive inverse, we conclude that $\phi(0_R) = 0_S$ by applying the inverse on both sides.

We have that
$$0_S = \phi(0_R) = \phi(a + (-a)) = \phi(a) + \phi(-a)$$
Hence $\phi(-a) = -\phi(a)$.

We first show ϕ sends units to units. Let $r \in R^\times$. Then we have
$$1_S = \phi(1_R) = \phi(r \cdot r^{-1}) = \phi(r)\phi(r^{-1})$$
Since multiplicative inverses are unique, we conclude that $\phi(r^{-1}) = \phi(r)^{-1}$. It follows that $\phi|_{R^\times}$ is a group homomorphism by the fact that $\phi(ab) = \phi(a)\phi(b)$. ∎

Lemma 9.1.4
Let R, S be rings. Let $\varphi : R \to S$ be a ring homomorphism. Then the following are true.
- If I is an ideal of R, then $\varphi(I)$ is an ideal of S if and only if φ is surjective.
- If J is an ideal of S, then $\varphi^{-1}(J)$ is an ideal of R.

Proof
Suppose that I is an ideal of R. Let $a, b \in \varphi(I)$. Then there exists $x, y \in I$ such that $\varphi(x) = a$ and $\varphi(y) = b$. Since I is an ideal, we have $x + y \in I$ and so $a + b = \varphi(x) + \varphi(y) = \varphi(x + y) \in \varphi(I)$. Now suppose that $s \in S$. Since φ is surjective, there exists $r \in R$ such that $\varphi(r) = s$. Then we have $sa = \varphi(r)\varphi(x) = \varphi(rx) \in \varphi(I)$ since I is an ideal. Hence we are done.

Conversely, if φ sends ideals to ideals, then $\varphi(R)$ is an ideal containing 1_S. Hence $\varphi(R) = S$ and so φ is surjective.

Now suppose that J is an ideal of S. Let $x, y \in \varphi^{-1}(J)$. Since J is an ideal of S, we have $\varphi(x) + \varphi(y) = \varphi(x + y) \in J$. Hence $x + y \in \varphi^{-1}(J)$. Now let $r \in R$. Then we have $\varphi(r)\varphi(x) = \varphi(rx) \in J$ since J is an ideal. Hence $rx \in \varphi^{-1}(J)$ and so φ^J is an ideal. ∎

Lemma 9.1.5
Let R, S be rings. Let $\varphi : R \to S$ be a ring homomorphism. Then the following are true.
- Let m be a maximal ideal of S. If φ is surjective, then $\varphi^{-1}(m)$ is a maximal ideal of R.
- Let P be a prime ideal of S. Then $\varphi^{-1}(P)$ is a prime ideal of R.

Proof
It is clear that $\varphi^{-1}(m)$ is an ideal. If it is not maximal, then there exists a proper ideal I such that $\varphi^{-1}(m) \subset I \subset R$. Then we have $m \subseteq \varphi(I) \subseteq \varphi(R) = S$. Since m is maximal, either $\varphi(I) = \varphi(R)$ or $\varphi(I) = m$. If it is the former, then

We already know that $\varphi^{-1}(P)$ is an ideal. Let $ab \in \varphi^{-1}(P)$. Then $\varphi(ab) = \varphi(a)\varphi(b) \in P$. Since P is a prime ideal, we have either $\varphi(a) \in P$ or $\varphi(b) \in P$. In both cases, this shows that either $a \in \varphi^{-1}(P)$ or $b \in \varphi^{-1}(P)$. Hence P is a prime ideal. ■

Proposition 9.1.6
Let R, S be a ring. Let $\phi : R \to S$ be a ring homomorphism. Then $\ker(\phi)$ is an ideal of R.

Proof
We already know that $\ker(\phi)$ is an abelian group. Let $r \in R$. Let $x \in \ker(\phi)$. Then $\phi(rx) = \phi(r)\phi(x) = 0$ and so $rx \in \ker(\phi)$. Hence $\ker(\phi)$ is an ideal. ■

9.2 The Quotient Ring

Definition 9.2.1 (Quotient Rings)
Let R be a ring. Let I be a two-sided ideal of R. Define the quotient ring of R by I to be the abelian group

$$\frac{R}{I} = \{r + I \mid r \in R\}$$

together with multiplication defined by

$$(r + I) \cdot (s + I) = rs + I$$

Proposition 9.2.2
Let R be a ring. Let I be a two-sided ideal of a ring R. Then R/I is a ring.

Proof
We already know that R/I is an abelian group. We show that multplication is well defined. Suppose that $x_1 + I = x_2 + I$ and $y_1 + I = y_2 + I$. Then

$$\begin{aligned}(x_1 + I)(y_1 + I) &= x_1y_1 + I \\ &= (x_1y_1 - x_1y_2 + x_1y_2 - x_2y_2 + x_2y_2) + I \\ &= (x_1(y_1 - y_2) + (x_1 - x_2)y_2 + x_2y_2) + I \\ &= x_2y_2 + I\end{aligned}$$

since $y_1 - y_2 \in I$ and $x_1 - x_2 \in I$. This shows that taking different represetatives of the cosets of an ideal does not matter for multiplication.
Now we show that $1 + I$ is the muplicative identity of R/I. Let $x + I \in R/I$. We have that $(1+I)(x+I) = x+I$ thus we are done. Associativity and distributivity follows from the fact that R has these properties. ■

Example 9.2.3
Let $n \in \mathbb{N}^+$. Then the following are true.
- $\mathbb{Z}/n\mathbb{Z}$ is a quotient ring.
- $m + n\mathbb{Z}$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$ if and only if m is prime.

- $\mathbb{Z}/n\mathbb{Z}$ is an field if and only if n is prime.

Proposition 9.2.4
Let R be a ring. Let I be a two-sided ideal of R. Then I is a maximal left ideal if and only if R/I is a division ring.

Proof
Suppose that I is a maximal ideal. Let $x \notin R$. We show that $x + I$ has an inverse. We know that $I + (x)$ is an ideal containing I. Since I is maximal, we must have $I + (x) = R$. This means that $1 \in I + (x)$ which means there exists $m \in I$ and $r \in R$ such that $1 = m + rx$. Now consider $r + M$. We have that

$$\begin{aligned}(x + I)(r + I) &= xr + I \\ &= (1 - m) + I \\ &= 1 + I\end{aligned}$$

Now suppose that R/I is a field. Let J be an ideal such that $I \subseteq J \subseteq R$ and $I \neq J$. Now let $x \in J \setminus I$. Then $I + x$ has a multiplicative inverse in R/I since $I + x \neq I$, the additive identity. Let the inverse be $I + y$. Then

$$I + 1 = (I + x)(I + y) = I + xy$$

Trivially $xy \in (x)$, thus $1 \in I + (x) \subseteq J$. But if the identity is in the ideal, the ideal is equal to the ring thus we are done. ∎

Corollary 9.2.5
Let R be a commutative ring with identity and M an ideal of R. Then M is maximal if and only if R/M is a field.

Proof
Immediate from the above. ∎

Proposition 9.2.6
Let R be a commutative ring with identity not equal to 0. Then P is a prime ideal in R if and only if R/P is an integral domain.

Proof
Suppose that P is a prime ideal. Then let $(a + P)(b + P) = P$. Since we also have that $(a + P)(b + P) = ab + P$. This means that $ab \in P$ thus either $a \in P$ or $b \in P$ which in turns leads to either $a + P = P$ or $b + P = P$ and thus the cancellation law applies.
Now suppose that R/P is an integral domain. Let $ab \in P$. Since R/P is an integral domain, we have that

$$(a + P)(b + P) = ab + P = P$$

means that either $a + P = P$ or $b + P = P$ which means that either $a \in P$ or $b \in P$ and we are done. ∎

Corollary 9.2.7
Let R be a commutative ring. Let I be an ideal of R. If I is a maximal ideal, then I is a prime ideal.

Proof
If M is maximal then R/I being a field implies that R/I is an integral domain thus I is prime. ∎

9.3 Isomophism Theorem for Rings

The isomorphism theorem for rings is a direct result that extends the group isomorphism theorems. Their proofs are mostly the same except that we also have to check that multiplication is preserved so that the isomorphisms inherited from groups is indeed a ring isomorphism.

Theorem 9.3.1 (The First Isomorphism Theorem for Rings)
Let R, S be rings. Let $\phi : R \to S$ be a ring homomorphism. Then the following are true.

- $\ker(\phi)$ is an ideal of R.
- $\mathrm{im}(\phi) \leq S$ is a subring of S.
- There is an isomorphism of rings
$$R/\ker(\phi) \cong \mathrm{im}(\phi)$$
given by the function $r + \ker(\phi) \mapsto \phi(r)$.

Proof
Since ϕ is a homomorphism of the underlying groups, the first isomorphism theorem for groups gives a group isomorphism
$$\overline{\phi} : \frac{R}{\ker(\phi)} \xrightarrow{\cong} \phi(R)$$
defined by $\overline{\phi}([r]) = \phi(r)$ for $r \in R$. It remains to show that $\overline{\phi}$ is a ring isomorphism. Let $[r], [s] \in R/\ker(\phi)$. Then we have
$$\overline{\phi}([r] \cdot [s]) = \overline{\phi}([rs]) = \phi(rs) = \phi(r)\phi(s) = \overline{\phi}([r])\overline{\phi}([s])$$
so we are done. ■

Theorem 9.3.2 (The Second Isomorphism Theorem for Rings)
Let R be a ring. Let $S \leq R$ be a subring. Let $I \subseteq R$ be an ideal of R. Then the following are true.

- $S + I = \{s + i \mid s \in S, i \in I\}$ is a subring of R.
- $S \cap I$ is an ideal of S.
- There is an isomorphism of rings
$$\frac{S + I}{I} \cong \frac{S}{S \cap I}$$
given by the function $s + S \cap I \mapsto s + I$.

Proof
From the second isomorphism theorem for groups, we know that $S + I$ is an abelian group. Also, we have $(s_1 + i_1)(s_2 + i_2) = s_1s_2 + i_1s_2 + i_2s_1 + i_1i_2 \in S + I$ since I is an ideal. Finally, $1 \in S$ and $0 \in I$ implies that $1 \in S + I$. Hence $S + I$ is a subring by the subring criterion.

From the second isomorphism theorem for groups, we know that $S \cap I$ is a subgroup. Let $s \in S$ and $i \in S \cap I$. Then $si \in S$ since S is a subring. Also $si \in I$ since I is an ideal. Hence $si \in S \cap I$ so that $S \cap I$ is an ideal.

From the second isomorphism theorem for groups, we know that the map $\phi : \frac{S}{S \cap I} \to \frac{S+I}{I}$ is a group isomorphism. Therefore it suffices to show that ϕ is a ring homomorphism. Let $[s_1], [s_2] \in \frac{S}{S \cap I}$. Then we have
$$\phi((s_1 + S \cap I)(s_2 + S \cap I)) = \phi(s_1s_2 + S \cap I) = s_1s_2 + I = (s_1 + I)(s_2 + I) = \phi(s_1 + S \cap I)\phi(s_2 + S \cap I)$$
so we are done. ■

Theorem 9.3.3 (The Third Isomorphism Theorem for Rings)
Let R be a ring. Let $I, J \subseteq R$ be two-sided ideals of R such that $I \subseteq J$. Then the following are true.

- J/I is an ideal of R/I.

- There is an isomorphism of rings
$$\frac{R/I}{J/I} \cong \frac{R}{J}$$
given by the function $(r+I)+J/I \mapsto r+J$.

Proof
From the third isomorphism theorem for groups, we know that J/I is a subgroup of R/I. Let $r \in R$. Let $j+I \in J/I$. Then we have $r(j+I) = rj+I \in J/I$ since J is an ideal. Hence J/I is an ideal.

From the third isomorphism theorem for groups, we know that the function $f : \frac{R/I}{J/I} \to \frac{R}{J}$ is an isomorphism of groups. It remains to show that the function$(r+I)+J/I \mapsto r+J$ is a ring homomorphism. Let $r_1, r_2 \in R$. Then we have

$$f((r_1+I+J/I)(r_2+I+J/I)) = f(r_1r_2+I+J/I) = r_1r_2+J = (r_1+J)(r_2+J) = f(r_1+I+J/I)f(r_2+I+J/I)$$

so we are done. ∎

Theorem 9.3.4 (The Correspondence Theorem)
Let R be a ring. Let I be a two-sided ideal of R. Then the following are true.
- There is an inclusion preserving bijection
$$\left\{ J \subseteq R \;\middle|\; \begin{array}{c} J \text{ is a left ideal of } R \\ \text{containing } J \end{array} \right\} \overset{1:1}{\longleftrightarrow} \left\{ J \subseteq R/I \;\middle|\; \begin{array}{c} J \text{ is a left ideal} \\ \text{of } R/I \end{array} \right\}$$
given by $J \mapsto J/I$.
- The above bijection restricts to a bijection
$$\left\{ S \subseteq R \;\middle|\; \begin{array}{c} S \text{ is a subring} \\ \text{of } R \text{ containing } I \end{array} \right\} \overset{1:1}{\longleftrightarrow} \left\{ S \subseteq R/I \;\middle|\; \begin{array}{c} S \text{ is a subring} \\ \text{of } R/I \end{array} \right\}$$

Example 9.3.5
Let $n \in \mathbb{N}^+$. Then the ideals of $\mathbb{Z}/n\mathbb{Z}$ are given by $m\mathbb{Z}/n\mathbb{Z}$ for $m \mid n$ together with the zero ideal.

9.4 Chinese Remainder Theorem

In this section we develop the necessary notions in order to illustrate the Chinese Remainder Theorem.

Definition 9.4.1 (Coprime Ideals)
Let R be a ring. Let I, J be left ideals of R. We say that I and J are coprime if
$$I + J = R$$

Notice that in $\mathbb{Z}$, the prime ideals are exactly the ideals (p) where p is a prime. We also have a nice inclusion of ideals whenever $p \mid a$ which is $(a) \subseteq (p)$, which we will prove later. This means that in general, the smaller the number a is, the larger the ideal (a) is and indeed, the smaller the number is in $\mathbb{Z}$, the more numbers it can possibly divide. Now recall that if a and b are coprime in $\mathbb{Z}$, then their gcd will be 1. Indeed we will develop the notion of gcd for ideals as well, which is to say that if $d = \gcd(a,b)$ in the usual sense, then $(a) \subseteq (d)$ and $(b) \subseteq (d)$. Then if a and b are coprime, their ideals are both subsets of (1), which is exactly R. This leads to why we say that two ideals are coprime.

Proposition 9.4.2 Let R be a commutative ring. Let $I_1, \dots, I_k$ be pairwise coprime ideals of R. Then we have
$$\prod_{i=1}^k I_i = \bigcap_{i=1}^k I_i$$

Proof
We first consider the case $k = 2$. Suppose that I and J are coprime ideals. Notice that we have $IJ \subseteq I, J$ hence $IJ \subseteq I \cap J$. On the other hand, we have

$$\begin{aligned} I \cap J &= (I + J)(I \cap J) && (I, J \text{ are coprime}) \\ &= I(I \cap J) + J(I \cap J) \\ &\subseteq IJ + JI \\ &= IJ \end{aligned}$$

Hence we are done.

We then induct on $k \geq 3$. For the base case, suppose that I_1, I_2, I_3 are pairwise coprime. I claim that $I_1 \cap I_2$ and I_3 are coprime. Since I and K are coprime, we can write $1 = i + k_1$ for $i \in I$ and $k_1 \in K$. Similarly, we can write $1 = j + k_2$ for $j \in J$ and $k_2 \in K$. Then $1 = (i + k_1)(j + k_2) = ij + ik_1 + jk_2 + k_1k_2$. The last three elements lie in K and the first element lie in $I \cap J$. Hence $I \cap J$ and K are coprime. Now assuming that the result is true for some $n \in \mathbb{N}$. Then we have

$$\bigcap_{i=1}^{n+1} I_i = I_1 \cap \bigcap_{i=2}^{n+1} I_i = I_1 \cap \prod_{i=2}^{n+1} = I_1 \cdot \prod_{i=2}^{n+1} I_i$$

since I_1 and $\prod_{i=2}^{n+1}$ are coprime by another set of induction. ■

Theorem 9.4.3 (Chinese Remainder Theorem)
Let R be a ring. Let $I_1, \dots, I_n$ be two sided ideals of R. Let $\phi : R \to \prod_{k=1}^n \frac{R}{I_k}$ be the ring homomorphism defined by $\phi(r) = (r + I_1, \dots, r + I_n)$. Then the following are true.

- We have $\ker(\phi) = \bigcap_{k=1}^n I_k$.
- If $I_1, \dots, I_n$ are pairwise coprime, then ϕ induces an isomorphism

$$\frac{R}{\bigcap_{k=1}^n I_k} \cong \prod_{k=1}^n \frac{R}{I_k}$$

Proof
Since ϕ is a collection of projections on to each factor, ϕ is a ring homomorphism. It is clear that I is the kernel of the homomorphism. Indeed given $x \in I$, then x lies in each and every I_k for $1 \leq k \leq n$. Thus $\phi(x) = (x + I_1, \dots, x + I_n) = 0$. Conversely, if $(x + I_1, \dots, x + I_n)$ is such that applying ϕ gives 0, then $x \in I_1, \dots, I_n$ so that $x \in I$.

When the ideals are pairwise corpime, consider the elements $e_k = (0, \dots, 0, x + I_k, 0, \dots, 0)$ for $1 \leq k \leq n$ (these are called full system of orthogonal central idempotents). Notice that they satisfy the equation $1 = e_1 + \cdots + e_n$ and $e_k^2 = e_k$ for $1 \leq k \leq n$ for $i \neq j$, we have $e_ie_j = 0$. Now since I_i and I_j are coprime, we have that $R = I_i + I_j$ so that $1 = a_j + z_j$ for $a_j \in I_i$ (note the subscript) and $z_j \in I_j$. Define $x_i = z_1 \cdots z_{i-1} \cdot z_{i+1} \cdots z_n$ for each $1 \leq i \leq n$. Write the projection homomorphism as $\psi : R \to R/I_j$. If $i \neq j$, then $\phi_j(x_i) = 0$ because x_i contains the element z_j that lies in I_j. Also, we have that

$$\psi_i(z_j) = \psi_i(1 - a_j) = \psi_i(1) - \psi_i(a_j) = \psi_i(1) = 1$$

in R/I_i since $a_j \in I_i$. Thus we have that

$$\psi_i(x_i) = \psi_i(z_1) \cdots \psi_i(z_n) = 1$$

All this means that $x_i \in R$ is such that $\psi(x_i) = (0, \dots, 0, 1 + I_i, 0, \dots, 0)$ for $1 \leq i \leq n$. Now to prove surjectivity, let $(r_1, \dots, r_n)$ be an element in the product. Then we have that

$$(r_1, \dots, r_n) = \sum_{j=1}^n \psi(r_i)e_i = \sum_{j=1}^n \psi(r_ix_i)$$

and so $r_1x_1 + \cdots + r_nx_n \in R$ is our desired element. And so we are done. ■

This is a more generalized version of the Chinese Remainder Theorem in number theory. Indeed, to recover the one in number theory, take $R = \mathbb{Z}$ and $I_k = (n_k)$ for some $n_k \in \mathbb{Z}$ so that I_k is the principal ideal on n_k. Moreover, by choosing each n_k to be pairwise coprime, we obtain the isomorphism which proves that the congruence relation can be solved.

Notice that the proof is instructive. Indeed we can simply follow the steps of the proof to find a solution. Namely, given $(r_1, \dots, r_n)$ in the product, we can find $r \in R$ such that $\phi(r) = (r_1, \dots, r_n)$. The steps are as follows.

1. Fix $1 \leq i \leq n$. Using $R = I_i + I_j$, find $a_j \in I_i$ and $z_j \in I_j$ such that $1 = a_j + z_j$ for each j. This can be done for example by Bezout's lemma in $R = \mathbb{Z}$.
2. Define $x_i = z_1 \cdots z_{i-1} \cdot z_{i+1} \cdots z_n$ for $1 \leq k \leq n$.
3. Do the same for each $1 \leq i \leq n$. The solution is then $r = r_1x_1 + \cdots + r_nx_n$.

Corollary 9.4.4 Let R be a commutative ring. Let $I_1, \dots, I_k$ be pairwise coprime ideals of R. Then we have a ring isomorphism

$$\frac{R}{\prod_{i=1}^k I_i} = \frac{R}{\bigcap_{i=1}^k I_i} \cong \prod_{i=1}^k \frac{R}{I_i}$$

Proof
Follows from the Chinese Remainder Theorem and the fact that $\prod_{i=1}^k I_i = \bigcap_{i=1}^k I_i$. ■

Example 9.4.5
Suppose that $x \in \mathbb{Z}$ satisfies $x \equiv 0 \pmod 3$, $x \equiv 3 \pmod 4$, $x \equiv 4 \pmod 5$. Then $x \equiv 39 \pmod{60}$.

Proof
In terms of ring theory, the problem can be translated to saying that $[x] = [0] \in \mathbb{Z}/3\mathbb{Z}$, $[x] = [3] \in \mathbb{Z}/4\mathbb{Z}$ and $[x] = [4] \in \mathbb{Z}/5\mathbb{Z}$. Consider the ideals $3\mathbb{Z}$, $4\mathbb{Z}$, $5\mathbb{Z}$ in $\mathbb{Z}$. They are pairwise coprime since their pairwise gcd is 1. By the Chinese remainder theorem, there is an isomorphism

$$\mathbb{Z}/60\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$$

given by $1 \mapsto (1, 1, 1)$. The goal is to find $x \in \mathbb{Z}/60\mathbb{Z}$ so that under the above isomorphism, $[x]$ corresponds to the element $(0, 3, 4)$. It suffices to solve for $(0, 1, 0)$ and $(0, 0, 1)$ since ring homomorphism preserves addition and multiplication. By inspection, we have that $(0, 1, 0)$ corresponds to 45 and $(0, 0, 1)$ corresponds to 36. Hence $[x] = [3 \cdot 45 + 4 \cdot 36] = [279] = [39] \in \mathbb{Z}/60\mathbb{Z}$. ■

Example 9.4.6
The idempotents of $\mathbb{Z}/60\mathbb{Z}$ are precisely $\{[0], [1], [16], [21], [25], [36], [40], [45]\}$.

Proof
Consider the ideals $3\mathbb{Z}$, $4\mathbb{Z}$, $5\mathbb{Z}$ in $\mathbb{Z}$. They are pairwise coprime since their pairwise gcd is 1. By the Chinese remainder theorem, there is an isomorphism

$$\mathbb{Z}/60\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$$

given by $1 \mapsto (1, 1, 1)$. Let $(a, b, c) \in \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ be an element. Then (a, b, c) is an idempotent if and only if a, b, c are each idempotents in the corresponding component in the product ring. The only idempotents in $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$ are 0 and 1 respectively. This gives eight idempotents in the product by considering all possible combinations of 0 and 1. Through the ring isomorphism, we conclude that $(1, 0, 0)$ corresponds to 40, $(0, 1, 0)$ corresponds to 45 and $(0, 0, 1)$ corresponds to 36. Hence we obtain the list of idempotents described above. ■

10 Integral Domains

10.1 Field of Fractions

Definition 10.1.1 (The Field of Fractions)
Let R be an integral domain. Define the set of fractions of R to be

$$\text{Frac}(R) = \{(a, b) \mid a, b \in R \text{ and } b \neq 0\}$$

Define a relation on Frac(R) by $(a, b) \sim (c, d)$ if $ad = bc$.

Lemma 10.1.2
Let R be an integral domain. The above relation is an equivalence relation.

Proof Since R is an integral domain, symmetry is satisfied. Clearly it is reflexive since $ab = ba$. For transitivity, suppose that $ad = bc$ and $cf = de$. Then $adcf = bcde$ and by cancellation law, $af = be$. ■

Proposition 10.1.3
Let R be an integral domain. The set of equivalence classes of Frac(R), under the equivalence relation $\sim$, together with the operations of addition and multiplication defined by

$$(a, b) + (c, d) = (ad + bc, bd)$$

and

$$(a, b) \cdot (c, d) = (ac, bd)$$

is a field.

Proof
Note that R is a field if and only if $(R, +)$ and $(R \setminus \{0\}, \cdot)$ are commutative groups and the distributive law holds.

- Let $(a, b), (c, d) \in S$. Then $ad + bc, bd \in R$ since R is closed thus $(ad + bc, bd) \in S$. Let (e, f) also be in S. Then

  $$\begin{aligned} ((a, b) + (c, d)) + (e, f) &= (ad + bc, bd) + (e, f) \\ &= ((ad + bc)f + (bd)e, bdf) \\ &= (adf + bcf + bde, bdf) \end{aligned}$$

  and

  $$\begin{aligned} (a, b) + ((c, d) + (e, f)) &= (a, b) + (cf + de, df) \\ &= (a(df) + b(cf + de), b(df)) \\ &= (adf + bcf + bde, bdf) \end{aligned}$$

  Thus associativity is satisfied. I claim that $(0, 1)$ is an identity. We have $(a, b) + (0, 1)(a \cdot 1 + b \cdot 0, b \cdot 1) = (a, b)$. If $(a, b) \in S$ then $(-a, b) \sim (a, -b)$ is an inverse. We have

  $$(a, b) + (-a, b) = (ab - ab, b^2) = (0, b^2) \sim (0, 1)$$

  Finally we have

  $$\begin{aligned} (a, b) + (c, d) &= (ad + bc, bd) && \\ &= (da + cb, db) && (R \text{ is an Integral Domain}) \\ &= (cb + da, db) \qquad = (c, d) + (a, b) && (R \text{ is an abelian group}) \end{aligned}$$

  Thus we have shown that $(S, +)$ is an abelian group.

- We now show that $(S, \cdot)$ is an abelian group. Let $(a, b), (c, d), (e, f) \in S$. Then $(a, b) \cdot (c, d) = (ac, bd) \in S$ sincve $ac, bd \in R$ by closure of rings. Thus the closure property is satisfied.

Associativity is inherited from R since elements in S are pairs of R. I claim that the identity is $(1,1) \sim (k,k)$ for any $k \in R$. We have

$$(a,b) \cdot (1,1) = (a \cdot 1, b \cdot 1) = (a,b)$$

If $(a,b) \in S$ then its inverse is (b,a). We have

$$(a,b) \cdot (b,a) = (ab, ba) = (ab, ab) = (1,1)$$

thus $(S, \cdot)$ is a group. Now to show abelian, we have

$$(a,b) \cdot (c,d) = (ac, bd) = (ca, db) = (c,d) \cdot (a,b)$$

since R is an integral domain. Thus we have shown that $(S, \cdot)$ is an abelian group.

- Finally we show distributivity. Let $(a,b), (c,d), (e,f) \in S$. Then

$$\begin{aligned}(a,b) \cdot ((c,d) + (e,f)) &= (a,b) \cdot (cf + de, df) \\ &= (acf + ade, bdf)\end{aligned}$$

and

$$\begin{aligned}(a,b) \cdot (c,d) + (a,b) \cdot (e,f) &= (ac, bd) + (ae, bf) \\ &= (acbf + bdae, bdbf) \\ &= (acf + ade, bdf) && \text{(equivalence relation)}\end{aligned}$$

Thus we are done.

■

Let R be an integral domain. An equivalence class $[(a,b)]$ in Frac(R) is commonly denoted as $\frac{a}{b}$.

Example 10.1.4
We have $\mathbb{Q} = \text{Frac}(\mathbb{Z})$.

Lemma 10.1.5
Let R be an integral domain. Then R injects as a subring of Frac(R) via the ring homomorphism $r \mapsto \frac{r}{1}$.

Proof
It suffices to show that the ring homoomorphism is injective. Suppose that $\frac{r}{1} = \frac{s}{1}$. This means that they lie in the same equivalence class in Frac(R). By definition of the equivalence relation we have $r = s$. Hence the homomorphism is injective. ■

10.2 Divisibility

Division is a property that only commutative rings enjoy.

Definition 10.2.1 (Division)
Let R be a commutative ring. Let $a, b \in R$ such that $b \neq 0$. We say that a is a multiple of b if there exists $x \in R$ such that $a = bx$. In this case, we say that b divides a and we write $b \mid a$.

Proposition 10.2.2
Let R be a commutative ring. Let $x, y \in R$ and $y \neq 0$. Then the following are equivalent.

- $x \mid y$
- $y \in (x)$
- $(y) \subseteq (x)$

Proof
(1) $\implies$ (2): Suppose that $y = kx = xk$ for some $k \in R$. Then $y \in (x) = \{ax \mid a \in R\}$ by definition.

(2) $\implies$ (3): Suppose that $y \in (x)$. Then there exists $k \in R$ such that $y = kx = xk$. To show that $(y) \subseteq (x)$, let $ry \in (y)$. Then $ry = rkx$ and $rk \in R$ thus $ry \in (x)$ thus we are done.

(3) $\implies$ (1): Suppose that $(y) \subseteq (x)$. Then there exists $k \in R$ such that $y = kx$. Then we are done. ■

Definition 10.2.3 (Greatest Common Divisor)
Let R be a commutative ring and let $a, b \in R$. A greatest common divisor of a and b is a non-zero element $d \in R$ such that
- $d \mid a$ and $d \mid b$
- If $c \mid a$ and $c \mid b$ then $c \mid d$

It is denoted $\gcd(a, b)$.

Definition 10.2.4 (Least Common Multiple)
Let R be a commutative ring and let $a, b \in R$. A least common multiple of a and b is a non-zero element $l \in R$ such that
- $a \mid l$ and $b \mid l$
- If $a \mid m$ and $b \mid m$ then $l \mid m$

It is denoted $\operatorname{lcm}(a, b)$.

Unfortunately, these numbers do not always exists for any a, b in a general ring. The existence of such a number depends on the ideal generated by the two given elements.

Proposition 10.2.5
Let R be a commutative ring. Let $x, y \in R$ such that they are nonzero. If the ideal generated by a and b, namely (a, b) is a principal ideal (d), then d is the gcd of a and b.

Proof
Suppose that $(a, b) = (d)$ for some $d \in R$. Then $a \in (d)$ and $b \in (d)$ already implies that $d \mid a$ and $d \mid b$. Suppose that $c \mid a$ and $c \mid b$. This means that $a \in (c)$ and $b \in (c)$. Since $d \in (a, b)$ there exists $r, s \in R$ such that $ra + sb = d$. This means that $d \in (c)$ thus $c \mid d$. ■

Definition 10.2.6 (Associates)
Let R be a commutative ring. Let $x, y \in R$. We say that x and y are associates if $x \mid y$ and $y \mid x$. We denote it as $x \sim y$.

Proposition 10.2.7
Let R be a integral domain. Let $x, y \in R$. Then the following are equivalent.
- $x \sim y$
- $(x) = (y)$
- There exists a unit $u \in R$ such that $x = qy$.

Proof
(1) $\implies$ (2): Suppose that $x \sim y$ then $x \mid y$ and $y \mid x$ which means that $(x) \subseteq (y)$ and $(y) \subseteq (x)$.

(2) $\implies$ (3): Suppose that $(x) = (y)$. Then since $x \in (y)$, there exists $s \in R$ such that $x = sy$ and likewise $y = tx$. Then $x = stx$ and since R is an integral domain, $st = 1$ which means that s, t are units.

$(3) \implies (1)$: Suppose that $x = qy$ for some unit q. Then clearly $y \mid x$. Since q is a unit, $q^{-1}x = y$ and thus $x \mid y$ which means that x and y are associates. ■

10.3 Primes and Irreducibles

Definition 10.3.1 (Primes)
Let R be a commutative ring. A nonzero element $p \in R$ that is not a unit is said to be a prime if $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Lemma 10.3.2
Let R be an integral domain. Let p be a non-unit. Then p is prime if and only if (p) is a prime ideal.

Proof
Suppose that p is prime. Suppose that $rp \in (p)$. Then $p \in P$ and we are done. Now suppose that (p) is a prime ideal. Suppose that $p \mid ab$. Then $pd = ab$ for some $d \in R$ thus $ab \in (p)$. WLOG take $a \in (p)$ by definition of prime ideal. Then we are done since $a \in (p)$ implies $p \mid a$. ■

Definition 10.3.3 (Irreducibles)
Let R be an integral domain. A nonzero element $p \in R$ that is not a unit is said to be irreducible if $p = ab$ implies a or b is a unit.

Proposition 10.3.4
Let R be an integral domain and $p \in R$. If p is a prime then p is irreducible.

Proof
Let p be a prime in R. Suppose that $p = ab$. Then trivially $p \mid ab$ thus $p \mid a$ or $p \mid b$. WLOG take $p \mid a$. Trivially $a \mid p$ since $a \mid ab$. This means that a and p are associates. Thus $p = aq$ for some unit q. Then since $aq = ab$ and integral domains have cancellation law, we must have $q = b$ which means that b is a unit. Thus p is irreducible. ■

10.4 Unique Factorization Domains

Definition 10.4.1 (Unique Factorization Domains)
Let R be an integral domain. We say that R is a unique factorization domain (UFD) if the following are true.

- Let $a \in R$ such that $a \neq 0$ and a is not a unit. Then a can be written as the product of irreducible elements in R.
- Let $a = p_1 \cdots p_r = q_1 \cdots q_s$, where p_i and q_j are irreducible. Then $r = s$ and there is a permutation such that $p_i = q_{\pi(i)}$ for $i \in \{1, \dots, r\}$.

Notice that in general integral domains, primes are not the same as irreducibles. But they have the nice property that they coincide in UFDs, which is why they are put in the chapter on UFDs here. Below gives a full converse to the relation between prime and irreducibles we gave above under the umbrealla that is UFDs.

Proposition 10.4.2
Let R be a UFD and $p \in R$. Then p is a prime if and only if p is irreducible.

Proof
We have already shown the forward implication. Now let p be irreducible. We show that (p) is prime. Let $ab \in (p)$. Then there exists $d \in D$ such that $pd = ab$. We can factorize ab and pd respectively into a product of irreducibles elements in D. But since they are equal, by uniqness of fatorization, p is exactly an associate of one of the irreducibles in the factorization of ab. If p is in the factorization of a then $p \mid a$ and we are done. Otherwise it is in b and we are also done. ■

Proposition 10.4.3
Let R be a UFD. Let $a, b \in R$. Then $\gcd(a, b)$ and $\operatorname{lcm}(a, b)$ exists.

Note: The gcd and lcm need not be unique (they are unique up to multiplication by a unit).

10.5 Principal Ideal Domains

Definition 10.5.1 (Principal Ideal Domains)
Let R be an integral domain. We say that R is a principal ideal domain (PID) if for ideal $I \subseteq R$, there exists $a \in I$ such that $I = (a)$.

Proposition 10.5.2
Let R be a ring. If R is a PID, then R is a UFD.

Proof
Suppose for a contradiction that x, a non-unit cannot be factorized into a product of irreducibles. Clearly x is not irreducible else a contradiction. Then there exists x_1, y_1 non unit such that $x = x_1 y_1$. Since x is assumed to be not a product of irreducibles, WLOG take x_1 to be not irreducible. Then we can repeat the process to get non units $x_2 y_2$ such that $x_1 = x_2 y_2$. Notice that $(x) \subset (x_1)$ is a proper containment of ideals if $x_1 \mid x$. Then we have a chain of ideals

$$(x) \subset (x_1) \subset (x_2) \subset \dots$$

I claim that

$$I = \bigcup_{k=0}^{\infty} (x_k)$$

is an ideal. Indeed if $r, s \in I$, then $r \in (x_m)$ and $s \in (x_n)$ for some $m, n \in \mathbb{N}$. WLOG take $m \leq n$. Then $(x_m) \subseteq (x_n)$ implies $r, s \in (x_n)$ thus $r + s \in (x_n) \subseteq I$. Also if $t \in R$, then $tr \in (x_m) \subseteq I$ thus I is indeed an ideal.
Since R is a PID, there exists some $d \in I$ such that $I = (d)$. This also means that $d \in (x_m)$ for some $m \in \mathbb{N}$. Then this means that $I = (d) \subseteq (x_m)$. This proves that the chain eventually stops and this is a contradiction since we assumed that the chain of ideals are properly contained.
This menas that x can indeed be factorized into a product of irreducibles. ■

Notice that the key in the proof is that the union of the countably finite principal ideals is again a principal ideals which allows the infinite chain of ideals to stop.

Lemma 10.5.3 (Bezout's Lemma)
Let R be a PID and $x, y \in R$. Then there exists $r, s \in R$ such that

$$\gcd(x, y) = rx + sy$$

Proof
Let $x, y \in R$. Then $(x) + (y)$ is an ideal of R, thus it must be prciniple, say $(d) = (x) + (y)$. Similarly, $(x) \cap (y)$ is also an ideal, say $(l) = (x) \cap (y)$.
We prove that d and l are the gcd and lcm respectively. Trivially, $(x) \subseteq (d)$ and $(y) \subseteq (d)$ implies $d \mid x$ and $d \mid y$. Also for any z such that z divides x and y, $(x) \subseteq (z)$ and $(y) \subseteq (z)$ thus $(d) \subseteq (z)$. The proof is similar for (l).
Since $(d) = (x) + (y)$, there exists $r, s \in (x), (y)$ respectively such that $d = rx + sy$ and we are done. ■

Proposition 10.5.4
Let R be a PID and (p) a nonzero ideal in R. Then the following are equivalent.
- (p) is maximal ideal.
- p is irreducible.
- p is prime.

- (p) is a prime ideal.

Proof
We have seen that every maximal ideal is a prime ideal. We have seen that if (p) is a prime ideal then p is prime. We have also seen that every prime is irreducible. Thus we have proved $(1) \implies (4) \iff (3) \implies (2)$.

We first show that $(2) \implies (3)$. Suppose that p is irreducible. Suppose that $p \mid ab$ for $a, b \in R$. By the above proposition, $d = \gcd(p, a)$ exists. Then for some $t \in R$, $p = dt$. Since p is irreducible, either d or t is a unit. We consider both cases. If t is a unit, then p and d are associates and thus $p \mid d$. Since $d \mid a$, we have that $p \mid a$ and we are done. Now if d is a unit, then $d = ra + sp$ for some $r, s \in R$. Multplying both sides with b gives $db = rab + spb$. Since $p \mid ab$ and $p \mid spb$, we have that $p \mid db$. Then $pu = db$ for some unit $u \in R$. Since d is a unit, we have that $d^{-1}pu = b$, which means that $p \mid b$ and we are done.

Now we show that if p is a prime then (p) is maximal. We know that (p) is a prime ideal. Suppose that $(p) \subseteq (q) \subseteq R$ for some ideal q of R. Since $p \in (q)$, $p = tq$ for some $t \in R$. Since $p \in (p)$, we have that $tq \in (p)$. Now (p) is prime implies that either $t \in (p)$ or $q \in (p)$. If $q \in (p)$ then $(q) \subseteq (p)$ and thus $(q) = (p)$ and we are done. If $t \in (p)$, then $t = rp$ for some $r \in R$, which means that $p = rpq$. Then $rq = 1$ by cancallation law in integral domains. This means that $1 \in (q)$ thus $(q) = R$ and we are done. ■

10.6 Euclidean Domains

Technically, division algorithms can exist in general domains so long that it has the notion of division. But without a measurement of size to guarantee division is taking larger numbers into smaller numbers, we cannot promise that division algorithms will halt eventually. Therefore we restrict the notion of division algorithm only to integral domains that has a notion of size.

Definition 10.6.1 (Euclidean Valuation)
Let R be an integral domain. A function $f : R \setminus \{0\} \to \mathbb{N}$ is said to be a Euclidean Valuation of R if

- $f(ab) \geq f(b)$ for all $a, b \in R \setminus \{0\}$
- For all $a, b \in R$ with $b \neq 0$, there exists q, r such that
$$a = qb + r$$
with $r = 0$ or $f(r) < f(b)$

In the above definition the second item is simply the division algorithm, with size of a number decided by the function f. Thus the Euclidean domain is simply an integral domain that possess a division algorithm.

Definition 10.6.2 (Euclidean Domain)
Let R be an integral domain. We say that R is a Euclidean domain if R admits a Euclidean valuation.

Proposition 10.6.3
Every Euclidean Domain is a PID.

Proof
Let R be a Euclidean domain. Let f be its Euclidean valuation. Let $I \subseteq R$ be an ideal. If $I = (0)$ then we are done. If $I \neq (0)$, then the set $\{f(x) \mid x \in I\} \subseteq \mathbb{N}$ has a minimal positive element by the well ordering principle. Choose $x \in I$ such that $f(x)$ is a minimal element of the set. I claim that $I = (x)$. The inclusion $(x) \subseteq I$ is clear. Conversely, let $0 \neq y \in I$. By the definition of the Euclidean domain, there exists $q, r \in R$ such that $y = qx + r$ where $r = 0$ or $f(r) < f(x)$. Since $f(x)$ is

assumed to be minimal positive, we must have $r = 0$. Hence $y = qx$ so that $y \in (x)$. Hence $I = (x)$.

Thus any arbitrary ideal is principle. ∎

Definition 10.6.4 (The Euclidean Algorithm)
Let R be a Euclidean domain. Let f be its Euclidean valuation. Let $x, y \in R$. The Euclidean algorithm is the algorithm defined as follows.

1. By definition of Euclidean domains, there exists $q_1, r_1 \in R$ such that
$$y = q_1 x + r_1$$
where $r_1 = 0$ or $f(r_1) < f(x)$.
2. Inductively, assume that that we have produced $q_1, \dots, q_k$ and $r_1, \dots, r_k$ such that $f(r_1) > \dots > f(r_k)$. By definition of Euclidean domains, there exists $q_{k+1}, r_{k+1} \in R$ such that
$$r_{k-1} = q_{k+1} r_k + r_{k+1}$$
where $r_{k+1} = 0$ or $f(r_{k+1}) < f(r_k)$.
3. The process terminates eventually since $f(r_i)$ is strictly decreasing in $\mathbb{N}$.

Proposition 10.6.5
Let R be a Euclidean domain. Let $x, y \in R$. Let d be the last non-zero remainder in the Euclidean algorithm. Then $d = \gcd(x, y)$.

In general, we have that

$$\text{Fields} \subset \begin{matrix}\text{Euclidean}\\\text{Domains}\end{matrix} \subset \begin{matrix}\text{Principle Ideal}\\\text{Domains}\end{matrix} \subset \begin{matrix}\text{Unique Factorization}\\\text{Domains}\end{matrix} \subset \begin{matrix}\text{Integral}\\\text{Domains}\end{matrix}$$

in which all containments are strict. The following summarizes the properties of the different types of domains.

Integral Domains:

- p is a prime if and only if (p) is prime.
- If p is prime the p is irreducible.

Unique Factorization Domains:

- Unique factorization into irreducibles.
- p is prime if and only if p is irreducible.
- $\gcd(a, b)$ and $\text{lcm}(a, b)$ exists.

Principle Ideal Domains:

- Bezout's lemma.
- I is a maximal ideal if and only if I is a prime ideal.

Euclidean Domains:

- The Euclidean Algorithm.
- $\gcd(a, b)$ can be computed using the Euclidean algorithm.

11 The Ring of Polynomials

11.1 Polynomials over General Rings

Definition 11.1.1 (The Ring of Polynomials)
Let R be a ring. Define the ring of polynomials over R to be the set

$$R[x] = \left\{ \sum_{i=0}^{n} a_i x^i \;\middle|\; a_0, \dots, a_n \in R, a_n \neq 0 \right\}$$

together with two binary operations defined as follows.

- For two elements $\sum_{i=0}^{n} a_i x^i$ and $\sum_{j=0}^{m} b_j x^j$, define their sum to be

$$\sum_{i=0}^{n} a_i x^i + \sum_{j=0}^{m} b_j x^j = \sum_{k=0}^{\max\{n,m\}} (a_k + b_k) x^k$$

- For two elements $\sum_{i=0}^{n} a_i x^i$ and $\sum_{j=0}^{m} b_j x^j$, define their product to be

$$\sum_{i=0}^{n} a_i x^i \cdot \sum_{j=0}^{m} b_j x^j = \sum_{k=0}^{n+m} \left(\sum_{s=0}^{k} a_s b_{k-s} \right) x^k$$

Elements of the set are called polynomials.

Lemma 11.1.2
Let R be a commutative ring with identity. Then $R[x]$ is a commutative ring with identity.

Proof
Since $R \leq R[x]$ by considering all the constant polynomials, the identity of R is also the identity of $R[x]$. Let $f, g \in R[x]$. Then the coefficient of x^n in $f(x)g(x)$ is

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

Since $a_k b_{n-k} = b_{n-k} a_k$, we have that $f(x)g(x) = g(x)f(x)$. ∎

Proposition 11.1.3
Let R be a commutative ring. Let I be an ideal of R. Then the following are true.

- $I[x]$ is an ideal of R
- There is an isomorphism $\frac{R[x]}{I[x]} \cong \frac{R}{I}[x]$ given by the map $\left(f = \sum_{k=0}^{n} a_k x^k + I[x]\right) \mapsto \left(\sum_{k=0}^{n} (a_k + I) x^k\right)$
- If I is a prime ideal of R, then $I[x]$ is a prime ideal of $R[x]$.

Definition 11.1.4 (The Evaluation Map)
Let R be a ring and let a be an element in the center $Z(R)$ of R. Define the evaluation map $\phi_a : R[x] \to R$ by the formula

$$\phi_a \left(\sum_{k=0}^{n} c_k x^k \right) = \sum_{k=0}^{n} c_k a^k$$

Proposition 11.1.5
Let R be a ring and let a be an element in the center $Z(R)$ of R. Then the evaluation homomorphism $\phi_a : R[x] \to R$ is a surjective ring homomorphism.

Proof
Let $f, g \in R[x]$ be given as $f(x) = \sum_{k=0}^{n} c_k x^k$ and $g(x) = \sum_{i=0}^{m} b_i x^i$. Then we have

$$\begin{aligned} \phi_a(f+g) &= \phi_a\left(\sum_{k=0}^{n} c_k x^k + \sum_{i=0}^{m} b_i x^i\right) \\ &= \phi_a\left(\sum_{k=0}^{\max\{m,n\}} (c_k + b_k) x^k\right) \\ &= \sum_{k=0}^{\max\{m,n\}} (c_k + b_k) a^k \\ &= \sum_{k=0}^{n} c_k a^k + \sum_{i=0}^{m} b_i a^i \\ &= \phi_a(f) + \phi_a(g) \end{aligned}$$

Moreover, we have that

$$\begin{aligned} \phi_a(fg) &= \phi_a\left(\sum_{k=0}^{n} c_k x^k \sum_{i=0}^{m} b_i x^i\right) \\ &= \phi_a\left(\sum_{k=0}^{m+n}\left(\sum_{i=0}^{k} c_i b_{k-i}\right) x^k\right) \\ &= \sum_{k=0}^{m+n}\left(\sum_{i=0}^{k} c_i b_{k-i}\right) a^k \\ &= \sum_{k=0}^{n} c_k a^k \sum_{i=0}^{m} b_i a^i && (a \in Z(R)) \\ &= \phi_a(f)\phi_a(g) \end{aligned}$$

Thus ϕ_a is a ring homomorphism. To show that ϕ_a is surjective, let $b \in R$. Then clearly $\phi_a(x - a + b) = b$. ∎

Definition 11.1.6 (Leading Term and Leading Coefficient)
Let R be a ring. Let $f = \sum_{k=0}^{n} a_k x^k \in R[x]$ be a polynomial over R.
- We say that the leading term of f is $a_n x^n$.
- We say that the leading coefficient of f is a_n.

Definition 11.1.7 (Monic Polynomials)
Let R be a ring. Let $f \in R[x]$ be a polynomial. We say that f is monic if the leading coefficient of f is 1.

Definition 11.1.8 (Degree of a Polynomial)
Let R be a ring. Let $f \in R[x]$ be written as $f = \sum_{k=0}^{n} a_k x^k$ so that $a_n \neq 0$. Then define the degree of f to be

$$\deg(f) = n$$

If $f = 0$, then by convention $\deg(f) = -\infty$.

Lemma 11.1.9
Let R be a ring. Let $f, g \in R[x]$. Then the following are true.
- $\deg(f+g) \leq \max\{\deg(f), \deg(g)\}$.
- $\deg(fg) \leq \deg(f) + \deg(g)$.

Proof
If $f = \sum_{i=0}^{n} a_i x^i$ and $g = \sum_{j=0}^{m} b_j x^j$, then $f + g = \sum_{k=0}^{\max\{n,m\}} (a_k + b_k)x^k$. Hence $\deg(f + g) \leq \deg(f) + \deg(g)$. By a similar method we can see that $\deg(fg) \leq \deg(f) + \deg(g)$. ∎

11.2 The Roots of a Polynomial

Theorem 11.2.1 (The Division Algorithm)
Let R be a commutative ring. Let $f, g \in R[x]$. Suppose that $\deg(f) \geq \deg(g)$. If the leading coefficient of g is a unit, then there exists unique polynomials $q, r \in R[x]$ such that

$$f(x) = q(x)g(x) + r(x)$$

and $\deg(r) < \deg(g)$.

Proof
We proceed by induction on $\deg(f)$. If $\deg(f) = 0$, then $\deg(g) = 0$ so that g is a unit. Then $q = fg^{-1}$ and $r = 0$ completes existence part of the base case.

Now suppose that the result is true for all polynomials f such that $\deg(f) < n$. Now let f be a polynomial of degree n. Let g be a polynomial of degree $m < n$. Let b be the leading coefficient of g. By assumption, it is a unit. Let a be the leading coefficient of f. Then the polynomial $ab^{-1}x^{n-m}g$ is a polynomial of degree n with leading coefficient a. We may apply the inductive hypothesis to $f - ab^{-1}x^{n-m}g$ to deduce $q', r \in R[x]$ such that

$$f - ab^{-1}x^{n-m}g = q'g + r$$

where $\deg(r) < \deg(g)$. Set $q = ab^{-1}X^{n-m} + q'$. Then we have $f = qg + r$. Thus the induction is complete.

For uniqueness, suppose that $f = q_1 g + r_1 = q_2 g + r_2$ for some $q_1, q_2, r_1, r_2 \in R[x]$ such that $\deg(r_1), \deg(r_2) < \deg(g)$. Suppose for a contradiction that $q_1 \neq q_2$ and $r_1 \neq r_2$. This implies that $\deg(q_1 - q_2) \geq 0$ and $\deg(r_1 - r_2) \geq 0$. Rewriting the equation gives $(q_1 - q_2)g = r_2 - r_1$. Now since the leading coefficient of g is a unit, by a simple expansion, we see that the coefficient of the $\deg(q_1 - q_2) + \deg(g)$ term is non-zero. Hence we have

$$\deg(q_1 - q_2) + \deg(g) = \deg((q_1 - q_2)g) = \deg(r_1 - r_2)$$

Since $\deg(r_2 - r_1) \leq \max\{\deg(r_2), \deg(r_1)\} < \deg(g)$, this is a contradiction. ∎

Definition 11.2.2 (The Roots of a Polynomial)
Let R be a commutative ring. Let $f \in R[x]$. We say that $a \in R$ is a root of f if $\phi_a(f) = 0$. In this case we write $f(a) = 0$.

Theorem 11.2.3 (Factor Theorem)
Let R be a commutative ring. Let $f \in R[x]$. Then $a \in R$ is a root of f if and only if $x - a$ divides f.

Proof
Let $a \in R$ be a root of f. By the division algorithm, $f(x) = q(x)(x - a) + r$ for some $q \in R[x]$ and $r \in R$. Then

$$0 = \phi_a(f) = \phi_a(q(x)(x - a) + r) = \phi_a(q(x))\phi_a(x - a) + \phi_a(r) = 0 + r$$

since $\phi_a(x - a) = a - a = 0$. Hence $r = 0$ and we have $f(x) = q(x)(x - a)$. Hence f is divisible by $x - a$.

Now suppose that $x - a$ divides f. Then there exists $q \in R[x]$ such that $f(x) = q(x)(x - a)$. Then it is clear that

$$\phi_a(f) = \phi_a(q)\phi_a(x-a) = 0$$

and so we conclude.

11.3 Polynomials over Integral Domains

Proposition 11.3.1
Let R be a commutative ring. Then R is an integral domain if and only if $R[x]$ is an integral domain.

Proof
Commutativity is clear since for $f, g \in R[x]$, coefficients of the product fg inherits commutativity from R thus $fg = gf$. Now we show that cancellation law exists in $R[X]$. Let $f, g \in R[x]$ such that $fg = 0$. Suppose for a contradiction that $f \neq 0$ and $g \neq 0$. This means that $\deg(f) = n$ and $\deg(g) = m$ for some $n, m \neq 0$. Consider the coefficient of x^{n+m} in fg, which is $a_n b_m$. Since $fg = 0$, $a_n b_m = 0$ and by cancellation law in R either $a_n = 0$ or $b_m = 0$. This contradicts the fact that $\deg(f) = n$ and $\deg(g) = m$ thus we are done.

Now suppose that $R[x]$ is an integral domain. Then R is a subring of $R[x]$ so that R is also an integral domain.

Proposition 11.3.2
Let R be an integral domain. Then if $f \neq 0$ and $g \neq 0$ in $R[x]$, then

$$\deg(fg) = \deg(f) + \deg(g)$$

Proof
Let the leading term of f be $a_n x^n$ and let the leading term of g be $b_m x^m$. By assumption a_n and b_m are non-zero. By definition of products in the polynomial ring, the coefficient of x^k is 0 for $k > n + m$. Since R is an integral domain, we have that a_n and $b_M$non-zero implies that $a_n b_m$ is non-zero. Hence the leading term of fg is $a_n b_m x^{n+m}$. Hence $\deg(fg) = n + m = \deg(f) + \deg(g)$.

Proposition 11.3.3
Let R be an integral domain. Let $f \in R[x]$. Then f has at most $\deg(f)$ roots in R.

Proof
Suppose for a contradiction that $a_1, \dots, a_{\deg(f)+1}$ are the distinct roots of f. Let $g(x) = (x - a_1) \cdots (x - a_{\deg(f)+1})$. By repeated application of the factor theorem, we have that g divides f. But this is a contradiction by 11.1.9.

Proposition 11.3.4
Let R be an integral domain. Then we have

$$R^\times = R[x]^\times$$

Proof
It is clear that $\subseteq$ is trivial. Now let $f \in R[x]$ such that f is a unit. Suppose that $fg = 1$ for some $g \in R[x]$. Write $f = \sum_{k=0}^n a_k x^k$ and $g = \sum_{i=0}^m b_i x^i$ for $a_n \neq 0$. comparing top coefficients with fg and 1 shows that $a_n b_m = 0$. Since R is an integral domain, we conclude that $b_m = 0$. We induct decreasingly on the coefficients of g. Suppose we showed that $b_n = \cdots = b_{k+1} = 0$ for some $k \geq 0$. Then the coefficient of x^{k+n} in fg is given by

$$a_{k+n-m} b_m + \cdots + a_{n-1} b_{k+1} + a_n b_k$$

Comparing coefficients with the polynomial 1 shows that the above expression is 0. Since $b_n = \cdots = b_{k+1} = 0$, we are left with $a_n b_k = 0$. Since $a_n \neq 0$, we have that $b_k = 0$. Thus $b_n = \cdots =$

$b_1 = 0$ by induction. We are left with the fact that g is a constant, say $g(x) = b$. But now $a_0 \cdot b = 1$ implies that b is a unit, hence the constant $g \in R$ is a unit. ■

11.4 Irreducibility over UFDs and Integral Domains

The primary goal of this chapter is to compute results relating to finding out irreducible polynomials in integral domains, rather than investigating the structure of polynomial rings.

Definition 11.4.1 (Primitive)
Let R be an integral domain. Let $f \in R[x]$ be non-zero and is given by $f = \sum_{k=0}^{n} a_k x^k$ for $a_n \neq 0$. We say that f is primitive if $\gcd(a_0, a_1, \dots, a_n)$ is a unit.

In particular, every monic polynomial is primitive.

Lemma 11.4.2
Let R be a UFD. Let $f, g \in R[x]$ be primitive. Then fg is also primitive.

Proof
Let $f = \sum_{k=0}^{n} a_k x^k$ and $g = \sum_{j=0}^{m} b_k x^k$ be polynomials. Then we have $fg = \sum_{i=0}^{n+m} \left(\sum_{s=0}^{i} a_s b_{i-s} \right) x^i$. Suppose for a contradiction that $r \in R$ is an irreducible such that r divides fg. Since f is primitive, r does not divide f. Let $0 \leq p \leq n$ be the smallest number such that r does not divide a_p. Similarly, since g is primitive, there exists a smallest number $0 \leq q \leq m$ such that r does not divide b_q. But the $(p+q)$th coefficient of fg is given by $\sum_{s=0}^{p+q} a_s b_{p+q-s}$ and so it contains $a_p b_q$. Since r does not divide a_p and b_q, it does not divide $a_p b_q$, but r divides all other terms in the sum by assumption. Hence r does not divide the $(p+q)$th coefficient and is a contradiction. ■

Theorem 11.4.3 (Eisentein's Criterion)
Let R be an integral domain. Let $f \in R[x]$ be primitive. Suppose that P is a prime ideal of R such that the following are true.
- $a_n \notin P$.
- $a_0, \dots, a_{n-1} \in P$.
- $a_0 \notin P^2$.

Then f is irreducible in $R[x]$.

Proof
Suppose for a contradiction that f is reducible in $R[x]$. Then there exists non-constant primitive polynomials $p, q \in R[x]$ such that $f = pq$. Then reducing the equation modulo P and using the assumptions, we deduce that $x^n = \overline{pq} \in R/P[x]$. Since R/P is an integral domain, the equation implies that $\overline{p}$ and $\overline{q}$ has 0 constant terms. Hence the constant term of p and q lie in P. But this implies that $a_0 \in P^2$, a contradiction. ■

Example 11.4.4 (
) he following polynomials are irreducible.
- $x^4 + 3x + 6 \in \mathbb{Z}[x]$.
- $x^4 + 1 \in \mathbb{Z}[x]$.

Proof
Apply Eisenstein's criterion with $P = (5)$.

Let $f(x) = x^4 + 1$. Apply Eisenstein's criterion to $f(x+1)$ with $P = (2)$. Since any factorization of $f(x+1)$ gives a factorization of $f(x)$ by replacing x with $x - 1$, we conclude that $f(x)$ is irreducible. ■

Theorem 11.4.5 (Generalized Gauss's Lemma)
Let R be a UFD. Let $f \in R[x]$ be primitive. Then f is irreducible in $R[x]$ if and only if f is irreducible in $\text{Frac}(R)[x]$.

Proof
Let $f \in R[x]$ be primitive. If $\deg(f) = 0$, then f is a unit and so is not irreducible in both $R[x]$ and $\text{Frac}(R)[x]$.

Now suppose that $\deg(f) > 0$. If f is reducible in $R[x]$, then there exists non-units $g, h \in R[x]$ such that $f = gh$. Since f is primitive and $\deg(f) > 0$, we must have that $\deg(g), \deg(h) > 0$. Hence $f = gh$ is a valid non-constant factorization of f in $\text{Frac}(R)[x]$. Hence f is reducible in $\text{Frac}(R)[x]$.

Assume that f is primitive and reducible in $\text{Frac}(R)[x]$. Then there exists non-units $g, h \in \text{Frac}(R)[x]$ such that $f = gh$. Write $g = \sum_{k=0}^n \frac{r_k}{s_k} x^k$ and $h = \sum_{j=0}^m \frac{p_j}{q_j} x^j$ for $r_k, s_k, p_j, q_j \in R$. Let $l_1 = \text{lcm}(s_1, \dots, s_n)$ and $l_2 = \text{lcm}(q_1, \dots, q_m)$. Then we have $l_1 g, l_2 h \in R[x]$. Let d_1 be the gcd of the coefficients of $l_1 g$ and let d_2 be the gcd of the coefficients of $l_2 h$. Then we have $\frac{l_1}{d_1} g, \frac{l_2}{d_2} h \in R[x]$. Moreover, they are primitive by assumption. Since the product of primitive polynomials is primitive, we know that $f' = \frac{l_1 l_2}{d_1 d_2} gh$ is primitive. Multiplying both sides with $d_1 d_2$ gives $d_1 d_2 f' = l_1 l_2 f$. Since f' is primitive, this equation shows that $d_1 d_2$ is the gcd of the coefficients of $l_1 l_2 f$. Also since f is primitive, the gcd of the coefficients of $l_1 l_2 f$ is given by $l_1 l_2$. Since gcd is unique up to multiplication of a unit, we know that $u = \frac{l_1 l_2}{d_1 d_2}$ is a unit in R. Hence we obtain a factorization $f = u \left(\frac{l_1}{d_1} g \right) \left(\frac{l_2}{d_2} h \right)$ of f in R. So f is reducible in $R[x]$. ■

The remainder of this section investigates irreducibility of polynomial rings over UFDs.

Proposition 11.4.6 (The Reduction Test)
Let R be an integral domain. Let $I \subseteq R$ be an ideal. Let $p \in R[x]$ be a non-constant monic polynomial. If $\overline{p} \in (R/I)[x]$ is irreducible, then p is irreducible in $R[x]$.

Proof
Suppose that p is reducible. Then there exists monic non-constant polynomials $f, g \in R[x]$ such that $p = fg$. Then we have $\overline{p} = \overline{fg} = \overline{f}\overline{g}$. Since f and g are monic, we must have $\deg(\overline{f}) = \deg(f)$ and similarly for g. Hence $\overline{p}$ is reducible. ■

The converse to the above proposition is in general false.

Example 11.4.7
The polynomial $x^2 + 1 \in \mathbb{Z}[x]$ is irreducible but is reducible in $\mathbb{Z}/2\mathbb{Z}[x]$.

Notice that even if the above tests does not show a positive result in irreducibility, it does not mean that the polynomial in question is irreducible.

Proposition 11.4.8 (Rational Root Test)
Let R bea UFD. Let $f = \sum_{k=0}^n a_k x^k \in R[x]$ be a polynomial. Let $r/s \in \text{Frac}(R)$ be a fraction such that $\gcd(r, s)$ is a unit. If r/s is a root of f, then $r \mid a_0$ and $s \mid a_n$.

Proof
If $f(r/s) = 0$, then we have

$$0 = a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_1 r^1 s^{n-1} + a_0 s^n$$

Since r does not divide s and r divides 0 and $a_n r^n + \cdots + a_1 r^1 s^{n-1}$, we must have $r \mid a_0$. Similarly we conclude that $s \mid a_n$. ■

11.5 Irreducibility over a Field

Proposition 11.5.1
Let F be a field. Then $F[x]$ is an Euclidean domain with valuation given by

$$\deg : F[x] \to \mathbb{N}$$

that sends f to $\deg(f)$.

Proof
Since F is also an integral domain, we know that $\deg(fg) = \deg(f) + \deg(g) \geq \deg(f), \deg(g)$. The division algorithm in this case is given by the division algorithm in thm11.2.2. ■

A number of results follows from this. For instance,

- $F[x]$ is a PID, a UFD and an integral domain.
- $f \in F[x]$ is irreducible if and only if (f) is maximal if and only if f is prime.
- Every $f \in F[x]$ has at most $\deg(f)$ roots.
- The gcd and hcf of two elements exists, Bezout's lemma is also applicable.
- Factor theorem and Remainder theorem both hold.
- The Euclidean algorithm is applicable.

Proposition 11.5.2
Let R be a commutative ring. Then R is a UFD if and only if $R[x]$ is an UFD.

Proof
If $R[x]$ is a UFD, then for any $r \in R$ the unique factorization of r in $R[x]$ also exists in R by degree considerations.

Let R be a UFD. Let $p(x) = \sum_{k=0}^{\infty} a_k x^k \in R[x]$ be non-zero and $a_n \neq 0$. Let $d = \gcd(a_0, \dots, a_n)$. Let $q = p/d$ so that $p = dq$. Since $d \in R$ and irreducibles in R are irreducibles in $R[x]$, it suffices to prove that q can be factored uniquely into irreducibles in $R[x]$. Since Frac$(R)[x]$ is a UFD, q can be factored uniquely into irreducibles in Frac$(R)[x]$. From the proof of Gauss's lemma, we know that q can be factored into irreducibles in $R[x]$.

It remains to show uniqueness of the factorization. Suppose that

$$q = p_1 \cdots p_n = q_1 \cdots q_m$$

are two factorizations of q into irreducibles in $R[x]$. By Gauss's lemma, each p_i, q_j are irreducible in Frac$(R)[x]$. By unique factorization in Frac$(R)[x]$, we must have $n = m$ and each p_i and q_i are associates in Frac(R). So there exists $a_i, b_i \in R$ with $b_i \neq 0$ such that $p_i = \frac{a_i}{b_i} q_i$. Rewriting the equation gives $b_i p_i = a_i q_i$. Since q is primitive, we know that p_i and q_i are primitive. Hence the gcd of the coefficients of $b_i p_i$ is b_i. Similarly, the gcd of the coefficients of $a_i q_i$ is a_i. Hence a_i and b_i are associates. This implies that p_i and q_i are associates, and that the two factorizations of q are multiples of each other by a unit. ■

Proposition 11.5.3
Let F be a field. Let $f \in F[x]$. If $\deg(f) = 1$ so that f is linear, then f irreducible.

Proof
Follows immediately from degree considerations. ■

Proposition 11.5.4
Let F be a field. Let $f \in F[x]$ such that $2 \leq \deg(f) \leq 3$. Then f is irreducible if and only if f has no roots in F.

Proof
Follows immediately from the Factor Theorem. ■

This criteria is useful in conjunction with the reduction test.

Example 11.5.5
The polynomial $f(x) = x^2 + x + 1 \in \mathbb{Z}[x]$ is irreducible.

Proof
We reduce it to modulo 2 and apply the above proposition to see that $\overline{f}\mathbb{Z}/2\mathbb{Z}[x]$ has no roots because $\mathbb{Z}/2\mathbb{Z}$ only has two elements. Hence by the reduction test, f is irreducible. ■

For polynomials of higher degrees, there is no general correspondence between roots and irreducibility.

Example 11.5.6
The polynomial $f(x) = (x^2 + 1)(x^2 + 2) \in \mathbb{R}[x]$ has no roots in $\mathbb{R}$ but is clearly reducible.

12 Field Theory

12.1 Properties of a Field

Recall that a field is a commutative division ring, denoted $(F, +, \times)$. In particular, it satisfies the following:

- $(F, +)$ is an abelian group with identity 0
- $(F \setminus \{0\}, \times)$ is an abelian group with identity 1
- (Distributivity) $a(b + c) = ab + ac$ for $a, b, c \in F$

Definition 12.1.1 (Subfields)
Let F be a field. A subfield of F is a subring $E \subseteq F$ such that E is a field.

Proposition 12.1.2
A commutative ring is a field if and only if its only ideals are the trivial ideal and the ring itself.

Proof Let F be a field. Suppose that I is a non trivial ideal of F. Then there exists $i \in I$. Since I is an ideal, then $i^{-1}I \subseteq I$ thus $i^{-1}i = 1 \in I$. But 1 in I implies that $I = F$ thus we are done. Suppose that the non trivial ideal of a commutative ring R is only R itself. We want to show that every element except 0 is a unit. Let $r \in R \setminus \{0\}$. Then the principal ideal (r) is nonzero and thus $(r) = R$. Then there exists $b \in R$ such that $rb = br = 1$. Thus b is an inverse of r. ■

Proposition 12.1.3 (Characteristic)
Let F be a field. Then either $\text{char}(F)$ is either 0 or a prime number.

Proof Suppose that $\text{char}(F) = a \cdot b \neq 0$. Then $ab \cdot 1 = 0$ implies that $a = 0$ or $b = 0$, a contradiction. ■

As usual in understanding different structures in algebra, it is important to study the homomorphisms betweens the structure.

Definition 12.1.4 (Field Homomorphisms)
Let F, K be fields. A field homomorphism from F to K is a ring homomorphism $\phi : F \to K$.

Lemma 12.1.5
Let F, K be fields and $\phi : F \to K$ a field homomorphism. Then ϕ is injective.

Proof Suppose that $a \in \ker(\phi)$. Then we have $\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = 1$ and $\phi(a)\phi(a^{-1}) = 0$. Thus there is a contradiction. ■

In particular, a field homomorphism $\varphi : F \to K$ means that F embeds into K since F is now isomorphic to $\varphi(F)$ which is a subfield of K. Instead of writing $\varphi(F)$ a subfield of K we will write $F \subset K$ instead. This embedding however depends on φ.

12.2 New Fields from Old

Definition 12.2.1 (Prime Subfields)
Let F be a field. Define the prime subfield F_0 of F to be the smallest subfield of F containing 1.

Lemma 12.2.2
Let F be a field. The prime subfield of F is equal to

$$F_0 = \bigcap_{K < F} K$$

Proof The fact that it is a field itself is trivial. Let $L = \bigcap_{K<F} K$. Then $L < F_0$ is trivial. Now $1 \in L$ since L is a subfield of F_0. Since F_0 is the smallest subfield containing 1, $F_0 < L$ and we are done. ■

Proposition 12.2.3
The prime subfield of a field F is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$

13 The Ring of Matrices

13.1 Matrices and its Ring Structure

Definition 13.1.1 (Matrix)
Let R be a ring. Let $m, n \in \mathbb{N}^+$. An $m \times n$ matrix A is a rectangular array of m rows and n columns of elements of R, denoted by

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

For $i = 1, \dots, m$ and $j = 1, \dots, n$, define

$$r_i = \begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix} \quad \text{and} \quad c_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

Then r_i is called the ith row of A and c_j is called the jth row of A. The element of A at the intersection of the ith row and jth column is called the (i, j)th entry of A. The set of all $m \times n$ matrices over R is denoted by $M_{m\times n}(R)$. We sometimes denote A as $(a_{i,j})_{m\times n}$

Definition 13.1.2 (Matrix Addition)
Let R be a ring. Let $m, n \in \mathbb{N}^+$. Define matrix addition to be the binary operation $+ : M_{m\times n}(R) \times M_{m\times n}(R) \to M_{m\times n}(R)$ given by

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$$

Definition 13.1.3 (Scalar Multiplication)
Let R be a ring. Let $m, n \in \mathbb{N}^+$. Define scalar multiplication to be the function $\cdot : \mathbb{R} \times M_{m\times n}(\mathbb{R}) \to M_{m\times n}(\mathbb{R})$ given by

$$\lambda \cdot (a_{ij}) = (\lambda a_{ij})$$

Proposition 13.1.4
Let R be a ring. Let $A, B \in M_{m\times n}(\mathbb{R})$ and $\lambda, \mu \in \mathbb{R}$. Then
- $(\lambda\mu)A = \lambda(\mu A)$
- $\lambda(A + B) = \lambda A + \lambda B$
- $(\lambda + \mu)A = \lambda A + \mu A$

Definition 13.1.5 (Matrix Multiplication)
Let R be a ring. Let $n \in \mathbb{N}^+$. Define matrix multiplication to be the binary operation $\cdot : \mathbb{R} \times M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R})$ given by

$$(a_{ij}) \cdot (b_{ij}) = (c_{ij})$$

where

$$c_{i,j} = \sum_{k=1}^{p} a_{ik} b_{kj}$$

Proposition 13.1.6
Let R be a ring. Let $n \in \mathbb{N}^+$. Then $(M_n(R), +, \cdot)$ is a ring.

Definition 13.1.7 (Invertible, Upper Triangular and Diagonal Matrices)
Let R be a ring. Let $n \in \mathbb{N}^+$. Let $A \in M_n(R)$.

- We say that A is invertible if A has a multiplicative inverse.
- We say that A is upper triangular if all entries below the diagonal of A is 0.
- We say that A is diagonal if all entries not on the diagonal is 0.

Definition 13.1.8 (Transpose)
Let R be a ring. Let $n, m \in \mathbb{N}^+$. Let $A = (a_{i,j}) \in M_{n\times m}(R)$ be a matrix. Define the transpose of A to be

$$A^T = (a_{j,i}) \in M_{m\times n}(R)$$

13.2 Determinants

Let $n \in \mathbb{N}^+$. Recall that S_n is the group of all bijections of the set $\{1, \dots, n\}$ together with functional composition.

Definition 13.2.1 (Determinants)
Let R be a ring. Let $A \in M_n(R)$. Define the determinant of A to be

$$\det(A) = \sum_{\phi \in S_n} \operatorname{sgn}(\phi) a_{1,\phi(1)} \cdots a_{n,\phi(n)}$$

Example 13.2.2
Let R be a ring. Let $I \in M_n(R)$ be the identity matrix. Then $\det(I) = 1$.

Proposition 13.2.3
Let R be a ring. Let $A \in M_n(R)$. Let $r_1, \dots, r_n$ be the rows of A. If $r_i = r_j$ for some $i \neq j$, then $\det(A) = 0$.

Proposition 13.2.4
Let R be a ring. Let $A = (a_{i,j}) \in M_n(R)$ be a matrix. If A is upper triangular, then

$$\det(A) = a_{1,1} \cdots a_{n,n}$$

Proposition 13.2.5
Let R be a ring. Let $A, B \in M_n(R)$. Then we have

$$\det(AB) = \det(A)\det(B)$$

Proposition 13.2.6
Let R be a ring. Let $A \in M_n(R)$. Then we have

$$\det\left(A^T\right) = \det(A)$$

13.3 Row Operations

Definition 13.3.1 (Scale Matrix)
Let R be a ring. Let $n \in \mathbb{N}^+$. Let $r \in R$. Define the scale matrix to be the matrix $S_i(r) \in M_n(R)$ given by

$$(S_i(r))_{p,q} = \begin{cases} 1 & \text{if } p = q \neq i \\ r & \text{if } p = q = i \\ 0 & \text{otherwise} \end{cases}$$

The scale matrix is of the form

$$S_i(r) = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & r & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix}$$

Multiplying a matrix $A \in M_n(R)$ with the scale matrix $S_i(r)$ has the effect of multiplying the ith row r_i of A by the scalar $a \in F \setminus \{0\}$.

Lemma 13.3.2
Let F be a field. Let $a \in F^\times$ be non-zero. Then the following are true.
- $S_i(a)^{-1} = S_i(1/a)$.
- $\det(S_i(a)) = a$.

Definition 13.3.3 (Transposition Matrix)
Let R be a ring. Let $n \in \mathbb{N}^+$. Define the transposition matrix to be the matrix $T_{i,j} \in M_n(R)$ given by

$$(T_{i,j})_{p,q} = \begin{cases} 1 & \text{if } i \neq p = q \neq j \text{ or } p = i, q = j \text{ or } p = j, q = i \\ 0 & \text{otherwise} \end{cases}$$

where the diagonals are all 1 except for the i,ith and j,jth element and other elements that are not shown is 0.

The transposition matrix is of the form

$$T_{i,j} = \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 0 & & & 1 & & \\ & & & 1 & & & & \\ & & & & \ddots & & & \\ & & & & & 1 & & \\ & & 1 & & & 0 & & \\ & & & & & & \ddots & \\ & & & & & & & 1 \end{pmatrix}$$

Multiplying a matrix $A \in M_n(R)$ with the transposition matrix $T_{i,j}$ has the effect of interchanging the ith row r_i of A with the jth row r_j of A.

Lemma 13.3.4
Let F be a field. Let $a \in F^\times$ be non-zero. Then the following are true.
- $T_{i,j}^{-1} = T_{i,j}$.
- $\det(T_{i,j}) = -1$.

Definition 13.3.5 (Recombine Matrix)
Let R be a ring. Let $n \in \mathbb{N}^+$. Let $r \in R$. Define the recombine matrix to be the matrix $A_{i,j}(r) \in M_n(R)$ given by

$$(A_{i,j}(r))_{p,q} = \begin{cases} 1 & \text{if } p = q \\ r & \text{if } p = i, q = j \\ 0 & \text{otherwise} \end{cases}$$

The recombine matrix is of the form

$$A_{i,j}(r) = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & & & \\ & & r & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix}$$

Multiplying a matrix $A \in M_n(R)$ with the recombine matrix $A_{i,j}$ has the effect of adding $a \cdot r_j$ to r_i, where $r_1, \dots, r_n$ are the rows of A.

Lemma 13.3.6
Let F be a field. Let $a \in F^\times$ be non-zero. Then the following are true.

- $A_{i,j}(a)^{-1} = A_{i,j}(-a)$.
- $\det(A_{i,j}(a)) = 1$.

Example 13.3.7 (The Vandermonde Matrix)
Let $\mathbb{F}$ be a field. The Vandermonde matrix is given by

$$V = \begin{pmatrix} 1 & a_1 & \cdots & a_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \cdots & a_n^{n-1} \end{pmatrix} \in M_n(\mathbb{F})$$

for $a_1, \dots, a_n \in \mathbb{F}$. Then the determinant of the Vandermonde matrix is given by

$$\det(V) = \prod_{1 \leq i < j \leq n} (a_j - a_i)$$

Proof
When $n = 1, 2$ the results are clear. Let $M \in M_n(\mathbb{F})$ be the matrix on the left. Using elementary row operations, we have

$$\begin{aligned} \det(M) &= \det(A_{2,1}(-a_1) \cdots A_{n,n-1}(-a_1)M) \\ &= \det \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & a_2 - a_1 & \cdots & a_2^{n-2}(a_2 - a_1) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n - a_1 & \cdots & a_n^{n-2}(a_n - a_1) \end{pmatrix} \\ &= \prod_{i=2}^{n} (a_i - a_1) \det \begin{pmatrix} 1 & a_2 & \cdots & a_2^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \cdots & a_n^{n-2} \end{pmatrix} \end{aligned}$$

so by induction we are done. ■

Definition 13.3.8 (Upper Echelon Form)
Let R be a ring. Let $M \in M_n(R)$ be a matrix. We say that M is in upper echelon form if the following are true.

- All zero rows are below all non-zero rows.
- The first non-zero entry of a row is to the right of the first non-zero entry of the row above.
- The first non-zero entry of every row is 1

Definition 13.3.9 (Row-reduced Form)
We say that a matrix in upper echelon form is in row-reduced form if above and below every first

non-zero entry of a row, all entries are 0.

Definition 13.3.10 (Reduction Procedure)
Let $\mathbb{F}$ be a field. Let $m, n \in \mathbb{N}^+$. Let $A = (a_{ij}) \in M_{m\times n}(\mathbb{F})$. Define the reduction procedure on A as follows. Start with $i = j = 1$.

1. If a_{ij} and all entries below are 0, move on pivot to the right to $a_{i,j+1}$ and repeat step 1, or terminate if $j = n$
2. If $a_{ij} = 0$ but $(a_{kj}) \neq 0$ for some $k > i$, apply $T_{i,k}$
3. If $a_{ij} \neq 1$, apply $S_i(a_{ij}^{-1})$
4. If for any $k \neq i$, $a_{kj} \neq 0$, apply $A_{k,i}(-a_{kj})$
5. If $i = m$ or $j = n$ then terminate, else move pivot to $i + 1, j + 1$ and go back to step 1.

13.4 Formula for the Inverse of an Invertible Matrix

Definition 13.4.1 (The Adjoint of a Matrix)
Let R be a field. Let $A \in M_n(R)$. Define the adjoint of A to be

$$\text{adj}(A) = (A_{ij})^T$$

where A_{ij} is given by the formula

$$A_{ij} = (-1)^{i+j}\det\begin{pmatrix} a_{1,1} & \cdots & a_{1,j-1} & a_{1,j+1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,j-1} & a_{i-1,j+1} & \cdots & a_{i-1,n} \\ a_{i+1,1} & \cdots & a_{i+1,j-1} & a_{i+1,j+1} & \cdots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,j-1} & a_{m,j+1} & \cdots & a_{m,n} \end{pmatrix}$$

Proposition 13.4.2
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$. Then we have

$$A \cdot \text{adj}(A) = \det(A) \cdot I_n = \text{adj}(A) \cdot A$$

Proposition 13.4.3
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$. Then A is invertible if and only if $\det(A) \neq 0$. In this case,

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A)$$

Proposition 13.4.4
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$. If A is invertible, then A^T is invertible and we have

$$(A^T)^{-1} = (A^{-1})^T$$

13.5 Direct Sum or Matrices

Definition 13.5.1 (Direct Sum)
Let R be a ring. Let $A \in M_n(R)$ in $B \in M_m(R)$. Define the direct sum of A and B to be

$$A \oplus B = \begin{pmatrix} A & 0_{n\times m} \\ 0_{m\times n} & B \end{pmatrix} \in M_{n+m}(R)$$

Lemma 13.5.2 Let $\mathbb{F}$ be a field. Let $n, m \in \mathbb{N}^+$. Let $\in M_n(\mathbb{F})$ in $B \in M_m(\mathbb{F})$. Then for any $k \in \mathbb{N}^+$, we have that

$$(A \oplus B)^n = A^n \oplus B^n$$

13.6 The General Linear Group and its Subgroups

Definition 13.6.1 (The General Linear Group) Let k be a field. Define the general linear group of $n \times n$ matrices over k to be

$$GL(n, k) = (M_n(k))^\times = \{M \in M_n(k) \mid M \text{ is invertible in } M_n(k)\}$$

Definition 13.6.2 (The Orthogonal Group) Let k be a field. Let $n \in \mathbb{N}^+$. Define the orthogonal group to be

$$O(n, k) = \{M \in GL(n, k) \mid M^T M = I_n\}$$

Definition 13.6.3 (The Special Linear Group) Let k be a field. Let $n \in \mathbb{N}^+$. Define the special linear group to be

$$SL(n, k) = \{M \in GL(n, k) \mid \det(M) = \pm 1\}$$

Definition 13.6.4 (The Special Orthogonal Group) Let k be a field. Let $n \in \mathbb{N}^+$. Define the special orthogonal group to be

$$SO(n, k) = \{M \in GL(n, k) \mid \det(M) = 1\}$$

Lemma 13.6.5 Let k be a field. Let $n \in \mathbb{N}^+$. Then

$$SO(n, k) = O(n, k) \cap SL(n, k)$$